HEISENBERG UNCERTAINTY AND TWISTED MÖBIUS DYNAMICS: ARITHMETIC COCYCLE CURVATURE, MERTENS RECORD LAWS, CRITICAL-SCALE LIMITS, AND GLOBAL QUADRATIC GEOMETRY

Abstract. We study an arithmetic specialization of the Weyl canonical commutation relations under the embedding $n \mapsto (\ell \log n, \kappa\Omega(n))$. The resulting Weyl multiplier induces a twisted Dirichlet convolution and a deformed Möbius inverse z_ϑ. We obtain an exact formula for its second deformation curvature, its prime-log variance representation, multiplicative superadditivity, and a Dirichlet generating function expressed through a prime-log covariance determinant. The associated ordered Euler defect is continued by grouped prime-zeta expansions with complete replica accounting, and the curvature admits a positive Gram polarization coupled to the classical Weil quadratic form.

On the Mertens side, writing $A^*(x) = \max_{n\le x} |M(n)|$ and $\Theta = \sup_{\zeta(\rho)=0} \Re\rho$, classical oscillation and upper-bound inputs give $A^*(x) = x^{\Theta+o(1)}$, uniform scale separation, logarithmic historical collapse, and the record-time law $r_k = k^{1/\Theta+o(1)}$. An auxiliary Weyl path produces the exact central area identity

$$2\mathcal{A}_0(N) = M(N)I(N) - \int_1^N \frac{M(t)^2}{t}\,dt, \qquad I(N) = \int_1^N \frac{M(t)}{t}\,dt,$$

and identifies the original arithmetic cocycle with the first transverse jet. Uniform Selberg–Delange theory for $P_x(z) = \sum_{n\le x} \mu(n)^2 z^{\Omega(n)}$ gives a unique critical root near -1, the reconstruction $1 + \xi_*(x) = M(x)(\log x)^2/x\,(1+o(1))$, and the compact-uniform limit

$$\frac{(\log x)^2 \log\log x}{x} F_{u/\log\log x}(x) \longrightarrow ue^{-u}.$$

Cauchy's formula yields uniform control of the full jet tower and its tails.

For the accumulated first jet we derive an exact Perron–Mellin representation with factors

$$U(s) = -\frac{1}{\zeta(s)} \sum_p \frac{1}{p^s - 1}, \qquad V(s) = \frac{\zeta'(s)}{\zeta(s)^2}.$$

The prime-power part admits the decomposition

$$\sum_p \frac{1}{p^s - 1} = \log\zeta(s) + H(s), \qquad H(s) = \sum_p \sum_{r\ge 2} \Big(1 - \frac{1}{r}\Big) p^{-rs},$$

with H holomorphic in $\Re s > 1/2$. This gives the exact centered factor $U_c(s) = -H(s)/\zeta(s)$, explicit prime-power-dilated Möbius coefficients, and channels $U_r(s) = -P(rs)/\zeta(s)$. The family $\{P(rs)\}_{r\ge 2}$ separates points in $\Re s > 1/2$, while the centered antisymmetric kernel cancels its diagonal contribution identically.

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1. Introduction and principal results

Let Q and P be the standard position and momentum operators on the Schwartz core. The canonical commutation relation $[Q,P]=i\hbar I$ and the Weyl product provide the phase-space structure used throughout. The arithmetic construction inserts $(\log n,\Omega(n))$ directly into this Weyl geometry and follows the induced cocycle through twisted Möbius inversion, Dirichlet summation, Mertens record geometry, and the critical squarefree deformation.

The arithmetic specialization assigns to an integer n the phase-space increments

$$x_n=\ell\log n,\qquad y_n=\kappa\Omega(n), \tag{1.1}$$

and the self-adjoint generator

$$G_n=y_nQ-x_nP. \tag{1.2}$$

The arithmetic specialization of the standard Robertson inequality gives, for two integers,

$$\boxed{\Delta_\psi G_m\,\Delta_\psi G_n\geq\frac{\hbar\ell\kappa}{2}\,|\Omega(m)\log n-\log m\,\Omega(n)|\,.} \tag{1.3}$$

For distinct primes p,q, this becomes

$$\boxed{\Delta_\psi G_p\,\Delta_\psi G_q\geq\frac{\hbar\ell\kappa}{2}\left|\log\frac{q}{p}\right|.} \tag{1.4}$$

Thus the same arithmetic symplectic area that later appears in the twisted Möbius product already appears as a lower bound forced directly by the Heisenberg uncertainty principle.

The associated Weyl operators are

$$W_n=\exp\left(\frac{i}{\hbar}G_n\right), \tag{1.5}$$

and satisfy

$$W_mW_n=\chi_\vartheta(m,n)W_{mn}, \tag{1.6}$$

where

$$\chi_\vartheta(m,n)=\exp\left[\frac{i\vartheta}{2}\big(\Omega(m)\log n-\log m\,\Omega(n)\big)\right],\qquad\vartheta=\frac{\ell\kappa}{\hbar}. \tag{1.7}$$

The cocycle is proved below to be cohomologically nontrivial for every $\vartheta \neq 0$. It induces an associative noncommutative Dirichlet product. Möbius inversion in that product defines coefficients $z_\vartheta(n)$. For prime powers and distinct prime pairs one obtains

$$z_\vartheta(p^r) = 1, \tag{1.8}$$

$$z_\vartheta(pq) = 2\cos\left(\frac{\vartheta}{2}\log\frac{q}{p}\right) - 1. \tag{1.9}$$

The same determinant-like arithmetic quantity therefore controls both the uncertainty lower bound (1.3) and the phase responsible for the deformed inverse.

The deformation also differentiates the twisted inverse itself at the commuting point. For

$$n = \prod_{j=1}^{r} p_j^{a_j}, \qquad T(a) = \frac{a(a+1)}{2}, \tag{1.10}$$

one obtains

$$\boxed{-z_0''(n) = \frac{1}{2}\sum_{1\le i<j\le r} T(a_i)T(a_j)\log^2\frac{p_j}{p_i}.} \tag{1.11}$$

For square-free $n = p_1\cdots p_r$,

$$\boxed{-z_0''(n) = \frac{1}{2}\left[r\sum_{p\mid n}(\log p)^2 - (\log n)^2\right] = \frac{r^2}{2}\operatorname{Var}_{p\mid n}(\log p).} \tag{1.12}$$

Summing this curvature does not return a two-variable Möbius correlation. For $\Re s > 1$,

$$\mathcal{K}(s) = -\sum_{n\ge 1}\frac{z_0''(n)}{n^s} = \frac{\zeta(s)}{2}\big(S_0(s)S_2(s) - S_1(s)^2\big), \tag{1.13}$$

where

$$S_j(s) = \sum_p \frac{p^{-s}}{(1-p^{-s})^2}(\log p)^j. \tag{1.14}$$

Section 12 rewrites these functions through $\log\zeta(rs)$ and $\zeta'(rs)/\zeta(rs)$, producing, after a branch is fixed by continuation from $\Re s > 1$, a local continued germ in $\Re s > 1/2$ and an exact singularity criterion for nontrivial zeta zeros to the right of the critical line.

The positivity layer has two branches. First, the exact cocyclic curvature is polarized into a Gram kernel $K_{\mathcal{C}} \succeq 0$. Second, the transverse operator $\Pi = i\partial_t|_{t=0}$ sends $B(s,t)$ exactly to $\zeta'(s)/\zeta(s)$, eliminating every grouped dilation sector except the primitive one. The corresponding zero kernel is then coupled to $K_{\mathcal{C}}$ entrywise. Positivity of the resulting kernel for the full Weil test class is proved equivalent to RH. The equivalence is not advertised as progress on RH, because a single positive diagonal curvature entry recovers Weil's classical criterion.

The Mertens layer is independent of the zero-kernel formulation. Write

$$M(x) = \sum_{n\le x}\mu(n), \qquad A^*(x) = \max_{n\le x}|M(n)|, \qquad \Theta = \sup_{\zeta(\rho)=0}\Re\rho. \tag{1.15}$$

Using the classical oscillation theorem of Kaczorowski–Pintz together with the standard upper estimate associated with the zero supremum, Section 27 proves

$$\boxed{A^*(x) = x^{\Theta+o(1)}} \tag{1.16}$$

and therefore, for every fixed $0 < \alpha < 1$,

$$\boxed{A^*(x^\alpha) = o(A^*(x)).} \tag{1.17}$$

If r_k is the first point at which $|M|$ attains the integer height k, then

$$\boxed{r_k = k^{1/\Theta+o(1)}} \tag{1.18}$$

and the full logarithmic past is negligible relative to the record height: for every $p > 0$,

$$\boxed{\frac{1}{\log r_k}\int_1^{r_k}\left|\frac{M(t)}{M(r_k)}\right|^p\frac{dt}{t}\longrightarrow 0.} \tag{1.19}$$

In particular, with $I(x) = \int_1^x M(t)dt/t$,

$$\boxed{I(r_k) = o(M(r_k)\log r_k).} \tag{1.20}$$

The same section constructs an auxiliary Weyl path whose ordered symplectic area is $\mathcal{A}_t(N)$. At order zero and first transverse order one obtains, respectively,

$$2\mathcal{A}_0(N) = M(N)I(N) - \int_1^N \frac{M(t)^2}{t}\,dt, \tag{1.21}$$

$$\mathcal{A}_0'(N) = \frac{1}{2}\sum_{m<n\le N}\mu(m)\mu(n)\big(\Omega(m)\log n - \log m\,\Omega(n)\big). \tag{1.22}$$

Thus the original arithmetic cocycle is exactly the first transverse jet of an area whose order-zero term contains the logarithmic history of Mertens. A standard Selberg–Delange input then yields the local reconstruction law

$$\boxed{\Delta\mathcal{A}_0'(N) = \frac{\mu(N)N}{2\log N} + o\Big(\frac{N}{\log N}\Big).} \tag{1.23}$$

2. Notation, domains, and symbol index

This section fixes the notation used throughout the paper.

2.1. **Basic arithmetic and analytic notation.**

$\mu(n)$:	the Möbius function.
$\Omega(n)$:	the completely additive number of prime factors of n, counted with multiplicity.
$\omega(n)$:	the number of distinct prime divisors of n.
$v_p(n)$:	the exponent of p in n.
$\mathrm{rad}(n)$:	the square-free kernel $\prod_{p\mid n} p$.
$\varphi(n)$:	Euler's totient and $J_2(n)$ the second Jordan totient.
$T(a)$:	the triangular weight $a(a+1)/2$.
$\zeta(s)$:	the Riemann zeta function and $L(s) = \zeta'(s)/\zeta(s)$ its logarithmic derivative.
$P(s)$:	the prime-zeta function $\sum_p p^{-s}$ in its initial half-plane $\Re s > 1$, followed by analytic continuation when explicitly stated.
ρ:	a nontrivial zero of ζ; m_ρ denotes its multiplicity.
s, t:	Write $s = \sigma + i\tau$. The real variable t is reserved for the Fourier/vertical shift in $B(s,t)$.

2.2. **Operator and deformation notation.**

$\mathcal{H}$:	$L^2(\mathbb{R}, dq)$.
Q, P:	position and momentum. Their commutator identity is asserted on the Schwartz core $\mathcal{S}(\mathbb{R})$, not on the whole intersection of maximal domains.
$W(x,y)$:	the Weyl displacement and W_n its arithmetic specialization.
ϑ:	the dimensionless deformation parameter $\ell\kappa/\hbar$.
$\chi_\vartheta(m,n)$:	the arithmetic Weyl 2-cocycle.
$\star_\vartheta$:	twisted Dirichlet convolution.

$z_\vartheta(n)$: the coefficientwise twisted inverse of μ.
$\mathcal{C}(n)$: the deformation curvature $-z_0''(n)$.
$\mathcal{K}(s)$: the Dirichlet series of $\mathcal{C}(n)$.

2.3. **Euler-locality and replica notation.**

E_ϑ: the increasing-prime ordered Euler object.
R_ϑ: the Euler-locality defect $z_\vartheta - E_\vartheta$.
$\Phi_{\mathrm{ord}}(n)$: the oriented prime-pair phase of the ordered Euler object.
$\mathcal{O}(s)$: the odd Euler-locality Dirichlet defect, initially holomorphic for $\Re s > 1$.
$u_p(s)$: $p^{-s}/(1-p^{-s}) = 1/(p^s-1)$.
$A(s)$: $\sum_p u_p(s) = B(s,0)$, initially for $\Re s > 1$.
$B(s,t)$: $\sum_p u_p(s)p^{it}$, initially for $\Re s > 1$.
$\mathcal{I}(s)$: the normalized Fourier energy $2\pi\mathcal{O}(s)/\zeta(s)$ in $\Re s > 1$. Its continued germ is the object whose leading local term is naturally of $z^{-1}\log z$-type.
s_0: a fixed point in the strip, and $z = s - s_0$ the local coordinate.
$\mathcal{R}(s_0)$: the finite set of dilation indices $N \geq 1$ for which $\zeta(Ns_0) = 0$.
q_N: the multiplicity of the zero Ns_0, for $N \in \mathcal{R}(s_0)$.
β_r: the strictly negative divisor energy attached to a single pure replica of index $r > 1$.
$\Phi_{pq}(n)$: the prime-pair curvature feature used to polarize $\mathcal{C}(n)$.
$K_{\mathcal{C}}(m,n)$: the positive-semidefinite curvature Gram kernel.
Π: the transverse primitive-replica filter $i\,\partial_t|_{t=0}$.
$G^{\#}(s)$: the Weil involution $\overline{G(1-\overline{s})}$.
$W_g(s)$: the Weil weight $G(s)G^{\#}(s)$ attached to a test function g.
$K_g^Z(m,n)$: the zero kernel built from primitive zeta zeros and the Weil weight.
$\mathscr{K}_g(m,n)$: the entrywise product $K_{\mathcal{C}}(m,n)K_g^Z(m,n)$.

2.4. **Squarefree-deformation notation.**

$P_x(z)$: the finite squarefree generating polynomial $\sum_{n\leq x}\mu(n)^2 z^{\Omega(n)}$.
$F_t(x)$: the signed deformation $P_x(-e^t) = \sum_{n\leq x}\mu(n)e^{t\Omega(n)}$.
$\xi_*(x)$: the unique zero of P_x converging to -1.
h_x: the critical displacement $1 + \xi_*(x)$.
L_x: $\log\log x$ in the critical-scale subsections.
$M_j(x)$: $\sum_{n\leq x}\mu(n)\Omega(n)^j$, the j-th transverse moment.
$\Phi_x(u)$: the critically renormalized deformation $(\log x)^2 L_x F_{u/L_x}(x)/x$.
$T_m(x)$: the root trace $\sum_j (1+\xi_{x,j})^{-m}$, defined when $M(x) \neq 0$.

Warning 2.1 (Normalized energy versus $\mathcal{O}$). The logarithmic-over-pole cancellation is a statement about $\mathcal{I} = 2\pi\mathcal{O}/\zeta$. At an actual zero $s = \rho$, the factor $\zeta(s)$ lowers the singular order of $\mathcal{O}$. Thus a formula containing $\log(s-\rho)/(s-\rho)$ must not be attributed to $\mathcal{O}$ itself at $r = 1$.

3. Exact canonical kinematics

3.1. **Hilbert space, position, and momentum.**

Definition 3.1 (Hilbert space)**.** Fix $\hbar > 0$ and set

$$\mathcal{H} = L^2(\mathbb{R}, dq) \tag{3.1}$$

with its standard complex Hilbert-space inner product.

Definition 3.2 (Position)**.** The position operator Q is

$$(Q\psi)(q) = q\psi(q) \tag{3.2}$$

with maximal domain

$$\mathcal{D}(Q) = \{\psi \in L^2(\mathbb{R}) : q\psi(q) \in L^2(\mathbb{R})\}. \tag{3.3}$$

Definition 3.3 (Momentum)**.** The momentum operator P is

$$(P\psi)(q) = -i\hbar\,\psi'(q) \tag{3.4}$$

with domain

$$\mathcal{D}(P) = H^1(\mathbb{R}). \tag{3.5}$$

Proposition 3.4 (Self-adjointness and common core)**.** *The operators Q and P are self-adjoint. The Schwartz space $\mathcal{S}(\mathbb{R})$ is a common invariant core for both.*

Proof. Q is the maximal multiplication operator by the real function q. Under the unitary Fourier transform with convention

$$(\mathcal{F}\psi)(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{\mathbb{R}} e^{-ipq/\hbar}\psi(q)\,dq, \tag{3.6}$$

P is unitarily equivalent to multiplication by the real variable p. The core statement follows by standard cutoff and mollification arguments. See, for example, [7]. □

Proposition 3.5 (Exact canonical commutation relation)**.** *For every $\psi \in \mathcal{S}(\mathbb{R})$,*

$$[Q, P]\psi = i\hbar\psi. \tag{3.7}$$

Proof. Direct differentiation gives

$$(QP\psi)(q) = -i\hbar q\psi'(q), \tag{3.8}$$

$$(PQ\psi)(q) = -i\hbar\psi(q) - i\hbar q\psi'(q). \tag{3.9}$$

Subtracting (3.9) from (3.8) proves (3.7). □

3.2. The Heisenberg uncertainty principle and exact incompatibility.

Definition 3.6 (Expectation and uncertainty)**.** Let A be self-adjoint and let $\psi \in \mathcal{D}(A)$ be normalized. Define

$$\langle A\rangle_\psi = \langle \psi, A\psi\rangle, \tag{3.10}$$

$$(\Delta_\psi A)^2 = \|(A - \langle A\rangle_\psi)\psi\|^2. \tag{3.11}$$

Theorem 3.7 (Heisenberg uncertainty principle)**.** *For every normalized $\psi \in \mathcal{S}(\mathbb{R})$,*

$$\boxed{\Delta_\psi Q\,\Delta_\psi P \geq \frac{\hbar}{2}.} \tag{3.12}$$

Proof. Set

$$\widetilde{Q} = Q - \langle Q\rangle_\psi I, \qquad \widetilde{P} = P - \langle P\rangle_\psi I. \tag{3.13}$$

By Cauchy–Schwarz,

$$\Delta_\psi Q\,\Delta_\psi P \geq \left|\langle \widetilde{Q}\psi, \widetilde{P}\psi\rangle\right|. \tag{3.14}$$

The modulus dominates the absolute value of the imaginary part, and on $\mathcal{S}(\mathbb{R})$

$$2i\ \mathrm{Im}\langle \widetilde{Q}\psi, \widetilde{P}\psi\rangle = \langle \psi, [Q, P]\psi\rangle \tag{3.15}$$

$$= i\hbar. \tag{3.16}$$

Combining (3.14)–(3.16) proves (3.12). □

Remark 3.8 (Meaning of the inequality). Equation (3.12) is the Heisenberg uncertainty principle used in this paper. The quantities $\Delta_\psi Q$ and $\Delta_\psi P$ are intrinsic standard deviations of a state. They are not measurement errors. Exact simultaneous localization would require both to vanish, which is prohibited by (3.12).

Theorem 3.9 (No simultaneous exact position and momentum)**.** *There is no nonzero state with simultaneous exact position and exact momentum. More precisely, both of the following are impossible for real q_0, p_0:*

(1) a nonzero vector ψ in a common invariant domain satisfying

$$Q\psi = q_0\psi, \qquad P\psi = p_0\psi; \tag{3.17}$$

(2) a nonzero tempered distribution $T \in \mathcal{S}'(\mathbb{R})$ satisfying

$$qT = q_0 T, \qquad -i\hbar T' = p_0 T. \tag{3.18}$$

Proof. For vectors, (3.17) gives $[Q,P]\psi = 0$, whereas the CCR gives $[Q,P]\psi = i\hbar\psi$; hence $\psi = 0$. For distributions define directly

$$QT = qT, \qquad PT = -i\hbar T'. \tag{3.19}$$

The distributional Leibniz rule gives

$$[Q,P]T = i\hbar T. \tag{3.20}$$

The simultaneous eigenvalue equations would make the left-hand side zero, so $T = 0$. □

4. Weyl displacements and the arithmetic phase space

4.1. **Weyl operators.**

Definition 4.1 (Weyl displacement)**.** For $(x,y) \in \mathbb{R}^2$, define

$$W(x,y) = \exp\left(\frac{i}{\hbar}(yQ - xP)\right). \tag{4.1}$$

This is a unitary operator on all of $\mathcal{H}$.

Proposition 4.2 (Coordinate action)**.** *For $\psi \in L^2(\mathbb{R})$,*

$$(W(x,y)\psi)(q) = \exp\left(\frac{i}{\hbar} y\left(q - \frac{x}{2}\right)\right)\psi(q-x) \tag{4.2}$$

almost everywhere.

Proof. Since $[iyQ/\hbar, -ixP/\hbar]$ is scalar, the Baker–Campbell–Hausdorff formula gives

$$W(x,y) = e^{-ixy/(2\hbar)} e^{iyQ/\hbar} e^{-ixP/\hbar}. \tag{4.3}$$

The last factor translates by x and the middle factor multiplies by $e^{iyq/\hbar}$, yielding (4.2). □

Proposition 4.3 (Weyl form of the CCR)**.** *Let $z = (x,y)$ and $z' = (x',y')$. Define*

$$\omega(z,z') = yx' - xy'. \tag{4.4}$$

Then

$$W(z)W(z') = \exp\left(\frac{i}{2\hbar}\omega(z,z')\right) W(z+z'). \tag{4.5}$$

Consequently,

$$[W(z), W(z')] = 2i\sin\left(\frac{\omega(z,z')}{2\hbar}\right) W(z+z'). \tag{4.6}$$

Proof. For

$$A_z = \frac{i}{\hbar}(yQ - xP), \tag{4.7}$$

the CCR gives

$$[A_z, A_{z'}] = \frac{i}{\hbar}\omega(z, z')I. \tag{4.8}$$

This commutator is central, so the Baker–Campbell–Hausdorff series stops after the commutator term and gives (4.5). Reversing z and z' changes the sign of ω, giving (4.6). The Weyl relations are standard; see [6]. □

Proposition 4.4 (Position and momentum displacement equations)**.** *On* $\mathcal{S}(\mathbb{R})$*,*

$$[Q, W(x, y)] = xW(x, y), \tag{4.9}$$

$$[P, W(x, y)] = yW(x, y). \tag{4.10}$$

Proof. Both identities follow by differentiating (4.2), or directly from (3.7) because the relevant commutators with the exponent are scalar. □

4.2. **Arithmetic embedding.**

Definition 4.5 (The additive prime-factor counter)**.** For

$$n = \prod_p p^{v_p(n)}, \tag{4.11}$$

define

$$\Omega(n) = \sum_p v_p(n). \tag{4.12}$$

Then

$$\Omega(mn) = \Omega(m) + \Omega(n). \tag{4.13}$$

Definition 4.6 (Arithmetic phase-space map)**.** Fix two positive scales $\ell > 0$ and $\kappa > 0$. For $n \in \mathbb{N}$, put

$$x_n = \ell \log n, \tag{4.14}$$

$$y_n = \kappa\Omega(n), \tag{4.15}$$

$$z_n = (x_n, y_n). \tag{4.16}$$

Define the dimensionless Heisenberg area parameter

$$\vartheta = \frac{\ell\kappa}{\hbar}. \tag{4.17}$$

Remark 4.7 (Why these two coordinates). The first coordinate is forced by multiplicative size because $\log(mn) = \log m + \log n$. The second coordinate is chosen as the simplest integer-valued completely additive function that directly controls the Möbius sign on square-free integers:

$$\mu(n) \neq 0 \quad \Longrightarrow \quad \mu(n) = (-1)^{\Omega(n)}. \tag{4.18}$$

The construction can be generalized by replacing Ω with another completely additive arithmetic function, but the present paper fixes Ω throughout.

Definition 4.8 (Arithmetic Weyl operators)**.** For $n \in \mathbb{N}$, define

$$W_n = W(x_n, y_n). \tag{4.19}$$

Theorem 4.9 (Exact Heisenberg cocycle on the integers)**.** *For $m, n \in \mathbb{N}$,*

$$W_m W_n = \chi_\vartheta(m,n) W_{mn}, \tag{4.20}$$

where

$$\chi_\vartheta(m,n) = \exp\left[\frac{i\vartheta}{2}\big(\Omega(m)\log n - \log m\,\Omega(n)\big)\right]. \tag{4.21}$$

Moreover,

$$[W_m, W_n] = 2i\sin\left[\frac{\vartheta}{2}\big(\Omega(m)\log n - \log m\,\Omega(n)\big)\right] W_{mn}. \tag{4.22}$$

Proof. Complete additivity gives

$$z_{mn} = z_m + z_n. \tag{4.23}$$

Substitution of (4.14)–(4.17) into (4.5) gives (4.20); (4.22) follows from (4.6). □

Corollary 4.10 (Prime commutator)**.** *For distinct primes p, q,*

$$[W_p, W_q] = 2i\sin\left(\frac{\vartheta}{2}\log\frac{q}{p}\right) W_{pq}. \tag{4.24}$$

Thus the prime operators commute only at the resonant values

$$\vartheta \log\frac{q}{p} \in 2\pi\mathbb{Z}. \tag{4.25}$$

Proof. For primes p, q, one has $\Omega(p) = \Omega(q) = 1$. Substituting these values into the integer Weyl commutator formula of Theorem 4.9 gives (4.24). The commutator vanishes exactly when its sine factor vanishes, which is equivalent to (4.25). □

Remark 4.11 (Where Heisenberg enters nontrivially). Equation (4.20) is the first point where ceases to be ordinary Dirichlet algebra in operator notation. If the canonical pair were commuting, the Weyl cocycle would be 1 and the arithmetic product below would reduce to ordinary Dirichlet convolution. For nonzero ϑ, the CCR creates order-dependent phases between different arithmetic factors.

Theorem 4.12 (The arithmetic cocycle is not gauge-trivial)**.** *Extend* log *and* Ω *additively to* $\mathbb{Q}_{>0}^\times$*. For every real $\vartheta \neq 0$, the cocycle χ_ϑ is not a scalar coboundary on $\mathbb{N}^\times$, and its extension is not a scalar coboundary on $\mathbb{Q}_{>0}^\times$.*

Proof. On a commutative semigroup, every scalar coboundary

$$(\delta a)(m,n) = a(m)a(n)a(mn)^{-1} \tag{4.26}$$

is symmetric in m, n. Hence a necessary condition for χ_ϑ to be a coboundary is

$$\frac{\chi_\vartheta(m,n)}{\chi_\vartheta(n,m)} = 1 \tag{4.27}$$

for all m, n. For distinct primes,

$$\frac{\chi_\vartheta(p,q)}{\chi_\vartheta(q,p)} = \exp\left(i\vartheta\log\frac{q}{p}\right). \tag{4.28}$$

If this were one simultaneously for $(p,q) = (2,3)$ and $(2,5)$, then there would be nonzero integers k, l such that

$$\vartheta\log(3/2) = 2\pi k, \qquad \vartheta\log(5/2) = 2\pi l. \tag{4.29}$$

Thus $\log(3/2)/\log(5/2) = k/l \in \mathbb{Q}$. Exponentiating a reduced rational relation would produce an equality between incompatible prime factorizations, contradicting unique factorization. Therefore (4.27) cannot hold for all primes. □

5. Structural filter: linear Möbius states cannot escape second order

Let $v_n \in \mathcal{H}$ be arbitrary vectors and define

$$\Psi_x = \sum_{n \le x} \mu(n) v_n. \tag{5.1}$$

Theorem 5.1 (Quadraticity of unnormalized sesquilinear moments)**.** *For every linear operator A for which the expression is defined,*

$$\langle \Psi_x, A\Psi_x \rangle = \sum_{m,n \le x} \mu(m)\mu(n) \langle v_m, A v_n \rangle. \tag{5.2}$$

Thus every unnormalized sesquilinear moment, including $\|\Psi_x\|^2$, is quadratic in the Möbius coefficients. Normalized expectations are quotients of such quadratic forms, and variances or Robertson expressions are rational combinations of them; they are not, in general, themselves quadratic forms.

Proof. Expand the inner product sesquilinearly and use that $\mu(n)$ is real. The assertions about normalized quantities follow directly by dividing by $\langle \Psi_x, \Psi_x \rangle$. □

Remark 5.2 (Scope of the linear-state filter)*.* Equation (5.2) is a structural warning, not a no-go theorem for every nonlinear observable constructed from a linear Möbius state. What is ruled out is only the claim that an unnormalized sesquilinear moment escapes second order in the coefficients.

6. Arithmetic uncertainty generated by the CCR

Definition 6.1 (Integer generator)**.** For $n \ge 1$, define on $\mathcal{S}(\mathbb{R})$ the symmetric operator

$$G_n^{(0)} = y_n Q - x_n P = \kappa \Omega(n) Q - \ell \log n \, P. \tag{6.1}$$

It is essentially self-adjoint on $\mathcal{S}(\mathbb{R})$; denote its self-adjoint closure by G_n. The corresponding unitary displacement is

$$W_n = \exp(i G_n / \hbar). \tag{6.2}$$

Proposition 6.2 (Arithmetic commutator)**.** *For all $m, n \ge 1$, the restrictions of G_m and G_n to $\mathcal{S}(\mathbb{R})$ satisfy*

$$[G_m, G_n] = i\hbar (x_m y_n - y_m x_n) I \tag{6.3}$$

$$= i\hbar\ell\kappa \big(\log m \, \Omega(n) - \Omega(m) \log n \big) I. \tag{6.4}$$

Proof. Expand the commutator and use $[Q, P] = i\hbar I$. Terms involving $[Q, Q]$ and $[P, P]$ vanish. □

Theorem 6.3 (Arithmetic Heisenberg uncertainty law)**.** *For every normalized $\psi \in \mathcal{S}(\mathbb{R})$ and all $m, n \ge 1$,*

$$\boxed{\Delta_\psi G_m \, \Delta_\psi G_n \ge \frac{\hbar\ell\kappa}{2} \left| \Omega(m) \log n - \log m \, \Omega(n) \right|.} \tag{6.5}$$

For distinct primes p, q,

$$\boxed{\Delta_\psi G_p \, \Delta_\psi G_q \ge \frac{\hbar\ell\kappa}{2} \left| \log \frac{q}{p} \right|.} \tag{6.6}$$

For two powers of the same prime the lower bound is zero.

Proof. Apply the Robertson inequality to the self-adjoint pair G_m, G_n:

$$\Delta_\psi G_m \, \Delta_\psi G_n \ge \frac{1}{2} \left| \langle \psi, [G_m, G_n] \psi \rangle \right|. \tag{6.7}$$

Substitution of (6.4) gives (6.5). If p, q are distinct primes then $\Omega(p) = \Omega(q) = 1$, which gives (6.6). If $m = p^a, n = p^b$, the determinant $\Omega(m) \log n - \log m \, \Omega(n)$ vanishes. □

Remark 6.4 (The common invariant). Define the arithmetic symplectic area

$$\mathcal{A}(m,n) = \Omega(m)\log n - \log m\,\Omega(n). \tag{6.8}$$

The uncertainty lower bound is proportional to $|\mathcal{A}(m,n)|$, while the twisted product below carries the phase $\exp(i\vartheta\mathcal{A}(m,n)/2)$. Thus uncertainty and twisted Möbius inversion are controlled by the same arithmetic invariant, although they are distinct consequences of the CCR.

6.1. **The semiclassical and arithmetic-commutative limits are distinct.** The parameter controlling the Weyl cocycle is

$$\vartheta(\hbar,\ell,\kappa) = \frac{\ell\kappa}{\hbar}. \tag{6.9}$$

The uncertainty scale and the deformation scale therefore depend on the same three parameters in different ways.

Proposition 6.5 (Opposite fixed-scale limits)**.** *Fix $\ell > 0$ and $\kappa > 0$. For every fixed pair m, n,*

$$\frac{\hbar\ell\kappa}{2}|\mathcal{A}(m,n)| \longrightarrow 0 \qquad (\hbar \to 0^+), \tag{6.10}$$

whereas

$$|\vartheta| = \frac{\ell\kappa}{\hbar} \longrightarrow \infty. \tag{6.11}$$

Consequently the ordinary semiclassical limit $\hbar \to 0^+$ with ℓ, κ fixed does not approach the untwisted Dirichlet product.

Proof. Equation (6.10) follows directly from Theorem 6.3, while (6.11) follows from (6.9). □

Proposition 6.6 (Joint semiclassical–commutative scaling)**.** *Let $\ell = \ell(\hbar) > 0$ and $\kappa = \kappa(\hbar) > 0$. Then*

$$\vartheta(\hbar) \longrightarrow 0 \qquad (\hbar \to 0^+) \tag{6.12}$$

if and only if

$$\ell(\hbar)\kappa(\hbar) = o(\hbar). \tag{6.13}$$

Under this scaling the twisted product converges coefficientwise to the ordinary Dirichlet product, and the arithmetic uncertainty lower bound obeys

$$\frac{\hbar\ell(\hbar)\kappa(\hbar)}{2}|\mathcal{A}(m,n)| = o(\hbar^2) \tag{6.14}$$

for every fixed m, n.

Proof. The equivalence of (6.12) and (6.13) is immediate from (6.9). For fixed m, n, $\chi_\vartheta(m,n) \to 1$ as $\vartheta \to 0$, so the finite divisor sum defining $f \star_\vartheta g$ converges to ordinary Dirichlet convolution coefficientwise. Multiplying $\ell(\hbar)\kappa(\hbar) = o(\hbar)$ by $\hbar/2$ gives (6.14). □

Remark 6.7 (Two independent notions of classicality). The limit $\hbar \to 0^+$ suppresses the Robertson–Heisenberg lower bound, whereas $\vartheta \to 0$ removes the arithmetic twist. They are not the same operation. With fixed ℓ, κ they move in opposite directions; simultaneous suppression requires the joint scaling (6.13).

7. The Heisenberg-twisted Dirichlet algebra

7.1. Definition and associativity.

Definition 7.1 (Heisenberg-twisted Dirichlet convolution)**.** For arithmetic functions $f, g : \mathbb{N} \to \mathbb{C}$, define

$$(f \star_\vartheta g)(k) = \sum_{d|k} f(d)g(k/d)\chi_\vartheta(d, k/d). \tag{7.1}$$

The sum is finite for every k.

Lemma 7.2 (Cocycle identity)**.** *For all $a, b, c \in \mathbb{N}$,*

$$\chi_\vartheta(a, b)\chi_\vartheta(ab, c) = \chi_\vartheta(b, c)\chi_\vartheta(a, bc). \tag{7.2}$$

Proof. The exponent in (4.21) is the alternating bilinear form obtained from the additive map $n \mapsto z_n$. Expanding both sides using $\log(ab) = \log a + \log b$ and $\Omega(ab) = \Omega(a) + \Omega(b)$ gives the same exponent. □

Proposition 7.3 (Associative noncommutative arithmetic algebra)**.** *The product $\star_\vartheta$ is associative and has identity δ_1, where*

$$\delta_1(1) = 1, \qquad \delta_1(n) = 0 \quad (n > 1). \tag{7.3}$$

For $\vartheta \neq 0$ it is generally noncommutative.

Proof. Associativity follows from Lemma 7.2 after grouping the finite divisor sums by triples $abc = k$. The identity property follows from $\chi_\vartheta(1, n) = \chi_\vartheta(n, 1) = 1$. Noncommutativity follows from Corollary 4.10; equivalently,

$$(\delta_p \star_\vartheta \delta_q)(pq) = \chi_\vartheta(p, q) \tag{7.4}$$

whereas

$$(\delta_q \star_\vartheta \delta_p)(pq) = \chi_\vartheta(q, p) = \overline{\chi_\vartheta(p, q)}. \tag{7.5}$$

□

7.2. Weighted Banach algebra and operator representation.

Definition 7.4 (Weighted arithmetic Banach space)**.** For $\sigma > 0$, set

$$\mathcal{A}_\sigma = \left\{ f : \mathbb{N} \to \mathbb{C} : \|f\|_\sigma := \sum_{n\geq 1} \frac{|f(n)|}{n^\sigma} < \infty \right\}. \tag{7.6}$$

Proposition 7.5 (Banach-algebra estimate)**.** *For $f, g \in \mathcal{A}_\sigma$,*

$$\|f \star_\vartheta g\|_\sigma \leq \|f\|_\sigma \|g\|_\sigma. \tag{7.7}$$

Hence $(\mathcal{A}_\sigma, \star_\vartheta)$ is a unital Banach algebra.

Proof. Since $|\chi_\vartheta| = 1$,

$$\|f \star_\vartheta g\|_\sigma \leq \sum_{k\geq 1} \frac{1}{k^\sigma} \sum_{d|k} |f(d)||g(k/d)| \tag{7.8}$$

$$= \sum_{d,e\geq 1} \frac{|f(d)|}{d^\sigma} \frac{|g(e)|}{e^\sigma} \tag{7.9}$$

$$= \|f\|_\sigma \|g\|_\sigma. \tag{7.10}$$

Completeness is the completeness of weighted ℓ^1. □

Definition 7.6 (Weyl–Dirichlet representation)**.** For $f \in \mathcal{A}_\sigma$, define

$$\Pi_{\sigma,\vartheta}(f) = \sum_{n\geq 1} \frac{f(n)}{n^\sigma} W_n, \tag{7.11}$$

where the series converges absolutely in operator norm.

Proposition 7.7 (Representation theorem)**.** *For $f, g \in \mathcal{A}_\sigma$,*

$$\|\Pi_{\sigma,\vartheta}(f)\| \leq \|f\|_\sigma, \tag{7.12}$$

$$\Pi_{\sigma,\vartheta}(f)\Pi_{\sigma,\vartheta}(g) = \Pi_{\sigma,\vartheta}(f \star_\vartheta g). \tag{7.13}$$

Proof. The norm bound follows from $\|W_n\| = 1$. Absolute norm convergence permits multiplication and regrouping of the double series. The coefficient of W_k is exactly (7.1) by Theorem 4.9. □

8. Möbius inversion deformed by the CCR

8.1. **The Möbius operator.**

Definition 8.1 (Möbius operator)**.** For $\sigma > 1$, define

$$\mathcal{M}_{\sigma,\vartheta} = \Pi_{\sigma,\vartheta}(\mu) = \sum_{n\geq 1} \frac{\mu(n)}{n^\sigma} W_n. \tag{8.1}$$

The series converges absolutely in operator norm, so $\mathcal{M}_{\sigma,\vartheta} \in \mathcal{B}(\mathcal{H})$ and

$$\mathcal{D}(\mathcal{M}_{\sigma,\vartheta}) = \mathcal{H}. \tag{8.2}$$

Remark 8.2 (Concrete coordinate action)*.* For $\psi \in L^2(\mathbb{R})$,

$$(\mathcal{M}_{\sigma,\vartheta}\psi)(q) = \sum_{n\geq 1} \frac{\mu(n)}{n^\sigma} \exp\left[\frac{i}{\hbar} y_n \left(q - \frac{x_n}{2}\right)\right] \psi(q - x_n), \tag{8.3}$$

with convergence in L^2. Thus the operator is a Möbius-weighted superposition of simultaneous translations in position and momentum; it is not a Fourier multiplier when $\kappa \neq 0$.

8.2. **Formal twisted inverse.**

Proposition 8.3 (Existence and uniqueness of arithmetic inverses)**.** *Let $f : \mathbb{N} \to \mathbb{C}$ satisfy $f(1) \neq 0$. Then there exists a unique arithmetic function g such that*

$$f \star_\vartheta g = g \star_\vartheta f = \delta_1. \tag{8.4}$$

Its coefficients are determined recursively from $g(1) = f(1)^{-1}$.

Proof. For $n > 1$, the equation $(f \star_\vartheta r)(n) = 0$ contains the term $f(1)r(n)$ and otherwise only values $r(d)$ with $d < n$. It therefore constructs a unique right inverse r satisfying $f \star_\vartheta r = \delta_1$. Separately, the equations $(\lambda \star_\vartheta f)(n) = 0$ construct a unique left inverse λ satisfying $\lambda \star_\vartheta f = \delta_1$. Associativity then gives

$$\lambda = \lambda \star_\vartheta (f \star_\vartheta r) = (\lambda \star_\vartheta f) \star_\vartheta r = r, \tag{8.5}$$

so the inverse is bilateral and unique. □

Definition 8.4 (Heisenberg-zeta coefficients)**.** Let $z_\vartheta : \mathbb{N} \to \mathbb{C}$ be the unique two-sided inverse of μ for $\star_\vartheta$:

$$\mu \star_\vartheta z_\vartheta = z_\vartheta \star_\vartheta \mu = \delta_1. \tag{8.6}$$

Thus

$$z_\vartheta(1) = 1, \tag{8.7}$$

and for $n > 1$,

$$z_\vartheta(n) = -\sum_{\substack{d|n \\ d>1}} \mu(d) z_\vartheta(n/d) \chi_\vartheta(d, n/d). \tag{8.8}$$

Proposition 8.5 (Commuting boundary)**.** *At $\vartheta = 0$,*

$$z_0(n) = 1 \tag{8.9}$$

for every $n \geq 1$.

Proof. When $\vartheta = 0$, $\chi_0 \equiv 1$ and $\star_\vartheta$ is ordinary Dirichlet convolution. Equation (8.6) becomes $\mu * z_0 = \delta_1$. Classical Möbius inversion gives $z_0 = \mathbf{1}$. See [10]. □

Proposition 8.6 (Ordered-factorization formula)**.** *For $n > 1$, the twisted inverse has the finite coefficient expansion*

$$\begin{aligned} z_\vartheta(n) = \sum_{r\geq 1} (-1)^r \sum_{\substack{n_1 \cdots n_r = n \\ n_j > 1}} \left(\prod_{j=1}^{r} \mu(n_j) \right) \\ \times \exp\left[\frac{i\vartheta}{2} \sum_{1 \leq i < j \leq r} \left(\Omega(n_i) \log n_j - \log n_i \, \Omega(n_j) \right) \right]. \end{aligned} \tag{8.10}$$

Only finitely many r and ordered factorizations contribute for fixed n.

Proof. Write $\mu = \delta_1 + h$ with $h(1) = 0$. Coefficientwise the formal Neumann expansion is

$$z_\vartheta = \sum_{r \geq 0} (-h)^{\star_\vartheta r}. \tag{8.11}$$

For fixed n, a product of more than $\log_2 n$ factors exceeding one cannot equal n, so the coefficient sum is finite. Iterating the bicharacter cocycle yields the pairwise phase in (8.10). □

8.3. **A bounded inverse in a rigorous half-plane.**

Definition 8.7 (Uniform Neumann threshold)**.** Let σ_* be the unique real number > 1 satisfying

$$\frac{\zeta(\sigma_*)}{\zeta(2\sigma_*)} = 2. \tag{8.12}$$

Numerically,

$$\sigma_* = 1.578017063034\ldots. \tag{8.13}$$

Theorem 8.8 (Uniform Neumann region)**.** *For every real ϑ and every $\sigma > \sigma_*$,*

$$z_\vartheta \in \mathcal{A}_\sigma. \tag{8.14}$$

Consequently

$$\mathcal{Z}_{\sigma,\vartheta} = \Pi_{\sigma,\vartheta}(z_\vartheta) = \sum_{n \geq 1} \frac{z_\vartheta(n)}{n^\sigma} W_n \tag{8.15}$$

converges absolutely in operator norm and satisfies

$$\mathcal{M}_{\sigma,\vartheta} \mathcal{Z}_{\sigma,\vartheta} = \mathcal{Z}_{\sigma,\vartheta} \mathcal{M}_{\sigma,\vartheta} = I. \tag{8.16}$$

Proof. Write $\mu = \delta_1 + h$, where $h(1) = 0$. For $\sigma > 1$,

$$\|h\|_\sigma = \sum_{n\geq 2} \frac{\mu(n)^2}{n^\sigma} \tag{8.17}$$

$$= \frac{\zeta(\sigma)}{\zeta(2\sigma)} - 1. \tag{8.18}$$

The right-hand side is continuous and strictly decreasing from $+\infty$ to 0 as σ runs from 1^+ to $+\infty$, so σ_* is unique and $\|h\|_\sigma < 1$ exactly when $\sigma > \sigma_*$. The Banach-algebra Neumann series

$$(\delta_1 + h)^{-1} = \sum_{r=0}^{\infty} (-h)^{\star_\vartheta r} \tag{8.19}$$

then converges in $\mathcal{A}_\sigma$. Uniqueness identifies it with z_ϑ, and the representation theorem gives (8.16). □

Warning 8.9 (Exact border of the proved inverse). The Möbius operator exists as a bounded operator for every $\sigma > 1$. The argument above proves a bounded inverse only for $\sigma > \sigma_*$. At $\sigma = \sigma_*$ the Neumann norm equals one and this argument is inconclusive. No claim is made here for $1 < \sigma \leq \sigma_*$.

9. The first Heisenberg arithmetic law

Theorem 9.1 (Prime rays are rigid; distinct primes interfere)**.** *Let p, q be distinct primes and let $r \geq 1$. Then*

$$z_\vartheta(p^r) = 1, \tag{9.1}$$

$$z_\vartheta(pq) = 2\cos\left(\frac{\vartheta}{2}\log\frac{q}{p}\right) - 1, \tag{9.2}$$

$$z_\vartheta(pq) - 1 = -4\sin^2\left(\frac{\vartheta}{4}\log\frac{q}{p}\right). \tag{9.3}$$

Proof. For p^r, only the divisors 1 and p contribute to the recursion because $\mu(p^j) = 0$ for $j \geq 2$. Moreover,

$$\Omega(p)\log(p^{r-1}) - \log p\,\Omega(p^{r-1}) = 0, \tag{9.4}$$

so $\chi_\vartheta(p, p^{r-1}) = 1$. Equation (8.8) gives

$$z_\vartheta(p^r) = z_\vartheta(p^{r-1}), \tag{9.5}$$

and (9.1) follows from $z_\vartheta(1) = 1$.

For pq, the four divisors $1, p, q, pq$ give

$$0 = z_\vartheta(pq) - \chi_\vartheta(p, q) - \chi_\vartheta(q, p) + 1. \tag{9.6}$$

Since

$$\chi_\vartheta(p, q) = \exp\left(\frac{i\vartheta}{2}\log\frac{q}{p}\right), \tag{9.7}$$

and $\chi_\vartheta(q, p)$ is its complex conjugate, (9.2) follows. The identity (9.3) is the elementary formula $2(\cos u - 1) = -4\sin^2(u/2)$. □

Corollary 9.2 (Loss of ordinary multiplicativity)**.** *Generically,*

$$z_\vartheta(pq) \neq z_\vartheta(p) z_\vartheta(q) = 1. \tag{9.8}$$

Thus the Heisenberg deformation preserves every one-prime ray but destroys ordinary multiplicativity through cross-prime interference.

Proof. The prime-pair formula gives $z_\vartheta(pq) = 2\cos(\frac{\vartheta}{2}\log(q/p)) - 1$. This equals one only on the stated resonance set; away from it the product of the two one-prime values, which is identically one, differs from the two-prime coefficient. □

Corollary 9.3 (Small-area expansion)**.** *For fixed distinct primes p, q,*

$$z_\vartheta(pq) = 1 - \frac{\vartheta^2}{4}\left(\log\frac{q}{p}\right)^2 + O(\vartheta^4) \qquad (\vartheta \to 0). \tag{9.9}$$

There is no linear correction.

Proof. Use $\cos x = 1 - x^2/2 + O(x^4)$ with $x = (\vartheta/2)\log(q/p)$ in the exact prime-pair formula. Multiplying out gives (9.9); evenness of the cosine excludes an odd term. □

Proposition 9.4 (Two-prime lattice recurrence)**.** *Let $p \neq q$ be primes, set*

$$Z_{a,b} = z_\vartheta(p^a q^b), \qquad \lambda = \frac{\vartheta}{2}\log\frac{q}{p}. \tag{9.10}$$

Then for $a, b \geq 1$,

$$Z_{a,b} = e^{ib\lambda}Z_{a-1,b} + e^{-ia\lambda}Z_{a,b-1} - e^{i(b-a)\lambda}Z_{a-1,b-1}, \tag{9.11}$$

with boundary values

$$Z_{a,0} = Z_{0,b} = 1. \tag{9.12}$$

In particular,

$$z_\vartheta(p^2 q) = 1 + 2\cos(2\lambda) - 2\cos\lambda. \tag{9.13}$$

Proof. Apply the right-inverse recursion (8.8) to $p^a q^b$. Since $\mu(d) \neq 0$ only when the exponents of p, q in d are at most one, only $d = p, q, pq$ contribute. Evaluation of the three cocycle factors gives (9.11); the boundary values are Theorem 9.1 on prime rays. □

Proposition 9.5 (Evenness of the full inverse)**.** *For every $n \geq 1$,*

$$z_{-\vartheta}(n) = z_\vartheta(n). \tag{9.14}$$

Proof. Changing ϑ to $-\vartheta$ reverses the cocycle:

$$(f \star_{-\vartheta} g)(n) = (g \star_\vartheta f)(n). \tag{9.15}$$

Since z_ϑ is a two-sided inverse of μ for $\star_\vartheta$, it is a two-sided inverse of μ for the opposite product $\star_{-\vartheta}$. Uniqueness of the inverse gives (9.14). □

Proposition 9.6 (Reality for real deformation)**.** *For real ϑ and every $n \geq 1$,*

$$z_\vartheta(n) \in \mathbb{R}. \tag{9.16}$$

Proof. Complex conjugation changes χ_ϑ to $\chi_{-\vartheta}$. Since μ is real, conjugating the inverse equation shows that $\overline{z_\vartheta}$ is the inverse for the $-\vartheta$ product. Proposition 9.5 therefore gives $\overline{z_\vartheta} = z_{-\vartheta} = z_\vartheta$. □

Remark 9.7 (Why this is the non-tautological step). At $\vartheta = 0$ the inverse coefficients are identically 1. At nonzero ϑ, equation (9.3) creates a new arithmetic coefficient depending on the logarithmic separation of two primes. This dependence is forced by the symplectic area in the Weyl product and disappears if the CCR is removed. It is therefore the first result in the paper where Heisenberg noncommutativity changes the arithmetic inverse itself rather than merely constraining an unrelated state space.

10. Curvature of the cocyclically deformed Möbius inverse

Set

$$\mathcal{A}(d,e) = \Omega(d)\log e - \log d\,\Omega(e), \tag{10.1}$$

so that

$$\chi_\vartheta(d,e) = \exp\!\left(\frac{i\vartheta}{2}\mathcal{A}(d,e)\right). \tag{10.2}$$

Proposition 10.1 (Coefficientwise entire dependence)**.** *For each fixed $n \geq 1$, the map $\vartheta \mapsto z_\vartheta(n)$ is entire. It is even, and it is real-valued on the real axis.*

Proof. Formula (8.10) is a finite sum for fixed n, each term being an exponential $e^{ic\vartheta}$. Evenness and reality follow from Propositions 9.5 and 9.6. □

Proposition 10.2 (Second derivative of the inverse)**.** *Define*

$$g(k) = \frac{1}{4}\sum_{d|k}\mu(d)\mathcal{A}(d,k/d)^2. \tag{10.3}$$

Then

$$\boxed{z_0''(n) = \sum_{k|n} g(k).} \tag{10.4}$$

Proof. Differentiate the inverse equation

$$0 = \sum_{d|n}\mu(d) z_\vartheta(n/d)\exp\!\left(\frac{i\vartheta}{2}\mathcal{A}(d,n/d)\right) \tag{10.5}$$

twice at $\vartheta = 0$. Since $z_0 \equiv 1$ and $z_0' \equiv 0$,

$$(\mu * z_0'')(n) = \frac{1}{4}\sum_{d|n}\mu(d)\mathcal{A}(d,n/d)^2 = g(n). \tag{10.6}$$

Convolving with $\mathbf{1} = \mu^{-1}$ gives (10.4). □

Lemma 10.3 (Support collapse)**.** *Let $k = \prod_{j=1}^r p_j^{a_j}$ with distinct primes p_j. Then*

$$g(k) = 0 \qquad \text{for } r = 0,\ r = 1,\ \text{or } r \geq 3. \tag{10.7}$$

For two distinct primes,

$$\boxed{g(p^\alpha q^\beta) = -\frac{\alpha\beta}{2}\log^2\frac{q}{p}.} \tag{10.8}$$

Proof. A contributing divisor is square-free and therefore corresponds to a subset $S \subseteq \{1,\dots,r\}$. Put

$$R = \Omega(k) = \sum_j a_j, \qquad L = \log k, \qquad X_S = \sum_{j\in S}\log p_j. \tag{10.9}$$

Then

$$\mathcal{A}(d,k/d) = |S|L - RX_S. \tag{10.10}$$

Its square has Boolean degree at most two, so its full alternating difference vanishes when $r \geq 3$. For $r = 1$ the area is zero. For $r = 2$, direct evaluation of the four subsets gives

$$\sum_{d|p^\alpha q^\beta}\mu(d)\mathcal{A}(d,p^\alpha q^\beta/d)^2 = -2\alpha\beta\log^2(q/p), \tag{10.11}$$

which yields (10.8). □

Theorem 10.4 (Exact deformation curvature)**.** *If*

$$n = \prod_{j=1}^{r} p_j^{a_j}, \qquad T(a) = \frac{a(a+1)}{2}, \tag{10.12}$$

then

$$\boxed{z_0''(n) = -\frac{1}{2} \sum_{1 \le i < j \le r} T(a_i) T(a_j) \log^2 \frac{p_j}{p_i}.} \tag{10.13}$$

Thus $-z_0''(n) \ge 0$, with equality exactly for $n = 1$ or a prime power.

Proof. Only divisors supported on exactly two primes survive by Lemma 10.3. Summing (10.8) over $1 \le \alpha \le a_i$, $1 \le \beta \le a_j$ gives the factor $T(a_i)T(a_j)$. □

Corollary 10.5 (Square-free prime-log variance)**.** *If $n = p_1 \cdots p_r$ is square-free, then*

$$-z_0''(n) = \frac{1}{2} \sum_{i<j} (\log p_j - \log p_i)^2 \tag{10.14}$$

$$= \frac{1}{2} \left[r \sum_{p \mid n} (\log p)^2 - (\log n)^2 \right] \tag{10.15}$$

$$= \frac{r^2}{2} \operatorname{Var}_{p \mid n}(\log p). \tag{10.16}$$

Proof. In Theorem 10.4, square-freeness gives $T(v_p(n)) = T(1) = 1$ for every $p \mid n$. The first equality follows. Expanding the pairwise squares and counting each $(\log p)^2$ exactly $r - 1$ times gives $\sum_{i<j}(a_j - a_i)^2 = r \sum_i a_i^2 - (\sum_i a_i)^2$, which yields the second equality. Dividing by r identifies the usual empirical variance and gives the third. □

Corollary 10.6 (Infinitesimal symplectic size)**.** *For square-free $n = p_1 \cdots p_r$, let*

$$B_{ij} = \log p_j - \log p_i. \tag{10.17}$$

Then B has rank at most two. If $r \ge 2$, its two nonzero singular values are both

$$s(n) = \sqrt{r \sum_{p \mid n} (\log p)^2 - (\log n)^2}, \tag{10.18}$$

and

$$\boxed{-z_0''(n) = \frac{1}{2} s(n)^2.} \tag{10.19}$$

Proof. Write $B = \mathbf{1}a^T - a\mathbf{1}^T$, $a_i = \log p_i$. The two nonzero singular values of $uv^T - vu^T$ equal

$$\sqrt{\|u\|^2 \|v\|^2 - (u \cdot v)^2}. \tag{10.20}$$

Apply this with $u = \mathbf{1}$, $v = a$, and then use Corollary 10.5. □

Corollary 10.7 (Exact Taylor law for each integer)**.** *For every fixed n,*

$$z_\vartheta(n) = 1 + \frac{\vartheta^2}{2} z_0''(n) + O_n(\vartheta^4), \tag{10.21}$$

and the Taylor series has infinite radius of convergence. For square-free n,

$$\boxed{z_\vartheta(n) = 1 - \frac{\vartheta^2}{4} \left[\omega(n) \sum_{p \mid n} (\log p)^2 - (\log n)^2 \right] + O_n(\vartheta^4).} \tag{10.22}$$

Proof. Coefficientwise entire dependence was proved in Proposition 10.1. Evenness gives the Taylor expansion with no odd powers. The coefficient of ϑ^2 is $z_0''(n)/2$, proving the first formula. Substitution of the square-free formula from Corollary 10.5 yields the boxed identity. □

11. Multiplicative geometry of the curvature

Define

$$\mathcal{C}(n) := -z_0''(n) = \frac{1}{2}\sum_{p<q} T(v_p(n))T(v_q(n))\log^2\frac{q}{p}. \tag{11.1}$$

This section studies $\mathcal{C}$ before any summation over n.

Proposition 11.1 (Prime-ray rigidity)**.**

$$\boxed{\mathcal{C}(n) = 0 \iff n = 1 \text{ or } n = p^a.} \tag{11.2}$$

Proof. Every summand in (11.1) is nonnegative. It is strictly positive exactly when two distinct primes occur with positive valuation. □

Theorem 11.2 (Multiplicative superadditivity)**.** *For all positive integers m, n,*

$$\boxed{\mathcal{C}(mn) \geq \mathcal{C}(m) + \mathcal{C}(n).} \tag{11.3}$$

If $m, n > 1$, equality holds exactly when $m = p^a$ and $n = p^b$ for the same prime p.

Proof. Put $a_p = v_p(m)$, $b_p = v_p(n)$, $A_p = T(a_p)$, $B_p = T(b_p)$. Since $T(a+b) = T(a) + T(b) + ab$, subtraction of $\mathcal{C}(m) + \mathcal{C}(n)$ from $\mathcal{C}(mn)$ gives

$$\mathcal{C}(mn) - \mathcal{C}(m) - \mathcal{C}(n) = \frac{1}{2}\sum_{p<q} \Xi_{p,q}(m,n)\log^2\frac{q}{p}, \tag{11.4}$$

where

$$\Xi_{p,q}(m,n) = A_pB_q + B_pA_q + (A_p + B_p)a_qb_q + (A_q + B_q)a_pb_p + a_pb_pa_qb_q. \tag{11.5}$$

All terms are nonnegative. If $m, n > 1$, all cross terms vanish exactly when both supports consist of the same single prime. □

Corollary 11.3 (Monotonicity on the divisibility poset)**.** *If $m \mid n$, then*

$$\boxed{\mathcal{C}(m) \leq \mathcal{C}(n).} \tag{11.6}$$

Proof. Write $n = mq$. Theorem 11.2 gives $\mathcal{C}(n) \geq \mathcal{C}(m) + \mathcal{C}(q) \geq \mathcal{C}(m)$, because $\mathcal{C}(q) \geq 0$. This is exactly (11.6). □

Proposition 11.4 (Exact power law)**.** *If $n = \prod_p p^{a_p}$, then for every integer $k \geq 1$,*

$$\boxed{\mathcal{C}(n^k) = \frac{1}{8}\sum_{p<q}\Big[k^4a_p^2a_q^2 + k^3(a_p^2a_q + a_pa_q^2) + k^2a_pa_q\Big]\log^2\frac{q}{p}.} \tag{11.7}$$

Hence, if n contains at least two distinct primes,

$$\boxed{\lim_{k\to\infty}\frac{\mathcal{C}(n^k)}{k^4} = \frac{1}{8}\sum_{p<q} a_p^2a_q^2\log^2\frac{q}{p} > 0.} \tag{11.8}$$

Proof. For every prime exponent a_p,

$$T(ka_p) = \frac{k^2 a_p^2 + k a_p}{2}. \tag{11.9}$$

Thus for each pair $p < q$,

$$\frac{1}{2} T(ka_p) T(ka_q) = \frac{1}{8}\left[k^4 a_p^2 a_q^2 + k^3(a_p^2 a_q + a_p a_q^2) + k^2 a_p a_q\right]. \tag{11.10}$$

Substitution into (11.1) gives (11.7). Division by k^4 and termwise passage to the limit is legitimate because the sum contains only finitely many prime pairs. If the support has at least two primes, at least one coefficient $a_p^2 a_q^2 \log^2(q/p)$ is strictly positive, proving (11.8). □

Proposition 11.5 (Coprime block decomposition)**.** *Assume* $(m, n) = 1$*. Define*

$$W(m) = \sum_{p|m} T(v_p(m)), \tag{11.11}$$

$$\overline{L}(m) = \frac{1}{W(m)} \sum_{p|m} T(v_p(m)) \log p, \tag{11.12}$$

$$V(m) = \frac{1}{W(m)} \sum_{p|m} T(v_p(m))(\log p - \overline{L}(m))^2, \tag{11.13}$$

with the analogous definitions for n*. Then*

$$\boxed{\mathcal{C}(mn) = \mathcal{C}(m) + \mathcal{C}(n) + \frac{1}{2} W(m) W(n) \left[V(m) + V(n) + (\overline{L}(m) - \overline{L}(n))^2\right].} \tag{11.14}$$

Proof. The cross-prime contribution is $\frac{1}{2}\sum_{p|m, q|n} A_p B_q (\log q - \log p)^2$. Expanding around the two weighted means gives (11.14). Abstractly this last algebraic identity is the standard weighted within/between variance decomposition; its use here is only to expose the multiplicative geometry of $\mathcal{C}$. □

12. Dirichlet summation of the curvature

Define

$$c(n) = -z_0''(n) \geq 0. \tag{12.1}$$

Theorem 12.1 (Exact curvature Dirichlet series)**.** *For* $\Re s > 1$*,*

$$\mathcal{K}(s) = \sum_{n \geq 1} \frac{c(n)}{n^s} \tag{12.2}$$

converges absolutely and

$$\boxed{\mathcal{K}(s) = \frac{\zeta(s)}{2}\left[S_0(s) S_2(s) - S_1(s)^2\right],} \tag{12.3}$$

where

$$S_j(s) = \sum_p \frac{p^{-s}}{(1 - p^{-s})^2} (\log p)^j, \qquad j = 0, 1, 2. \tag{12.4}$$

Equivalently,

$$S_0 S_2 - S_1^2 = \sum_{p<q} \frac{p^{-s} q^{-s}}{(1 - p^{-s})^2 (1 - q^{-s})^2} \log^2 \frac{q}{p}. \tag{12.5}$$

Proof. Use Theorem 10.4 and

$$\sum_{a\geq 1} T(a)x^a = \frac{x}{(1-x)^3}. \tag{12.6}$$

Factoring the unrestricted Euler factors gives $\zeta(s)$, while a selected prime contributes

$$\frac{p^{-s}}{(1-p^{-s})^2}. \tag{12.7}$$

The pair sum is exactly the weighted variance determinant $S_0S_2 - S_1^2$. Absolute convergence follows for $\Re s > 1$. □

Corollary 12.2 (Prime-zeta reduction)**.** *Let $P(s) = \sum_p p^{-s}$ for $\Re s > 1$. Then*

$$S_0(s) = \sum_{m\geq 1} mP(ms), \tag{12.8}$$

$$S_1(s) = -\sum_{m\geq 1} mP'(ms), \tag{12.9}$$

$$S_2(s) = \sum_{m\geq 1} mP''(ms). \tag{12.10}$$

Using

$$P(s) = \sum_{k\geq 1} \frac{\mu(k)}{k} \log \zeta(ks), \tag{12.11}$$

gives

$$S_0(s) = \sum_{r\geq 1} \frac{J_2(r)}{r} \log \zeta(rs), \tag{12.12}$$

$$S_1(s) = -\sum_{r\geq 1} \varphi(r) \frac{\zeta'(rs)}{\zeta(rs)}, \tag{12.13}$$

$$S_2(s) = \left(\frac{\zeta'}{\zeta}\right)'(s), \tag{12.14}$$

where $J_2(r) = r^2 \prod_{p|r}(1-p^{-2})$.

Proof. Expand $x/(1-x)^2 = \sum_{m\geq 1} mx^m$, substitute (12.11), and group by $r = mk$. The needed divisor identities are

$$r\sum_{k|r} \frac{\mu(k)}{k^2} = \frac{J_2(r)}{r}, \tag{12.15}$$

$$r\sum_{k|r} \frac{\mu(k)}{k} = \varphi(r), \tag{12.16}$$

$$r\sum_{k|r} \mu(k) = \begin{cases} 1, & r = 1, \\ 0, & r > 1. \end{cases} \tag{12.17}$$

The last identity collapses S_2 to (12.14). □

Proposition 12.3 (Singularity at $s = 1$)**.** *As $s \to 1^+$,*

$$\boxed{\mathcal{K}(s) \sim \frac{1}{2(s-1)^3} \log \frac{1}{s-1}.} \tag{12.18}$$

Proof. From (12.11),

$$P(s) = \log \frac{1}{s-1} + O(1). \tag{12.19}$$

Hence

$$S_0(s) = \log \frac{1}{s-1} + O(1), \quad S_1(s) = \frac{1}{s-1} + O(1), \quad S_2(s) = \frac{1}{(s-1)^2} + O(1). \tag{12.20}$$

Combine with $\zeta(s) \sim (s-1)^{-1}$. □

13. Continuation of the curvature and detection of zeta zeros

Theorem 13.1 (Local continuation to $\Re s > 1/2$)**.** *Let $U \subset \{\Re s > 1/2\}$ be simply connected and contain no zero or pole of $\zeta(s)$. Choose a branch of $\log \zeta(s)$ on U. Then* (12.12)–(12.14)*, substituted into* (12.3)*, continue $\mathcal{K}$ holomorphically to U.*

Proof. For $r \geq 2$, $\Re(rs) > 1$, so the corresponding terms are holomorphic and converge normally on compact subsets. The $r = 1$ terms are holomorphic by the hypothesis on U. □

Theorem 13.2 (Zeta zeros are logarithmic curvature singularities)**.** *Let ρ be a zero of multiplicity $m \geq 1$ with $1/2 < \Re\rho < 1$, and write*

$$\zeta(s) = c_\rho (s-\rho)^m (1 + O(s-\rho)), \qquad c_\rho \neq 0. \tag{13.1}$$

Then

$$\boxed{\mathcal{K}(s) = -\frac{c_\rho m^2}{2}(s-\rho)^{m-2}\log(s-\rho) + O\Big((s-\rho)^{m-2}\Big).} \tag{13.2}$$

For a simple zero,

$$\boxed{\mathcal{K}(s) = -\frac{\zeta'(\rho)}{2}\frac{\log(s-\rho)}{s-\rho} + O\left(\frac{1}{s-\rho}\right).} \tag{13.3}$$

Proof. For $r \geq 2$, the terms in (12.12)–(12.13) are holomorphic near ρ. With $t = s - \rho$, the singular pieces are

$$S_0 = m \log t + O(1), \qquad S_1 = -\frac{m}{t} + O(1), \qquad S_2 = -\frac{m}{t^2} + O(1). \tag{13.4}$$

Thus

$$S_0 S_2 - S_1^2 = -\frac{m^2 \log t}{t^2} + O(t^{-2}). \tag{13.5}$$

Multiply by $\zeta(s)/2$. □

Corollary 13.3 (Curvature criterion equivalent to RH)**.** *RH is equivalent to the absence of logarithmic branch singularities of $\mathcal{K}$ in*

$$\boxed{\frac{1}{2} < \Re s < 1.} \tag{13.6}$$

Proof. Theorem 13.2 detects every zero in the right half of the critical strip, while Theorem 13.1 gives holomorphy on zero-free neighbourhoods. The functional equation makes the absence of zeros in this half-strip equivalent to RH. □

14. Where the logarithmic branching comes from

The logarithmic singularity in Theorem 13.2 does not come from a meromorphic function such as ζ, ζ', or ζ''. It comes from the prime-zeta component of S_0.

Proposition 14.1 (Exact source of the logarithm)**.** *For $\Re s > 1$,*

$$S_0(s) = \sum_{m\geq 1} mP(ms), \qquad P(s) = \sum_{r\geq 1} \frac{\mu(r)}{r} \log\zeta(rs). \tag{14.1}$$

Consequently

$$S_0(s) = \sum_{k\geq 1} \frac{J_2(k)}{k} \log\zeta(ks). \tag{14.2}$$

If ρ is a zero of ζ of multiplicity m and $1/2 < \Re\rho < 1$, then in a punctured simply connected neighbourhood of ρ,

$$S_0(s) = m\log(s-\rho) + O(1). \tag{14.3}$$

Proof. The first identity follows by expanding $p^{-s}(1-p^{-s})^{-2} = \sum_{m\geq 1} mp^{-ms}$ and summing over primes. The second follows from Möbius inversion for the prime zeta function and collecting the coefficient of $\log\zeta(ks)$, exactly as in Corollary 12.2. If $1/2 < \Re\rho < 1$, every term with $k \geq 2$ is holomorphic near ρ, because $\Re(k\rho) > 1$. The $k = 1$ term is $\log\zeta(s) = m\log(s-\rho) + O(1)$. □

Remark 14.2 (Global branch structure). For the classical prime-zeta function, only square-free dilation indices occur directly because of the factor $\mu(r)$. By contrast, S_0 carries the nonzero weight $J_2(r)/r$ for every r. The classical prime-zeta continuation generated by

$$P(s) = \sum_{r\geq 1} \frac{\mu(r)}{r} \log\zeta(rs) \tag{14.4}$$

has logarithmic branch points at scaled poles and zeros, including points $1/r$ and ρ/r, and the line $\Re s = 0$ is a natural boundary for P; see Fröberg [8]. This global phenomenon does not invalidate Corollary 13.3: in the open half-strip $1/2 < \Re s < 1$, scaled points ρ/r with $r \geq 2$ cannot occur, because $\Re(\rho/r) \leq 1/2$. Hence the only logarithmic branches there are the unscaled $r = 1$ branches attached to zeros of ζ itself.

15. Local monodromy and the Jordan-totient weight

The branch structure is used here only locally. No globally reordered double sum over zeta zeros is asserted.

Proposition 15.1 (Local monodromy at an isolated scaled zero)**.** *Let ρ be a zero of ζ of multiplicity m_ρ, let $r \geq 1$, and set $s_0 = \rho/r$. Suppose a sufficiently small punctured neighbourhood of s_0 contains no other branch point contributing to* (14.2)*. Analytic continuation once positively around s_0 changes S_0 by*

$$\boxed{\Delta_{\rho/r} S_0 = 2\pi i\, m_\rho \frac{J_2(r)}{r}.} \tag{15.1}$$

Since S_1 and S_2 are meromorphic and hence single-valued, the corresponding monodromy of $\mathcal{K}$ is

$$\boxed{\Delta_{\rho/r}\mathcal{K}(s) = \pi i\, m_\rho \frac{J_2(r)}{r}\, \zeta(s) \left(\frac{\zeta'}{\zeta}\right)'(s).} \tag{15.2}$$

If several scaled singularities collide at the same point, their local monodromy contributions are added before applying this formula.

Proof. In (14.2), the branch of $\log\zeta(rs)$ changes by $2\pi i m_\rho$ around $s_0=\rho/r$. Multiplication by its coefficient $J_2(r)/r$ gives (15.1). The functions S_1 and S_2 are built from logarithmic derivatives of ζ, so they are meromorphic and carry no logarithmic monodromy. Substitution in $\mathcal{K}=\zeta(S_0S_2-S_1^2)/2$ gives (15.2). □

Proposition 15.2 (Growth of the local branch weight)**.** *For every $r\ge 1$,*

$$\frac{r}{\zeta(2)} \le \frac{J_2(r)}{r} = r\prod_{p\mid r}\left(1-\frac{1}{p^2}\right) \le r. \tag{15.3}$$

Consequently the individual local weight is of linear size in r. In particular, the unregularized cumulative weight cannot be finite; more precisely,

$$\sum_{r\le R}\frac{J_2(r)}{r} = \frac{R^2}{2\zeta(3)} + O(R\log R). \tag{15.4}$$

Proof. The product in (15.3) is bounded below by $\prod_p(1-p^{-2})=1/\zeta(2)$. For the average, use $J_2(r)/r = r\sum_{d\mid r}\mu(d)d^{-2}$, interchange the finite sums, and sum m up to R/d. The main term is $R^2\sum_{d\ge1}\mu(d)d^{-3}/2 = R^2/(2\zeta(3))$; elementary estimation of the remaining terms gives the displayed error. □

Remark 15.3 (Why S_0 nevertheless converges away from its branches). For fixed s with $\Re s>0$, once $r\Re s>1$, $\log\zeta(rs)=O(2^{-r\Re s})$. This exponential decay dominates the linear factor $J_2(r)/r$. Thus the growth of the local monodromy weight is compatible with convergence of (14.2) away from its singular points.

16. Gauge invariance of the arithmetic cocycle

The previous sections use the completely additive coordinate $A(n)=\Omega(n)$. It is useful to record an exact invariance that holds for any completely additive A.

Proposition 16.1 (Exact gauge invariance)**.** *Let*

$$\mathcal{A}_A(m,n) = A(m)\log n - \log m\, A(n) \tag{16.1}$$

and let $\chi_{\vartheta,A}=\exp(i\vartheta\mathcal{A}_A/2)$. For every $c\in\mathbb{R}$,

$$\boxed{\mathcal{A}_{A+c\log}(m,n)=\mathcal{A}_A(m,n), \qquad \chi_{\vartheta,A+c\log}=\chi_{\vartheta,A}.} \tag{16.2}$$

Consequently every twisted product and twisted inverse constructed only from this cocycle depends on A only through its class modulo $\mathbb{R}\log$.

Proof. The additional contribution is

$$c\log m\log n - c\log m\log n = 0. \tag{16.3}$$

□

Remark 16.2. This is an obstruction as well as an invariance: the deformation cannot recover the component of an additive coordinate parallel to log from the cocycle alone. The statement concerns the chosen cocycle itself and should not be confused with invariance of the coefficient curvature under arbitrary replacement by a cohomologous cocycle.

17. Cohomological obstruction to scalar Euler locality

Write $e_n = \delta_n$ for the canonical arithmetic basis. For distinct primes $p < q$,

$$e_p \star_\vartheta e_q = e^{\,i\vartheta \log(q/p)/2} e_{pq}, \qquad e_q \star_\vartheta e_p = e^{-\,i\vartheta \log(q/p)/2} e_{pq}. \tag{17.1}$$

Theorem 17.1 (No scalar prime-local Euler factorization)**.** *Fix $\vartheta \in \mathbb{R}$. Suppose that there exist scalars $a_p(\vartheta)$ and a fixed increasing order of the primes such that the square-free coefficients of μ are produced by an ordered twisted product*

$$\mu = \overrightarrow{\bigstar_{p}^{\vartheta}} (1 - a_p e_p). \tag{17.2}$$

Then $\vartheta = 0$. Hence, for every $\vartheta \neq 0$, the nontrivial arithmetic cocycle obstructs any scalar Euler factorization into independent one-prime factors of this form.

Proof. For primes $p < q$, comparison of the pq-coefficient gives

$$a_p a_q e^{\,i\vartheta \log(q/p)/2} = 1. \tag{17.3}$$

For $p < q < r$, the three corresponding equations imply

$$a_p^2 = e^{-i\vartheta \log(q/p)}. \tag{17.4}$$

The left side is independent of q. Taking two primes $q_1, q_2 > p$ therefore yields

$$e^{-i\vartheta \log(q_1/q_2)} = 1. \tag{17.5}$$

If $\vartheta \neq 0$, applying this simultaneously to independent prime ratios forces a nontrivial rational linear relation among logarithms of distinct primes, contradicting unique factorization. □

18. Ordered Euler object and the global Euler-locality defect

For $\sigma > 1$, let $\mathcal{A}_\sigma$ be the Banach space of arithmetic functions with

$$\|f\|_\sigma = \sum_{n \geq 1} |f(n)| n^{-\sigma}. \tag{18.1}$$

Since $|\chi_\vartheta| = 1$,

$$\|f \star_\vartheta g\|_\sigma \leq \|f\|_\sigma \|g\|_\sigma. \tag{18.2}$$

For a prime p, define

$$L_p = \sum_{a \geq 0} e_{p^a}. \tag{18.3}$$

The cocycle is trivial on a single prime ray, so

$$\boxed{L_p = (1 - e_p)^{-1}_{\star_\vartheta}.} \tag{18.4}$$

Moreover,

$$\sum_p \|L_p - 1\|_\sigma = \sum_p \frac{p^{-\sigma}}{1 - p^{-\sigma}} < \infty. \tag{18.5}$$

Proposition 18.1 (Norm-convergent ordered Euler object)**.** *For $\sigma > 1$, the increasing-prime product*

$$\boxed{E_\vartheta = \overrightarrow{\bigstar_{p}^{\vartheta}} L_p} \tag{18.6}$$

converges in $\mathcal{A}_\sigma$. *If* $n = \prod_{j=1}^r p_j^{a_j}$ *with* $p_1 < \cdots < p_r$, *then*

$$\boxed{E_\vartheta(n) = \exp\left[\frac{i\vartheta}{2}\sum_{i<j} a_i a_j \log\frac{p_j}{p_i}\right].} \tag{18.7}$$

Proof. The norm convergence follows from (18.5) and the standard convergence criterion for ordered infinite products in a Banach algebra. The coefficient formula follows by repeated use of the cocycle identity; all self-ray phases vanish. □

For $\sigma > \sigma_*$, the twisted Möbius inverse z_ϑ is already an element of $\mathcal{A}_\sigma$. Define

$$\boxed{R_\vartheta = z_\vartheta - E_\vartheta.} \tag{18.8}$$

Theorem 18.2 (Exact vanishing criterion for the Euler-locality defect)**.** *For real* ϑ,

$$\boxed{R_\vartheta \equiv 0 \iff \vartheta = 0.} \tag{18.9}$$

In fact, products of two primes already detect nonvanishing.

Proof. At $\vartheta = 0$, both coefficient systems equal 1. For $p < q$, put $\lambda = \frac{\vartheta}{2}\log(q/p)$. Then

$$R_\vartheta(pq) = 2\cos\lambda - 1 - e^{i\lambda} = e^{-i\lambda} - 1. \tag{18.10}$$

If this vanished for every prime pair, then $\vartheta\log(q/p) \in 4\pi\mathbb{Z}$ for every $p < q$, contradicting the same universal-resonance argument used in Theorem 17.1 unless $\vartheta = 0$. □

19. The odd Euler-locality defect: existence and exact initial domain

Because $z_\vartheta = z_{-\vartheta}$, one has $z_0'(n) = 0$. Define

$$\Phi_{\mathrm{ord}}(n) = \sum_{p<q} v_p(n) v_q(n) \log\frac{q}{p}. \tag{19.1}$$

Differentiating the coefficient formula for the ordered Euler object gives

$$R_0'(n) = -\frac{i}{2}\Phi_{\mathrm{ord}}(n). \tag{19.2}$$

Definition 19.1 (Odd Euler-locality Dirichlet defect)**.** For $\Re s > 1$, set

$$\mathcal{O}(s) = \frac{1}{2}\sum_{n\geq 1} \frac{\Phi_{\mathrm{ord}}(n)}{n^s}. \tag{19.3}$$

Whenever the differentiated defect series is used, it is understood through this absolutely convergent Dirichlet series; no interchange with ∂_ϑ is made outside the domain justified below.

Put

$$u_p(s) = \frac{p^{-s}}{1-p^{-s}} = \frac{1}{p^s - 1}, \qquad A(s) = \sum_p u_p(s), \qquad B(s,t) = \sum_p u_p(s) p^{it}. \tag{19.4}$$

Theorem 19.2 (Absolute convergence, holomorphy, and differentiation)**.** *The series* (19.3)*,* $A(s)$*, and* $B(s,t)$ *converge absolutely and locally uniformly for* $\Re s > 1$*, uniformly in real* t *on compact sets. The function* $\mathcal{O}$ *is holomorphic there. For every nonnegative integers* a, b*, the derivatives* $\partial_s^a \partial_t^b B(s,t)$ *are obtained by termwise differentiation on compact subsets of* $\Re s > 1$ *and bounded real* t*-intervals.*

Proof. Fix $\varepsilon > 0$ and restrict to $\Re s \geq 1 + \varepsilon$. Since $|1 - p^{-s}| \geq 1 - p^{-1-\varepsilon}$,

$$|u_p(s)| \leq \frac{p^{-1-\varepsilon}}{1 - p^{-1-\varepsilon}} \leq C_\varepsilon p^{-1-\varepsilon}, \tag{19.5}$$

with a constant independent of p. Therefore $\sum_p |u_p(s)|(\log p)^j$ converges uniformly on the stated compacta for every fixed $j \geq 0$, by comparison with $\sum_{n\geq 2} n^{-1-\varepsilon}(\log n)^j$. This proves normal convergence of A, B, and all the displayed termwise derivatives.

For $\mathcal{O}$, use

$$0 \leq \Phi_{\mathrm{ord}}(n) \leq \sum_{p<q} v_p(n) v_q(n)(\log p + \log q). \tag{19.6}$$

Summing first over n with fixed distinct p, q, absolute convergence of Euler products gives

$$\sum_{n\geq 1} \frac{v_p(n) v_q(n)}{n^s} = \zeta(s) u_p(s) u_q(s), \qquad \Re s > 1. \tag{19.7}$$

Consequently the absolute majorant for (19.3) is bounded by a constant multiple of

$$\left(\sum_p |u_p(s)| \right) \left(\sum_q |u_q(s)| \log q \right), \tag{19.8}$$

which is locally uniformly finite by (19.5). The Weierstrass theorem then gives holomorphy and termwise differentiation in s. □

Theorem 19.3 (Prime-pair and Fourier representations)**.** *For $\Re s > 1$,*

$$\mathcal{O}(s) = \frac{\zeta(s)}{2} \sum_{p<q} u_p(s) u_q(s) \log \frac{q}{p}, \tag{19.9}$$

$$= \frac{\zeta(s)}{2\pi} \int_0^\infty \frac{A(s)^2 - B(s,t)B(s,-t)}{t^2}\, dt. \tag{19.10}$$

The integral is absolutely convergent. Moreover,

$$B(s,t) = \sum_{k\geq 1} P(ks - it), \qquad \Re s > 1, \tag{19.11}$$

with absolute and locally uniform convergence for real t.

Proof. Identity (19.7) follows by multiplying the local geometric series at p and q:

$$\sum_{a\geq 0} a p^{-as} = \frac{p^{-s}}{(1 - p^{-s})^2}, \tag{19.12}$$

and division by the two ordinary local Euler factors leaves one factor $u_p(s)u_q(s)$. Inserting this identity into the absolutely convergent series of Theorem 19.2 proves (19.9).

Next, normal convergence permits multiplication of the two prime sums:

$$A(s)^2 - B(s,t)B(s,-t) = 2 \sum_{p<q} u_p(s) u_q(s) \big[1 - \cos(t \log(q/p))\big]. \tag{19.13}$$

For real a, integration by parts gives

$$\int_0^\infty \frac{1 - \cos(at)}{t^2}\, dt = \frac{\pi}{2}|a|. \tag{19.14}$$

The boundary term vanishes because $(1-\cos(at))/t = O(t)$ at zero and is $O(t^{-1})$ at infinity; the remaining integral is the classical Dirichlet integral. Furthermore,

$$\sum_{p<q} |u_p(s)u_q(s)| \int_0^\infty \frac{|1-\cos(t\log(q/p))|}{t^2}\,dt$$

$$\le \frac{\pi}{2} \sum_{p<q} |u_p(s)u_q(s)|(\log p + \log q) < \infty, \tag{19.15}$$

by (19.8). Fubini's theorem is therefore applicable, and (19.10) follows.

Finally, $u_p(s) = \sum_{k\ge1} p^{-ks}$ absolutely for $\Re s > 1$. Thus

$$\sum_p \sum_{k\ge1} |p^{-ks}| \le \sum_p \frac{p^{-\sigma}}{1-p^{-\sigma}} < \infty, \tag{19.16}$$

so Tonelli–Fubini gives (19.11). The same majorant is independent of real t, proving the asserted local uniformity. □

Definition 19.4 (Normalized Fourier energy)**.** In $\Re s > 1$, define

$$\boxed{\mathcal{I}(s) := \int_0^\infty \frac{A(s)^2 - B(s,t)B(s,-t)}{t^2}\,dt = \frac{2\pi\mathcal{O}(s)}{\zeta(s)}.} \tag{19.17}$$

19.1. **Endpoint regularity and transverse derivatives.** For

$$M_j(s) = \sum_p u_p(s)(\log p)^j, \tag{19.18}$$

normal convergence from Theorem 19.2 yields

$$B(s,t) = A(s) + iM_1(s)t - \frac{1}{2}M_2(s)t^2 + O_K(t^3) \tag{19.19}$$

uniformly for s in every compact $K \subset \{\Re s > 1\}$. Hence

$$\lim_{t\to0} \frac{A(s)^2 - B(s,t)B(s,-t)}{t^2} = A(s)M_2(s) - M_1(s)^2. \tag{19.20}$$

At infinity, $|B(s,t)| \le \sum_p |u_p(s)|$, so the integrand is $O_K(t^{-2})$. These two estimates give a second direct proof of absolute integrability in (19.17).

Proposition 19.5 (Exact first transverse derivative)**.** *For $\Re s > 1$,*

$$\boxed{\partial_t B(s,0) = -i\frac{\zeta'}{\zeta}(s).} \tag{19.21}$$

For every integer $a \ge 0$,

$$\boxed{\partial_s^a \partial_t B(s,0) = -i\left(\frac{\zeta'}{\zeta}\right)^{(a)}(s).} \tag{19.22}$$

Both identities extend meromorphically wherever the right-hand side does.

Proof. Termwise differentiation is justified by Theorem 19.2; hence

$$\partial_t B(s,0) = i\sum_p \frac{\log p}{p^s - 1}. \tag{19.23}$$

The absolutely convergent Euler-product logarithmic derivative is

$$-\frac{\zeta'}{\zeta}(s) = \sum_p \sum_{a\ge1} (\log p)p^{-as} = \sum_p \frac{\log p}{p^s - 1}, \tag{19.24}$$

which proves (19.21). Differentiating in s termwise on compacta gives (19.22). The final statement follows from uniqueness of meromorphic continuation. □

Proposition 19.6 (Higher pure transverse derivatives)**.** *For every integer $j \geq 1$ and $\Re s > 1$,*

$$\partial_t^j B(s,0) = (-i)^j \sum_{n\geq 1} c_j(n) \left(\frac{\zeta'}{\zeta}\right)^{(j-1)}(ns), \tag{19.25}$$

where

$$c_j(n) = \sum_{d|n} \mu(d) d^{j-1} = \prod_{p|n} (1 - p^{j-1}). \tag{19.26}$$

In particular, $c_1(n) = \delta_{n1}$, explaining why only the first transverse derivative is a complete replica filter.

Proof. Insert the prime-zeta Möbius inversion

$$P(w) = \sum_{m\geq 1} \frac{\mu(m)}{m} \operatorname{Log} \zeta(mw), \qquad \Re w > 1, \tag{19.27}$$

into (19.11). Absolute convergence in $\Re s > 1$ permits termwise differentiation, giving

$$\partial_t^j B(s,0) = (-i)^j \sum_{k,m\geq 1} \mu(m) m^{j-1} L^{(j-1)}(mks), \tag{19.28}$$

with $L = \zeta'/\zeta$. Grouping by $n = mk$ is legitimate because the same differentiated series is absolutely convergent on compacta. The inner divisor sum is $c_j(n)$, proving (19.25). Since μ is supported on square-free integers, multiplicativity gives the Euler factor $1 - p^{j-1}$ in (19.26). For $j = 1$, the standard identity $\sum_{d|n} \mu(d) = \delta_{n1}$ completes the proof. □

20. Continuation of the Fourier energy and complete replica accounting

The prime-zeta identity (19.27) is classical [8]. What is needed here is not merely the location of its branch points, but a proof that the energy integral can be continued while retaining control of all sums, derivatives, contour singularities, and boundary terms.

20.1. **Grouped log-zeta expansion and its normal tail.** For $\Re s > 1$, regrouping the absolutely convergent double series gives

$$\boxed{B(s,t) = \sum_{n\geq 1} \sum_{d|n} \frac{\mu(d)}{d} \operatorname{Log} \zeta(ns - idt).} \tag{20.1}$$

Lemma 20.1 (Normal convergence of the large-dilation tail)**.** *Fix s_0 with $\sigma_0 := \Re s_0 > 0$. There are a neighbourhood U of s_0 and an integer N_0 such that the part of (20.1) with $n > N_0$ is single-valued and holomorphic in $s \in U$, is C^∞ in real t, and all mixed derivatives of fixed finite order converge normally on $U \times \mathbb{R}$ after multiplication by any fixed polynomially decaying weight in t.*

Proof. Choose U so that $\Re s \geq \sigma_0/2$ on U, and choose N_0 with $(N_0+1)\sigma_0/2 > 1+\delta$ for some $\delta > 0$. Then for $n > N_0$,

$$\Re(ns - idt) = n \Re s \geq 1 + \delta. \tag{20.2}$$

In that half-plane the Euler product defines the principal holomorphic logarithm and

$$|\operatorname{Log} \zeta(w)| \leq \sum_{m\geq 2} \frac{C_\delta}{m^{\Re w}} \leq C'_\delta 2^{-\Re w}. \tag{20.3}$$

Every fixed derivative of $\operatorname{Log}\zeta(w)$ satisfies the same estimate with an additional polynomial factor in $\Re w$, obtained by differentiating the absolutely convergent von Mangoldt series. The divisor coefficient satisfies

$$\sum_{d|n}\frac{1}{d}\le\tau(n), \tag{20.4}$$

and $\tau(n)$ has subpolynomial growth. Hence the resulting tail is majorized by a convergent series of the form $\sum_{n>N_0} n^C\tau(n)2^{-n\sigma_0/2}$, uniformly in real t. The Weierstrass theorem proves normal convergence and permits all asserted finite-order differentiations. □

Definition 20.2 (Continued branch of $\mathcal{I}$)**.** A branch $\mathcal{I}^\sharp$ near a point s_0 with $\Re s_0>0$ is obtained by analytic continuation from $\Re s>1$ of the finitely many head terms $n\le N_0$ in (20.1), while the normal tail of Lemma 20.1 is kept on its single-valued branch. Whenever a moving logarithmic singularity meets the real t-axis, the t-contour is deformed continuously around it. Different continuation paths may change lower-order branch constants; the principal coefficients computed below are independent of those choices.

Lemma 20.3 (Convergence of the continued contour integral)**.** *Every local branch from Definition 20.2 can be represented by a locally finite deformed t-contour on which the continued integrand is absolutely integrable at infinity and locally integrable at every logarithmic branch point. The contour may be chosen so that the contribution of all indentations above height T tends to zero as $T\to\infty$.*

Proof. Only the finite head $n\le N_0$ can contribute zeros or the pole of ζ to the t-contour. For each fixed head pair (n,d), those points are obtained from

$$ns-idt=\rho' \tag{20.5}$$

with ρ' a zero or pole of ζ. The classical zero-counting estimate

$$N_\zeta(T)=O(T\log T) \tag{20.6}$$

is unconditional; see [17]. Hence the number of contour singularities of height at most T is $O(T\log T)$.

Near a zero of multiplicity m, local factorization gives

$$\operatorname{Log}\zeta(\rho'+w)=m\operatorname{Log}w+h(w), \tag{20.7}$$

with h holomorphic. Thus each singularity is logarithmic and $|\operatorname{Log}|t-t_0||^2$ is locally integrable. Choose pairwise compatible indentations with radii no larger than $(1+|t_0|)^{-4}$, shrinking them further when necessary to avoid overlaps. On the complementary contour, standard bounds obtained from the Hadamard product for ξ imply

$$|\operatorname{Log}\zeta(\sigma+it)|\le C_K\log^2(2+|t|) \tag{20.8}$$

for the finitely many vertical strips involved, after the chosen disks have been removed; this standard estimate is sufficient here and follows from the zero count and local factorization, again as in [17]. Consequently

$$\frac{|A(s)^2-B(s,t)B(s,-t)|}{|t|^2}\le C_K\frac{1+\log^4(2+|t|)}{|t|^2} \tag{20.9}$$

away from the indentation arcs. The right-hand side is integrable.

On an indentation of radius r_0 around height t_0, the logarithmic factor is $O(|\log r_0|+\log(2+|t_0|))$, the arc length is $O(r_0)$, and the weight is $O(t_0^{-2})$ for $|t_0|\ge1$. The sum of arc contributions is therefore dominated by

$$\sum_{t_0}\frac{r_0\,[1+|\log r_0|+\log^2(2+|t_0|)]}{1+t_0^2}, \tag{20.10}$$

which converges because $r_0 \le (1+|t_0|)^{-4}$ and the number of t_0's in a unit interval is $O(\log(2+|t_0|))$ on average by (20.6). Local logarithmic integrability handles the finitely many points with $|t_0| < 1$. This proves absolute convergence and the vanishing of the indentation tail. □

Theorem 20.4 (Existence of the continued energy germ)**.** *For every s_0 with $\Re s_0 > 0$, every branch obtained from Definition 20.2 defines a sectorial analytic germ $\mathcal{I}^\sharp$ at s_0 away from its local logarithmic cuts. It coincides with $\mathcal{I}$ in the common domain $\Re s > 1$. No RH, zero-simplicity, or nonresonance assumption is used.*

Proof. Lemma 20.1 reduces the continuation to finitely many logarithms of meromorphic functions. On every compact subarc of the deformed contour not meeting an indentation, these logarithms are holomorphic in s, and their products are dominated by the integrable majorant (20.9). On each indentation, use the local factorization (20.7); after translating the local contour, the integral is a finite linear combination of parameter integrals of $\mathrm{Log}(a(s-s_*)+b\tau)$ and products of two such logarithms. These are holomorphic in a slit sector by differentiation under the finite local integral. The normal convergence of the sum of indentation contributions follows from (20.10). Therefore the sum of the compact-contour part, indentation parts, and tail is analytic in the slit sector. Continuation from $\Re s > 1$ fixes the branch on overlaps, so uniqueness of analytic continuation gives agreement with (19.17) in the initial half-plane. □

20.2. **All contributions arriving at a scaled point.** Fix s_0 with $0 < \Re s_0 < 1$, and define

$$\mathcal{R}(s_0) = \{N \in \mathbb{N} : \zeta(Ns_0) = 0\}, \qquad q_N = \mathrm{ord}_{w=Ns_0}\,\zeta(w). \tag{20.11}$$

The set $\mathcal{R}(s_0)$ is finite, because a nontrivial zero has real part in $(0,1)$, so $N\Re s_0 < 1$. Define

$$C(s_0) = \sum_{N\in\mathcal{R}(s_0)} q_N \frac{\varphi(N)}{N}, \tag{20.12}$$

$$D(s_0) = \sum_{N\in\mathcal{R}(s_0)} q_N \frac{\varphi(N)}{N}\log N, \tag{20.13}$$

and, for real u,

$$G_{s_0}(u) = \sum_{N\in\mathcal{R}(s_0)} q_N \sum_{d\mid N} \frac{\mu(d)}{d}\,\mathrm{Log}\left(1 - i\frac{d}{N}u\right). \tag{20.14}$$

Lemma 20.5 (Complete endpoint singular block)**.** *With $z = s - s_0$, the complete contribution of all log-zeta terms singular at $(s,t) = (s_0, 0)$ is*

$$S(z,t) = \sum_{N\in\mathcal{R}(s_0)} q_N \sum_{d\mid N} \frac{\mu(d)}{d}\,\mathrm{Log}(Nz - idt). \tag{20.15}$$

After subtracting S, the remainder $R(z,t) = B(s_0+z,t) - S(z,t)$ is holomorphic in z and twice continuously differentiable in real t in some product neighbourhood of $(0,0)$.

Proof. In the finite head of (20.1), a term indexed by (n,d) is singular at $(s_0,0)$ precisely when $\zeta(ns_0) = 0$. Thus $n = N \in \mathcal{R}(s_0)$, while every divisor $d \mid N$ contributes the same zero at $t = 0$. Local factorization (20.7) then yields exactly (20.15). Since the head is finite and all endpoint singular factors have been removed, discreteness of the zero set permits a small product neighbourhood containing no further head singularity. The normal tail is smooth by Lemma 20.1. Hence the remainder has the asserted regularity. □

Lemma 20.6 (Interior contour singularities are lower order)**.** *Let $t_0 > 0$ be any logarithmic singularity of a continued head term. Its local contribution to $\mathcal{I}^\sharp(s_0+z)$, including products of two logarithmic factors, is $O(1) + O(z\log^2 z)$ in a slit sector. In particular, no interior singularity can contribute either $z^{-1}\log z$ or z^{-1}.*

Proof. Write $t = t_0 + x$. The weight is analytic there:

$$\frac{1}{t^2} = \frac{1}{t_0^2} + O(x). \tag{20.16}$$

A local logarithmic factor has the form $\mathrm{Log}(az - ibx) + h(z, x)$, with h holomorphic. On a symmetric local contour of fixed radius, set $x = zu$ only in the central part $|x| \le c|z|$; the Jacobian contributes a factor z, while the weight has no inverse power of z. This region is therefore $O(z \log^2 z)$. On the complementary part $c|z| < |x| < \delta$, expand

$$\mathrm{Log}(az - ibx) = \mathrm{Log}(-ibx) + O\Big(\frac{|z|}{|x|}\Big) \tag{20.17}$$

uniformly on the chosen slit sector. The error integrates to $O(z|\log z|)$, and products of two logarithms to $O(z \log^2 z)$. The zeroth-order terms give a finite constant because $\mathrm{Log}\,|x|$ and $\mathrm{Log}^2\,|x|$ are locally integrable. Therefore no negative power of z is produced. □

Theorem 20.7 (Exact coefficient of the logarithmic-over-pole term)**.** *For every continued branch $\mathcal{I}^\sharp$ at s_0, the coefficient of $\log z/z$, $z = s - s_0$, is*

$$\boxed{[\log z/z]\,\mathcal{I}^\sharp(s_0 + z) = -\pi q_1 C(s_0),} \tag{20.18}$$

where $q_1 = 0$ when $1 \notin \mathcal{R}(s_0)$. Consequently,

$$\boxed{[\log z/z]\,\mathcal{I}^\sharp = 0 \iff \zeta(s_0) \neq 0.} \tag{20.19}$$

No singularity with $t_0 > 0$ can change this coefficient.

Proof. From (20.15), for $t = zu$ one has

$$S(z, zu) = C(s_0)\,\mathrm{Log}\,z + D(s_0) + G_{s_0}(u), \qquad S(z, 0) = C(s_0)\,\mathrm{Log}\,z + D(s_0). \tag{20.20}$$

The singular–singular numerator equals

$$\begin{aligned} &S(z,0)^2 - S(z,zu)S(z,-zu) \\ &= -\big(C\,\mathrm{Log}\,z + D\big)\big(G(u) + G(-u)\big) - G(u)G(-u). \end{aligned} \tag{20.21}$$

The contour rotation from the ray $u = t/z$ to the positive real axis is legitimate in every closed subsector because the branch points $u = \pm iN/d$ lie on the imaginary axis; the large circular arc contributes zero since the integrand is $O(\log^2 |u|/|u|^2)$. Therefore the coefficient of $\mathrm{Log}\,z/z$ is

$$-C(s_0) \int_0^\infty \frac{G(u) + G(-u)}{u^2}\, du. \tag{20.22}$$

For real u,

$$G(u) + G(-u) = \sum_{N \in \mathcal{R}(s_0)} q_N \sum_{d|N} \frac{\mu(d)}{d} \log\left(1 + \frac{d^2u^2}{N^2}\right). \tag{20.23}$$

Integration by parts gives, for $a > 0$,

$$\int_0^\infty \frac{\log(1 + a^2u^2)}{u^2}\, du = \pi a. \tag{20.24}$$

The boundary term vanishes at zero and infinity. Thus

$$\begin{aligned} \int_0^\infty \frac{G(u) + G(-u)}{u^2}\, du &= \pi \sum_{N \in \mathcal{R}(s_0)} \frac{q_N}{N} \sum_{d|N} \mu(d) \\ &= \pi q_1, \end{aligned} \tag{20.25}$$

because $\sum_{d|N}\mu(d)=\delta_{N1}$. This proves (20.18). Since all summands in $C(s_0)$ are positive, $C(s_0)>0$ whenever $\mathcal{R}(s_0)\neq\varnothing$, so the coefficient vanishes exactly when $q_1=0$, equivalently $\zeta(s_0)\neq 0$. Finally, Lemma 20.6 excludes any restoration of the same order by interior contour singularities. □

Corollary 20.8 (Blockwise Möbius cancellation)**.** *Let ρ be a zero of multiplicity m_ρ, let $r\geq 1$, and set $s_0=\rho/r$. The singular–singular block attached specifically to (ρ,r) contributes*

$$\boxed{-\pi m_\rho^2\frac{\varphi(r)}{r^2}\left(\sum_{d|r}\mu(d)\right)\frac{\operatorname{Log}(s-s_0)}{s-s_0}} \tag{20.26}$$

to the logarithmic-over-pole scale of $\mathcal{I}^\sharp$. Hence this block is zero for every $r>1$, whereas for $r=1$ it equals

$$-\pi m_\rho^2\frac{\operatorname{Log}(s-\rho)}{s-\rho}. \tag{20.27}$$

Proof. Restrict (20.15) to a single dilation index r. Then $C=m_\rho\varphi(r)/r$, while the integral (20.25) for that single block is $\pi m_\rho r^{-1}\sum_{d|r}\mu(d)$. Their negative product gives (20.26). The divisor identity $\sum_{d|r}\mu(d)=0$ for $r>1$ and equals one for $r=1$, proving both conclusions. □

20.3. **Pure replicas: the next coefficient and the complete local germ.** Assume now that $\mathcal{R}(s_0)\neq\varnothing$ but $\zeta(s_0)\neq 0$, so $q_1=0$. Define

$$\beta(s_0):=-\int_0^\infty\frac{|G_{s_0}(u)|^2}{u^2}\,du. \tag{20.28}$$

Lemma 20.9 (Convergence and strict negativity of the collision energy)**.** *Under $q_1=0$, the integral* (20.28) *converges and*

$$\boxed{\beta(s_0)<0.} \tag{20.29}$$

Proof. At zero,

$$G'_{s_0}(0)=-i\sum_{N\in\mathcal{R}(s_0)}\frac{q_N}{N}\sum_{d|N}\mu(d)=0, \tag{20.30}$$

because every $N\in\mathcal{R}(s_0)$ is greater than one. Since $G(0)=0$, it follows that $G(u)=O(u^2)$ and the integrand is $O(u^2)$ at zero. At infinity,

$$G(u)=C(s_0)\log u+O(1), \tag{20.31}$$

so the integrand is $O(\log^2 u/u^2)$. Thus the integral converges.

It remains to exclude $G\equiv 0$. Let $N_{\max}=\max\mathcal{R}(s_0)$. Differentiating (20.14) meromorphically in u, the $d=1$ term belonging to $N_{\max}$ has a simple pole at $u=-iN_{\max}$. Every pole from a term with $M<N_{\max}$ has modulus $M/d\leq M<N_{\max}$, and terms with $N=N_{\max}$, $d>1$, have modulus $N_{\max}/d<N_{\max}$. Hence that pole cannot cancel. Therefore $G\not\equiv 0$, its squared modulus is positive on a set of positive measure, and (20.29) follows. □

Theorem 20.10 (Complete local expansion at a pure replica point)**.** *Suppose $\mathcal{R}(s_0)\neq\varnothing$ and $\zeta(s_0)\neq 0$. Every continued branch satisfies, in every closed slit subsector,*

$$\boxed{\mathcal{I}^\sharp(s_0+z)=\frac{\beta(s_0)}{z}+\pi C(s_0)\frac{\zeta'}{\zeta}(s_0)\operatorname{Log}z+O(1).} \tag{20.32}$$

In particular the pole coefficient is nonzero and no term $\operatorname{Log}z/z$ is present.

Proof. Decompose $B = S + R$ as in Lemma 20.5. The singular–singular part is given by (20.21). Since $q_1 = 0$, equation (20.25) makes the entire term linear in $C \operatorname{Log} z + D$ vanish. Therefore its negative-power contribution is exactly

$$\frac{1}{z}\left(-\int_0^\infty \frac{G(u)G(-u)}{u^2}\,du\right) = \frac{\beta(s_0)}{z}, \tag{20.33}$$

because on the positive real axis $G(-u) = \overline{G(u)}$. The contour rotation used in Theorem 20.7 shows the same coefficient in every slit subsector.

Near $(0,0)$, write

$$R(z, \pm t) = R_0(z) \pm h_1 t + O(t^2 + |z|t), \tag{20.34}$$

with $h_1 = \partial_t R(0,0)$. The constant cross term is proportional to

$$\int_0^\infty \frac{G(u) + G(-u)}{u^2}\,du = 0, \tag{20.35}$$

so it produces neither z^{-1} nor $z^{-1}\operatorname{Log} z$. For the odd linear term, one obtains

$$h_1 \int \frac{S(z,t) - S(z,-t)}{t}\,dt. \tag{20.36}$$

As $u \to +\infty$,

$$G(u) - G(-u) = -i\pi C(s_0) + O(u^{-1}), \tag{20.37}$$

so integration from $u = 1$ to $u = \delta/|z|$ gives

$$i\pi C(s_0) h_1 \operatorname{Log} z + O(1). \tag{20.38}$$

All Taylor terms of order t^2 or higher are integrable at the endpoint and contribute $O(1)$; terms involving z are also $O(1)$. By Lemma 20.6, every interior contour singularity is bounded up to $O(z \log^2 z)$, and the contour tail is normally convergent by Lemma 20.3. Thus no additional unbounded term of order z^{-1}, $z^{-1}\operatorname{Log} z$, or $\operatorname{Log} z$ is present.

Finally, differentiating S at $t = 0$ gives

$$\partial_t S(z,0) = -\frac{i}{z} \sum_{N \in \mathcal{R}(s_0)} \frac{q_N}{N} \sum_{d \mid N} \mu(d) = 0, \tag{20.39}$$

so $h_1 = \partial_t B(s_0, 0)$. Proposition 19.5 and meromorphic continuation give

$$h_1 = -i\frac{\zeta'}{\zeta}(s_0), \tag{20.40}$$

because $\zeta(s_0) \neq 0$. Substitution into (20.38) proves (20.32). □

20.4. **Single-replica divisor energy.** For a single zero ρ of multiplicity m_ρ and one dilation index $r > 1$, define

$$G_r(u) = \sum_{d \mid r} \frac{\mu(d)}{d} \operatorname{Log}\left(1 - i\frac{d}{r}u\right). \tag{20.41}$$

Proposition 20.11 (Replica energy, finite formula, and sign)**.** *Define*

$$\boxed{\beta_r = -\int_0^\infty \frac{|G_r(u)|^2}{u^2}\,du.} \tag{20.42}$$

Then $\beta_r < 0$ for every $r > 1$, and

$$\boxed{\beta_r = -\frac{\pi}{r} \sum_{d,e \mid r} \frac{\mu(d)\mu(e)}{de}\Big[(d+e)\log(d+e) - d\log d - e\log e\Big].} \tag{20.43}$$

For an isolated pure replica point $s_0 = \rho/r$, the pole coefficient in (20.32) *is $m_\rho^2 \beta_r$.*

Proof. The convergence and strict sign follow from the proof of Lemma 20.9; for a single $r > 1$, $G_r(0) = G_r'(0) = 0$ and $G_r \not\equiv 0$.

For the finite formula, set

$$F(a,b) = \Re \int_0^\infty \frac{\operatorname{Log}(1 - iau)\operatorname{Log}(1 + ibu)}{u^2}\, du, \qquad a, b > 0. \tag{20.44}$$

The integral converges absolutely after taking the real part of the paired expression. Differentiate first with respect to a and b, justified on compact parameter rectangles by dominated convergence after splitting $(0,1) \cup (1,\infty)$. One obtains

$$\frac{\partial^2 F}{\partial a \partial b} = \Re \int_0^\infty \frac{du}{(1 - iau)(1 + ibu)} = \frac{\pi}{a + b}. \tag{20.45}$$

The last equality follows from partial fractions and evaluation of the logarithms at a finite cutoff before taking the limit. Since $F(a,0) = F(0,b) = 0$, two integrations give

$$F(a,b) = \pi\big[(a+b)\log(a+b) - a\log a - b\log b\big]. \tag{20.46}$$

Expanding $|G_r(u)|^2$, using $a = d/r$, $b = e/r$, and the homogeneity of (20.46), gives (20.43). □

Corollary 20.12 (Radical law)**.** *For every $r > 1$,*

$$\boxed{\beta_r = \frac{\operatorname{rad}(r)}{r}\beta_{\operatorname{rad}(r)}.} \tag{20.47}$$

Equivalently, $r\beta_r$ depends only on the square-free kernel of r.

Proof. Put $q = \operatorname{rad}(r)$. In the rescaled form of (20.42), let $u = rx$. Then

$$\beta_r = -\frac{1}{r}\int_0^\infty \frac{1}{x^2}\left|\sum_{d \mid r} \frac{\mu(d)}{d}\operatorname{Log}(1 - idx)\right|^2 dx. \tag{20.48}$$

Only square-free divisors contribute, and those are exactly the divisors of q. The integral in (20.48) is therefore the same as the corresponding integral for q, while the exterior factors are $1/r$ and $1/q$. Hence $r\beta_r = q\beta_q$, which is (20.47). □

Corollary 20.13 (Prime and prime-power replica asymptotics)**.** *For a prime p,*

$$\boxed{\beta_p = -\frac{2\pi}{p^2}\left[p\log p - (p+1)\log(p+1) + (p+1)\log 2\right].} \tag{20.49}$$

As $p \to \infty$,

$$\beta_p = -\frac{2\pi\log 2}{p} + O\Big(\frac{\log p}{p^2}\Big). \tag{20.50}$$

For every integer $a \geq 1$,

$$\boxed{\beta_{p^a} = p^{1-a}\beta_p.} \tag{20.51}$$

Proof. In (20.43), for $r = p$ the only divisors are $1, p$. Writing out the four terms and collecting logarithms gives (20.49). Expanding $(p+1)\log(p+1) = p\log p + \log p + 1 + O(p^{-1})$ yields (20.50); the weaker displayed error is sufficient and uniform. Finally, $\operatorname{rad}(p^a) = p$, so Corollary 20.12 gives (20.51). □

20.5. **Actual zeta zeros, $\mathcal{O}$, and monodromy.**

Theorem 20.14 (Survival at a genuine zero)**.** *Let ρ be a zero of multiplicity m_ρ. For every continued branch,*

$$\mathcal{I}^\sharp(\rho+z) = -\pi m_\rho C(\rho)\frac{\operatorname{Log} z}{z} + O\!\left(\frac{1}{z}\right), \tag{20.52}$$

where

$$C(\rho) = \sum_{N\in\mathcal{R}(\rho)} q_N \frac{\varphi(N)}{N} \geq m_\rho. \tag{20.53}$$

If no scaled zero other than $N=1$ collides at ρ, then

$$\boxed{\mathcal{I}^\sharp(\rho+z) = -\pi m_\rho^2 \frac{\operatorname{Log} z}{z} + O(z^{-1}).} \tag{20.54}$$

Proof. At $s_0=\rho$, one has $q_1 = m_\rho > 0$. Theorem 20.7 gives the coefficient $-\pi q_1 C(\rho)$, proving (20.52). All other terms are at most $O(z^{-1})$: the singular–singular expansion (20.21) has no stronger scale, cross terms contain at most one singular logarithm, and interior singularities are lower order by Lemma 20.6. In the absence of collisions, $C(\rho)=m_\rho$, yielding (20.54). □

Corollary 20.15 (Correct singular orders for $\mathcal{O}$)**.** *Let $\zeta(\rho+z) = a_m z^m + O(z^{m+1})$, $a_m \neq 0$, with $m = m_\rho$. In the noncolliding case,*

$$\boxed{\mathcal{O}(\rho+z) = -\frac{a_m m^2}{2} z^{m-1}\operatorname{Log} z + O(z^{m-1}).} \tag{20.55}$$

For a simple zero,

$$\boxed{\mathcal{O}(\rho+z) = -\frac{\zeta'(\rho)}{2}\operatorname{Log} z + O(1).} \tag{20.56}$$

At a pure replica $s_0 = \rho/r$, $r>1$, with no collision with an actual zero at s_0,

$$\boxed{\mathcal{O}(s_0+z) = \frac{\zeta(s_0)\,\beta(s_0)}{2\pi z} + \frac{C(s_0)\zeta'(s_0)}{2}\operatorname{Log} z + O(1).} \tag{20.57}$$

Proof. Use the identity $\mathcal{O} = \zeta\mathcal{I}/(2\pi)$, initially valid for $\Re s > 1$ and then on every continued branch by analytic continuation. Multiplying (20.54) by $a_m z^m + O(z^{m+1})$ gives (20.55); the case $m=1$ is (20.56). At a pure replica, $\zeta(s_0)\neq 0$, so multiplication of (20.32) gives (20.57). □

Warning 20.16 (The requested $\log z/z$ formula belongs to $\mathcal{I}$). A formula of the form

$$C_{\rho,r}\left(\sum_{d\mid r}\mu(d)\right)\frac{\operatorname{Log}(s-\rho/r)}{s-\rho/r} \tag{20.58}$$

is correct for the normalized energy $\mathcal{I} = 2\pi\mathcal{O}/\zeta$, with

$$C_{\rho,r} = -\pi m_\rho^2 \frac{\varphi(r)}{r^2} \tag{20.59}$$

for the individual singular–singular block. It is not the correct local order of $\mathcal{O}$ itself at $r=1$, because ζ vanishes there and cancels the pole. Equations (20.55) and (20.57) are the correct statements for $\mathcal{O}$.

Corollary 20.17 (Monodromy detector for genuine zeros)**.** *Let $\Delta\mathcal{I}^\sharp$ denote one positive turn of the local branch. Then*

$$\operatorname*{Res}_{s=s_0} \Delta\mathcal{I}^\sharp(s) = -2\pi^2 i\, q_1 C(s_0). \tag{20.60}$$

Consequently

$$\boxed{\zeta(s_0) = 0 \iff \Delta\mathcal{I}^\sharp \text{ has a pole at } s_0.} \tag{20.61}$$

Proof. Only the term $-\pi q_1 C(s_0)\operatorname{Log} z/z$ changes by a term of order z^{-1} after $\operatorname{Log} z \mapsto \operatorname{Log} z + 2\pi i$. Its change is $-2\pi^2 i q_1 C(s_0)/z$. All pole terms without a logarithm are single-valued, and all lower logarithmic terms have bounded monodromy. Since $C(s_0) > 0$, the residue is nonzero exactly when $q_1 > 0$, i.e. when $\zeta(s_0) = 0$. □

20.6. **Analytic checklist for the replica theorem.**

Proposition 20.18 (Eight-point analytic closure)**.** *The replica analysis above establishes the following items without RH, zero-simplicity, or a nonresonance assumption:*

1. *$\mathcal{O}(s)$ and $\mathcal{I}(s)$ exist absolutely for $\Re s > 1$ and are holomorphic there.*
2. *Every local branch required for the replica analysis is obtained by continuation from that half-plane, with a normally convergent large-dilation tail.*
3. *All exchanges among prime sums, dilation sums, the t-integral, and the differentiations used in the initial half-plane are justified by the majorants displayed above.*
4. *The complete singular expansion through the unbounded terms is* (20.32) *at pure replica points and* (20.52) *at actual zeros.*
5. *Every contribution reaching s_0 at the endpoint is included by the finite set $\mathcal{R}(s_0)$; every interior contour contribution is covered by Lemma 20.6.*
6. *For each individual index $r > 1$, the coefficient of $z^{-1}\operatorname{Log} z$ vanishes by $\sum_{d|r}\mu(d) = 0$.*
7. *No other endpoint or interior term can restore a $z^{-1}\operatorname{Log} z$ contribution when $\zeta(s_0) \neq 0$.*
8. *The corresponding $r = 1$ term survives with nonzero coefficient $-\pi m_\rho^2$ in the noncolliding case, and more generally with $-\pi q_1 C(s_0)$.*

Proof. Items (1)–(3) are Theorems 19.2, 19.3, Lemmas 20.1 and 20.3, and Theorem 20.4. Item (4) is Theorems 20.10 and 20.14. Item (5) is Lemmas 20.5 and 20.6. Item (6) is Corollary 20.8. Item (7) follows from Theorem 20.7 together with Lemma 20.6. Item (8) is Theorem 20.14. Thus every item is a consequence of a proved statement in the preceding subsections. □

21. A two-variable Euler product and the covariance determinant

The determinant in (12.3) has an intrinsic interpretation that does not involve a linearly Möbius-weighted quantum state. For $\Re s > 1$ and u in a neighbourhood of zero define

$$\mathfrak{Z}(s,u) = \prod_p \left(1 - e^u p^{-s}\right)^{-1}, \qquad F(s,u) = \log \mathfrak{Z}(s,u). \tag{21.1}$$

Then $\mathfrak{Z}(s,0) = \zeta(s)$.

Theorem 21.1 (Hessian form of the curvature)**.** *For $\Re s > 1$,*

$$F_{uu}(s,0) = S_0(s), \tag{21.2}$$

$$-F_{su}(s,0) = S_1(s), \tag{21.3}$$

$$F_{ss}(s,0) = S_2(s). \tag{21.4}$$

Consequently

$$\boxed{\frac{2\mathcal{K}(s)}{\zeta(s)} = \det\begin{pmatrix} F_{uu}(s,0) & F_{us}(s,0) \\ F_{su}(s,0) & F_{ss}(s,0) \end{pmatrix}.} \tag{21.5}$$

Proof. Differentiate

$$F(s,u) = -\sum_p \log(1 - e^u p^{-s}) \tag{21.6}$$

termwise in a neighbourhood of $u = 0$, justified by normal convergence for $\Re s > 1$. The three derivatives are exactly (12.4). Equation (21.5) is then Theorem 12.1. □

Theorem 21.2 (Zeta-distribution covariance identity)**.** *Let $\sigma > 1$ be real and let N_σ be the zeta-distributed integer*

$$\mathbb{P}_\sigma(N_\sigma = n) = \frac{n^{-\sigma}}{\zeta(\sigma)}. \tag{21.7}$$

Then

$$\mathrm{Var}_\sigma(\Omega(N_\sigma)) = S_0(\sigma), \tag{21.8}$$

$$\mathrm{Cov}_\sigma(\Omega(N_\sigma), \log N_\sigma) = S_1(\sigma), \tag{21.9}$$

$$\mathrm{Var}_\sigma(\log N_\sigma) = S_2(\sigma). \tag{21.10}$$

Therefore

$$\boxed{\frac{2\mathcal{K}(\sigma)}{\zeta(\sigma)} = \det \mathrm{Cov}_\sigma \begin{pmatrix} \Omega(N_\sigma) \\ \log N_\sigma \end{pmatrix} > 0.} \tag{21.11}$$

Proof. Under (21.7), the prime valuations $v_p(N_\sigma)$ are independent geometric random variables with

$$\mathrm{Var}(v_p) = \frac{p^{-\sigma}}{(1 - p^{-\sigma})^2}. \tag{21.12}$$

Since

$$\Omega(N_\sigma) = \sum_p v_p(N_\sigma), \qquad \log N_\sigma = \sum_p v_p(N_\sigma) \log p, \tag{21.13}$$

independence gives (21.8)–(21.10). Strict positivity follows from

$$S_0 S_2 - S_1^2 = \sum_{p<q} w_p w_q \log^2(q/p), \qquad w_p = \frac{p^{-\sigma}}{(1 - p^{-\sigma})^2} > 0. \tag{21.14}$$

□

22. The positivity route and its exact obstruction

The coefficients $c(n) = -z_0''(n)$ are nonnegative. It is therefore natural to ask whether positivity itself forces the absence of the logarithmic branches in $1/2 < \Re s < 1$. It does not.

Proposition 22.1 (Complete monotonicity on the real axis)**.** *For every real $\sigma > 1$ and every integer $r \geq 0$,*

$$\boxed{(-1)^r \mathcal{K}^{(r)}(\sigma) = \sum_{n \geq 1} \frac{c(n)(\log n)^r}{n^\sigma} \geq 0.} \tag{22.1}$$

In particular $\mathcal{K}$ is completely monotone on $(1, \infty)$.

Proof. Differentiate the absolutely convergent Dirichlet series termwise. □

Proposition 22.2 (Positivity does not remove complex logarithmic branches)**.** *Nonnegative Dirichlet coefficients, complete monotonicity on the real axis, and the strict covariance inequality* (21.11) *do not imply a branch-free continuation of $\mathcal{K}$ to $1/2 < \Re s < 1$.*

Proof. The source term $P(s) = \sum_p p^{-s}$ itself has nonnegative Dirichlet coefficients for $\Re s > 1$, yet its continuation is expressed through $\log \zeta(rs)$ and has logarithmic branch points at scaled zeros of ζ; see (14.4). More directly, the positive-coefficient function S_0 in (14.2) is exactly the factor that contributes $m \log(s - \rho)$ at a hypothetical zero ρ with $1/2 < \Re\rho < 1$. Thus real-axis positivity and complete monotonicity do not control the monodromy of the analytic continuation. □

Corollary 22.3 (What would still be needed to prove RH)**.** *To deduce Corollary 13.3 from the curvature side, one needs an additional property that controls the complex continuation of $\mathcal{K}$, not merely the signs of its Dirichlet coefficients or the positive-semidefiniteness of the real covariance matrix in* (21.11)*.*

Proof. Proposition 22.1 and Theorem 21.2 control only real-axis signs and positive semidefiniteness. Proposition 22.2 exhibits logarithmic branches compatible with those same properties. Therefore any proof using this route must add information that controls the complex continuation of $\mathcal{K}$, as asserted. □

Remark 22.4 (Status of the attempted step)*.* The expected coefficient-positivity step has therefore been carried out and closed: $c(n) \geq 0$ gives complete monotonicity and a positive covariance Hessian, but those properties do not forbid the logarithmic branches whose absence is equivalent to RH. The obstruction is visible inside the same formula: S_0 has positive coefficients and is simultaneously the carrier of the prime-zeta logarithmic monodromy.

23. A curvature–Weil positive kernel after primitive-replica filtering

The underlying positivity criterion is Weil's classical criterion for RH [18, 19]. What is specific to the present paper is the way the cocyclic curvature supplies a canonical positive Gram factor and the way the transverse variable in $B(s,t)$ removes prime-zeta replicas before the zero kernel is formed.

23.1. **Polarization of the arithmetic curvature.** For an integer $n \geq 1$, write

$$w_p(n) := T(v_p(n)) = \frac{v_p(n)(v_p(n)+1)}{2}. \tag{23.1}$$

For primes $p < q$, define the real feature

$$\Phi_{pq}(n) := \frac{1}{\sqrt{2}} \sqrt{w_p(n) w_q(n)}\, \log \frac{q}{p}. \tag{23.2}$$

Only finitely many of these features are nonzero for each fixed n.

Proposition 23.1 (Exact Gram polarization of the curvature)**.** *For every integer $n \geq 1$,*

$$\boxed{\mathcal{C}(n) = \sum_{p<q} |\Phi_{pq}(n)|^2.} \tag{23.3}$$

Consequently

$$K_{\mathcal{C}}(m,n) := \sum_{p<q} \Phi_{pq}(m)\Phi_{pq}(n) \tag{23.4}$$

is a well-defined positive-semidefinite kernel on $\mathbb{N}$, and

$$K_{\mathcal{C}}(n,n) = \mathcal{C}(n). \tag{23.5}$$

Proof. The closed curvature formula proved in Theorem 10.4 is

$$\mathcal{C}(n) = \frac{1}{2} \sum_{p<q} T(v_p(n)) T(v_q(n)) \log^2 \frac{q}{p}. \tag{23.6}$$

Substitution of (23.1) and (23.2) into the right-hand side of (23.3) gives exactly (23.6). For any finitely supported complex family (a_n), every sum below is finite and therefore may be rearranged without a convergence issue:

$$\sum_{m,n} \overline{a_m} a_n K_{\mathcal{C}}(m,n) = \sum_{p<q} \sum_{m,n} \overline{a_m} a_n \Phi_{pq}(m)\Phi_{pq}(n) \tag{23.7}$$

$$= \sum_{p<q} \left| \sum_n a_n \Phi_{pq}(n) \right|^2 \geq 0. \tag{23.8}$$

Thus $K_{\mathcal{C}}$ is positive semidefinite. Setting $m = n$ in (23.4) and using (23.3) proves (23.5). □

Remark 23.2 (Why the raw squared cocycle is not the kernel). The seemingly direct kernel $\mathscr{A}(m,n)^2$, where $\mathscr{A}(m,n) = \Omega(m)\log n - \log m\,\Omega(n)$, is not positive semidefinite in general. If $\mathscr{A}(m,n) \neq 0$, its restriction to the two indices m, n has matrix

$$\begin{pmatrix} 0 & \mathscr{A}(m,n)^2 \\ \mathscr{A}(m,n)^2 & 0 \end{pmatrix}, \tag{23.9}$$

whose eigenvalues are $\pm\mathscr{A}(m,n)^2$. The polarization in Proposition 23.1 is therefore essential.

23.2. **The exact primitive-replica filter.** Recall from Theorem 19.2 that, for $\Re s > 1$,

$$B(s,t) = \sum_p \frac{p^{it}}{p^s - 1}. \tag{23.10}$$

Proposition 23.3 (Exact transverse primitivization)**.** *For $\Re s > 1$, differentiation in t is legitimate term by term and*

$$\boxed{\Pi B(s) := i\,\frac{\partial B}{\partial t}(s,t)\Big|_{t=0} = \frac{\zeta'}{\zeta}(s).} \tag{23.11}$$

Moreover, if the grouped log-zeta representation is decomposed according to the dilation index $N = mk$, then Π annihilates every sector $N > 1$ and retains the primitive sector $N = 1$.

Proof. For any fixed $\sigma_0 > 1$ and $|t| \leq T$,

$$\left| \frac{\partial}{\partial t} \frac{p^{it}}{p^s - 1} \right| \leq \frac{\log p}{p^{\sigma_0} - 1}. \tag{23.12}$$

The series on the right converges because

$$\sum_p \frac{\log p}{p^{\sigma_0} - 1} \leq \sum_p \sum_{a \geq 1} (\log p) p^{-a\sigma_0} = -\frac{\zeta'}{\zeta}(\sigma_0) < \infty. \tag{23.13}$$

The Weierstrass theorem therefore permits termwise differentiation of (23.10), uniformly on compact subsets of $\Re s > 1$ and bounded t-intervals. At $t = 0$,

$$\partial_t B(s,t)|_{t=0} = i \sum_p \frac{\log p}{p^s - 1} \tag{23.14}$$

$$= -i\frac{\zeta'}{\zeta}(s), \tag{23.15}$$

which proves (23.11).

For the sector statement, use the absolutely convergent representation in $\Re s > 1$

$$B(s,t) = \sum_{k,m \geq 1} \frac{\mu(m)}{m} \log \zeta(mks - imt). \tag{23.16}$$

Differentiating as above and grouping by $N = mk$ gives

$$\partial_t B(s,t)|_{t=0} = -i \sum_{N\geq 1} \frac{\zeta'}{\zeta}(Ns) \sum_{m|N} \mu(m). \tag{23.17}$$

All rearrangements are justified by absolute convergence in $\Re s > 1$. The elementary identity

$$\sum_{m|N} \mu(m) = \begin{cases} 1, & N = 1, \\ 0, & N > 1 \end{cases} \tag{23.18}$$

proves the asserted annihilation of every replica sector. □

Remark 23.4 (Completion). The completed logarithmic derivative is recovered from (23.11) by the standard identity

$$\frac{\xi'}{\xi}(s) = \Pi B(s) + \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi\left(\frac{s}{2}\right), \tag{23.19}$$

where $\psi = \Gamma'/\Gamma$. Thus the nontrivial poles of the completed primitive object occur at the genuine zeros ρ, not at the prime-zeta replicas ρ/r with $r > 1$.

23.3. **A zero kernel and Weil positivity.** Let $g \in C_c^\infty((0,\infty))$, and use the Mellin transform

$$G(s) := \int_0^\infty g(x) x^{s-1}\, dx. \tag{23.20}$$

Define

$$G^{\#}(s) := \overline{G(1-\overline{s})}, \qquad W_g(s) := G(s)G^{\#}(s). \tag{23.21}$$

For every fixed vertical strip, $G(s)$ decreases faster than every inverse power of $|\Im s|$; this follows by repeated integration by parts after the change of variable $x = e^u$. Consequently the zero sums below converge absolutely, because the classical zero-counting estimate is $N(T+1)-N(T) = O(\log(2+T))$; see [17].

Definition 23.5 (Primitive-zero kernel)**.** For positive integers m, n, define

$$K_g^Z(m,n) := \sum_\rho W_g(\rho) \left(\frac{m}{n}\right)^{\rho-1/2}, \tag{23.22}$$

where every nontrivial zero is counted with multiplicity and the positive real logarithm is used in the power. The absolute convergence just noted makes (23.22) independent of the order of summation.

Lemma 23.6 (Quadratic form of the zero kernel)**.** *For every finitely supported complex family (a_n), set*

$$A_a(s) := \sum_n a_n n^{1/2-s}, \qquad A_a^{\#}(s) := \overline{A_a(1-\overline{s})}. \tag{23.23}$$

Then

$$\boxed{\sum_{m,n} \overline{a_m} a_n K_g^Z(m,n) = \sum_\rho W_g(\rho) A_a(\rho) A_a^{\#}(\rho).} \tag{23.24}$$

Proof. Because (a_n) has finite support and (23.22) is absolutely convergent, the finite sums in m, n may be interchanged with the zero sum. For a fixed zero ρ,

$$\sum_{m,n} \overline{a_m} a_n \left(\frac{m}{n}\right)^{\rho-1/2} = \left(\sum_m \overline{a_m} m^{\rho-1/2}\right)\left(\sum_n a_n n^{1/2-\rho}\right) \tag{23.25}$$

$$= A_a(\rho) A_a^{\#}(\rho), \tag{23.26}$$

where the final identity is exactly the definition in (23.23). Substitution proves (23.24). □

Theorem 23.7 (RH implies positivity of the primitive-zero kernel)**.** *If RH holds, then for every $g \in C_c^\infty((0,\infty))$, the kernel K_g^Z is positive semidefinite.*

Proof. Under RH, every nontrivial zero has the form $\rho = 1/2 + i\gamma$, whence

$$G^{\#}(\rho) = \overline{G(\rho)}, \qquad W_g(\rho) = |G(\rho)|^2. \tag{23.27}$$

Likewise

$$A_a^{\#}(\rho) = \overline{A_a(\rho)}. \tag{23.28}$$

Lemma 23.6 therefore gives

$$\sum_{m,n} \overline{a_m} a_n K_g^Z(m,n) = \sum_{\rho} |G(\rho)|^2 |A_a(\rho)|^2 \geq 0. \tag{23.29}$$

The zero sum converges absolutely by the decay stated before Definition 23.5. □

Theorem 23.8 (Weil equivalence for the primitive-zero kernel)**.** *The following are equivalent:*

(i) RH holds.

(ii) For every $g \in C_c^\infty((0,\infty))$, the kernel K_g^Z is positive semidefinite on finitely supported sequences.

Proof. The implication (i)⇒(ii) is Theorem 23.7. Conversely, assume (ii). Choose a single index N and a sequence supported only at N, with $a_N = 1$. Then (23.24) gives

$$K_g^Z(N,N) = \sum_{\rho} W_g(\rho) \geq 0 \tag{23.30}$$

for every $g \in C_c^\infty((0,\infty))$. Bombieri's formulation of Weil's criterion states that RH is equivalent to

$$\sum_{\rho} G(\rho)\overline{G(1-\overline{\rho})} \geq 0 \quad \text{for every } g \in C_c^\infty((0,\infty)), \tag{23.31}$$

with zeros counted with multiplicity [18, 19]. The summand in (23.31) is precisely $W_g(\rho)$ by (23.21). Thus (23.30) for every g implies RH. □

23.4. **The curvature–zero Schur kernel.**

Definition 23.9 (Curvature–zero kernel)**.** For $g \in C_c^\infty((0,\infty))$, define

$$\mathscr{K}_g(m,n) := K_{\mathcal{C}}(m,n) K_g^Z(m,n). \tag{23.32}$$

Theorem 23.10 (Exact positivity criterion for the combined kernel)**.** *The following are equivalent:*

(i) RH holds.

(ii) For every $g \in C_c^\infty((0,\infty))$, the kernel $\mathscr{K}_g$ is positive semidefinite on finitely supported sequences.

Proof. Assume RH. Proposition 23.1 and Theorem 23.7 show that both factors in (23.32) are positive semidefinite. A direct proof of positivity of their entrywise product avoids invoking the Schur product theorem. Under RH, insert the Gram expansions (23.4) and (23.29). For every finitely supported (a_n), absolute convergence of the zero sum and finiteness of the support permit rearrangement, giving

$$\sum_{m,n} \overline{a_m} a_n \mathscr{K}_g(m,n) = \sum_{\rho} |G(\rho)|^2 \sum_{p<q} \left| \sum_n a_n \Phi_{pq}(n) n^{1/2-\rho} \right|^2 \geq 0. \tag{23.33}$$

Thus (i) implies (ii).

Conversely, assume (ii). Choose any integer N with at least two distinct prime factors; then Theorem 10.4 gives $\mathcal{C}(N) > 0$. Test the quadratic form on the one-point sequence $a_N = 1$. By (23.5),

$$\mathscr{K}_g(N, N) = \mathcal{C}(N) \sum_{\rho} W_g(\rho) \geq 0. \tag{23.34}$$

Division by the positive number $\mathcal{C}(N)$ gives Weil positivity for every g. Theorem 23.8 then yields RH. □

24. The Heisenberg-deformed zeta series

Definition 24.1 (Heisenberg-deformed zeta series)**.** For $\Re s > \sigma_*$, define

$$\zeta_{\mathrm{H}}(s; \vartheta) = \sum_{n \geq 1} \frac{z_\vartheta(n)}{n^s}. \tag{24.1}$$

Theorem 8.8 guarantees absolute convergence for every $\Re s > \sigma_*$.

Proposition 24.2 (Classical boundary)**.** *For $\Re s > 1$,*

$$\zeta_{\mathrm{H}}(s; 0) = \zeta(s). \tag{24.2}$$

Proof. By Proposition 8.5, $z_0(n) = 1$ for all n. Substitution into (24.1) gives the ordinary Dirichlet series for $\zeta(s)$. □

Proposition 24.3 (Even Heisenberg deformation)**.** *For $\Re s > \sigma_*$,*

$$\zeta_{\mathrm{H}}(s; -\vartheta) = \zeta_{\mathrm{H}}(s; \vartheta). \tag{24.3}$$

Proof. This follows termwise from Proposition 9.5 and absolute convergence. □

Warning 24.4 (No analytic continuation is asserted). The series (24.1) is absolutely convergent and holomorphic on the open half-plane $\Re s > \sigma_*$. No meromorphic continuation, functional equation, scalar Euler product, or critical-strip theory for $\zeta_{\mathrm{H}}(s; \vartheta)$ is asserted here.

Proposition 24.5 (Conditional first-order rigidity of simple zeros)**.** *Assume that, for some simple zero ρ of the ordinary zeta function, there exists a function $F(s, \vartheta)$ analytic in a neighbourhood of $(\rho, 0)$ such that*

$$F(s, 0) = \zeta(s), \tag{24.4}$$

$$F(s, -\vartheta) = F(s, \vartheta), \tag{24.5}$$

and that F agrees with ζ_{H} wherever both are defined. Then the local zero branch $\rho(\vartheta)$ through ρ satisfies

$$\rho'(0) = 0. \tag{24.6}$$

Proof. The implicit-function theorem applies because $\zeta'(\rho) \neq 0$. Evenness (24.5) makes the unique local zero branch even, so its first derivative at the origin vanishes. □

Warning 24.6. Proposition 24.5 is conditional on an analytic continuation in both variables and does not supply such a continuation. It is included because it identifies a precise way in which the exact $\vartheta \mapsto -\vartheta$ symmetry would constrain the motion of classical Riemann zeros under the Heisenberg deformation if that continuation exists.

25. A genuine Möbius Hamiltonian

25.1. Definition and domain.

Definition 25.1 (Cosine and sine Weyl quadratures)**.** For $n \geq 1$, set

$$C_n = \frac{W_n + W_n^*}{2}, \tag{25.1}$$

$$J_n = \frac{W_n - W_n^*}{2i}. \tag{25.2}$$

Both are bounded self-adjoint operators with norm at most 1.

Definition 25.2 (Möbius Hamiltonian)**.** Fix an energy scale $E_0 > 0$ and $\sigma > 1$. Define

$$H_{\mu,\sigma,\vartheta} = E_0 \left(\operatorname{Re} \mathcal{M}_{\sigma,\vartheta} - I\right) = E_0 \sum_{n\geq 2} \frac{\mu(n)}{n^\sigma} C_n. \tag{25.3}$$

The series converges absolutely in operator norm.

Proposition 25.3 (Self-adjoint bounded dynamics)**.** *For $\sigma > 1$, $H_{\mu,\sigma,\vartheta}$ is bounded and self-adjoint on all of $\mathcal{H}$:*

$$\mathcal{D}(H_{\mu,\sigma,\vartheta}) = \mathcal{H}. \tag{25.4}$$

It therefore generates the unitary group

$$U(t) = \exp\left(-\frac{i}{\hbar} t H_{\mu,\sigma,\vartheta}\right). \tag{25.5}$$

Proof. The norm series in (25.3) is dominated by $E_0 \sum_{n\geq 2} |\mu(n)| n^{-\sigma}$, which converges for $\sigma > 1$. Every C_n is self-adjoint, hence so is the norm limit. The spectral theorem then defines (25.5). □

Proposition 25.4 (Explicit Schrödinger equation)**.** *For $\psi_0 \in \mathcal{H}$, the evolution $\psi(t) = U(t)\psi_0$ satisfies*

$$i\hbar\, \partial_t \psi(t,q) = (H_{\mu,\sigma,\vartheta}\psi(t))(q) \tag{25.6}$$

in the strong L^2 sense, where

$$(H_{\mu,\sigma,\vartheta}\psi)(q) = \frac{E_0}{2} \sum_{n\geq 2} \frac{\mu(n)}{n^\sigma} \Bigg[e^{\frac{i}{\hbar} y_n (q - x_n/2)} \psi(q - x_n) + e^{-\frac{i}{\hbar} y_n (q + x_n/2)} \psi(q + x_n) \Bigg]. \tag{25.7}$$

Proof. Equation (25.7) follows from Proposition 4.2 applied to W_n and W_n^*. The series converges in operator norm, so the bounded Hamiltonian gives a strong Schrödinger evolution for every initial vector. □

25.2. Exact equations of motion for position and momentum.

Lemma 25.5 (Bounded first commutators)**.** *For $\sigma > 1$, the series*

$$\sum_{n\geq 2} \frac{|\mu(n)| x_n}{n^\sigma}, \tag{25.8}$$

$$\sum_{n\geq 2} \frac{|\mu(n)| y_n}{n^\sigma} \tag{25.9}$$

converge. Hence $[H_{\mu,\sigma,\vartheta}, Q]$ and $[H_{\mu,\sigma,\vartheta}, P]$, initially defined on $\mathcal{S}(\mathbb{R})$, extend to bounded operators.

Proof. The first series is bounded by a constant multiple of $\sum n^{-\sigma}\log n$. Since

$$\Omega(n) \le \frac{\log n}{\log 2}, \tag{25.10}$$

the second is bounded by another constant multiple of the same convergent series. □

Theorem 25.6 (Exact Heisenberg equations of motion)**.** *Let*

$$Q(t) = U(t)^* Q U(t), \tag{25.11}$$

$$P(t) = U(t)^* P U(t). \tag{25.12}$$

The bounded commutators in Lemma 25.5 imply invariance of $\mathcal{D}(Q)$ and $\mathcal{D}(P)$ under $U(t)$. Indeed, for $A = Q$ or P and $\psi \in \mathcal{D}(A)$, Duhamel's formula first on a core and then by graph-norm closure gives

$$AU(t)\psi = U(t)A\psi + \frac{i}{\hbar}\int_0^t U(t-\tau)[H,A]U(\tau)\psi\, d\tau. \tag{25.13}$$

whose right-hand side lies in L^2 because $[H, A]$ is bounded. On these domains, and equivalently as quadratic-form identities, the Heisenberg equations are

$$\dot{Q}(t) = \frac{E_0}{\hbar} U(t)^* \left[\sum_{n\ge 2} \frac{\mu(n) x_n}{n^\sigma} J_n\right] U(t), \tag{25.14}$$

$$\dot{P}(t) = \frac{E_0}{\hbar} U(t)^* \left[\sum_{n\ge 2} \frac{\mu(n) y_n}{n^\sigma} J_n\right] U(t). \tag{25.15}$$

At $t = 0$ this becomes

$$\dot{Q}(0) = \frac{E_0 \ell}{\hbar} \sum_{n\ge 2} \frac{\mu(n)\log n}{n^\sigma} J_n, \tag{25.16}$$

$$\dot{P}(0) = \frac{E_0 \kappa}{\hbar} \sum_{n\ge 2} \frac{\mu(n)\Omega(n)}{n^\sigma} J_n. \tag{25.17}$$

Proof. For $A = Q$ or P, the bounded-commutator theorem for unitary dynamics gives

$$\dot{A}(t) = \frac{i}{\hbar} U(t)^* [H_{\mu,\sigma,\vartheta}, A] U(t). \tag{25.18}$$

From Proposition 4.4,

$$[Q, C_n] = i x_n J_n, \tag{25.19}$$

$$[P, C_n] = i y_n J_n. \tag{25.20}$$

Therefore

$$[H_{\mu,\sigma,\vartheta}, Q] = -iE_0 \sum_{n\ge 2} \frac{\mu(n) x_n}{n^\sigma} J_n, \tag{25.21}$$

$$[H_{\mu,\sigma,\vartheta}, P] = -iE_0 \sum_{n\ge 2} \frac{\mu(n) y_n}{n^\sigma} J_n. \tag{25.22}$$

Lemma 25.5 justifies the sums as bounded operators. Substitution into (25.18) gives (25.14) and (25.15). □

Remark 25.7 (Physical and arithmetic meaning of the two equations). The position variable is Q, the logarithmic-scale coordinate. The momentum variable is $P = -i\hbar\partial_q$, the generator conjugate to that coordinate. Multiplication by an integer n is represented by a phase-space displacement of size $\ell \log n$ in Q and $\kappa\Omega(n)$ in P. Equations (25.16) and (25.17) therefore say that the Möbius

Hamiltonian drives position by logarithmic size and momentum by prime-factor count, with the same quantum current operators J_n.

25.3. **The CCR is visible in the dynamics.**

Proposition 25.8 (Pairwise dynamical noncommutativity)**.** *Extend* log *and* Ω *additively to positive rationals and define* W_r *by* (4.19) *for* $r \in \mathbb{Q}_{>0}$*. For* $m, n \in \mathbb{N}$*, put*

$$\Phi_{m,n} = \vartheta(\Omega(m)\log n - \log m\,\Omega(n)). \tag{25.23}$$

Then

$$[C_m, J_n] = \frac{1}{2}\sin\Big(\frac{\Phi_{m,n}}{2}\Big)\Big(W_{mn} + W_{m/n} - W_{n/m} - W_{1/(mn)}\Big). \tag{25.24}$$

Proof. Expand C_m and J_n using (25.1)–(25.2). Apply (4.6) to the four pairs (z_m, z_n), $(z_m, -z_n)$, $(-z_m, z_n)$, and $(-z_m, -z_n)$. □

Corollary 25.9 (Second-order arithmetic commutator)**.** *At* $t = 0$*,*

$$\begin{aligned}\ddot{Q}(0) = \frac{iE_0^2}{2\hbar^2}\sum_{m,n\geq 2}\frac{\mu(m)\mu(n)x_n}{(mn)^\sigma}\sin\Big(\frac{\Phi_{m,n}}{2}\Big)\\ \times\Big(W_{mn} + W_{m/n} - W_{n/m} - W_{1/(mn)}\Big),\end{aligned} \tag{25.25}$$

and the analogous momentum equation is

$$\begin{aligned}\ddot{P}(0) = \frac{iE_0^2}{2\hbar^2}\sum_{m,n\geq 2}\frac{\mu(m)\mu(n)y_n}{(mn)^\sigma}\sin\Big(\frac{\Phi_{m,n}}{2}\Big)\\ \times\Big(W_{mn} + W_{m/n} - W_{n/m} - W_{1/(mn)}\Big).\end{aligned} \tag{25.26}$$

The double sums converge in operator norm for $\sigma > 1$*.*

Proof. Differentiate (25.14) and (25.15) once more and use Proposition 25.8. Polynomial logarithmic weights are summable against $(mn)^{-\sigma}$ for $\sigma > 1$. □

Corollary 25.10 (The interaction disappears with the Weyl cocycle)**.** *If* $\vartheta = 0$*, then*

$$\Phi_{m,n} = 0 \tag{25.27}$$

for all m, n*, and all pairwise terms in* (25.25)–(25.26) *vanish. Thus the second-order cross-arithmetic second-order term displayed above is generated entirely by the exact Heisenberg cocycle.*

Proof. If $\vartheta = 0$, the phase $\Phi_{m,n}$ in Proposition 25.8 is zero for every pair, so every summand in the displayed second-order cross terms contains a factor $\sin(\Phi_{m,n}/2) = 0$ or the equivalent commutator factor. The pairwise contributions therefore vanish identically. □

26. Exact non-localization inside the arithmetic dynamics

Proposition 26.1 (Schwartz action and distributional transpose)**.** *For* $\sigma > 1$*, the Hamiltonian* $H_{\mu,\sigma,\vartheta}$ *maps* $\mathcal{S}(\mathbb{R})$ *continuously into itself. Its continuous transpose* $H^\times : \mathcal{S}'(\mathbb{R}) \to \mathcal{S}'(\mathbb{R})$ *is defined by*

$$\langle H^\times T, \phi\rangle = \langle T, H_{\mu,\sigma,\vartheta}\phi\rangle, \qquad T \in \mathcal{S}'(\mathbb{R}), \quad \phi \in \mathcal{S}(\mathbb{R}). \tag{26.1}$$

Then $H^\times$ *is continuous on* $\mathcal{S}'(\mathbb{R})$*.*

Proof. For every Schwartz seminorm $p_{a,b}(\phi) = \sup_q |q^a \phi^{(b)}(q)|$, the coordinate action of a Weyl operator gives

$$p_{a,b}(W_n\phi) \le C_{a,b}(1 + |x_n| + |y_n|)^{N_{a,b}} \sum_{j\le b} p_{a+j,b-j}(\phi) \tag{26.2}$$

for suitable constants. Since $x_n, y_n = O(\log n)$ and

$$\sum_{n\ge 2} n^{-\sigma}(1+\log n)^N < \infty \qquad (\sigma > 1) \tag{26.3}$$

for every fixed N, the Hamiltonian series converges in every Schwartz seminorm and defines a continuous map $\mathcal{S} \to \mathcal{S}$. Therefore (26.1) defines the standard continuous dual action on $\mathcal{S}'$. No termwise extension to all tempered distributions is assumed without this transpose construction. □

Proposition 26.2 (No exact position point is a stationary generalized mode)**.** *Let $\sigma > 1$, $\ell > 0$, and $E_0 > 0$. For no $q_0 \in \mathbb{R}$ and no scalar $E \in \mathbb{C}$ does $\delta_{q_0} \in \mathcal{S}'(\mathbb{R})$ satisfy*

$$H^{\times}_{\mu,\sigma,\vartheta}\delta_{q_0} = E\delta_{q_0}. \tag{26.4}$$

Proof. The transpose action of a Weyl displacement on a point mass is

$$W(x,y)^{\times}\delta_{q_0} = \exp\left(\frac{i}{\hbar}y(q_0 - x/2)\right)\delta_{q_0-x}, \tag{26.5}$$

$$(W(x,y)^*)^{\times}\delta_{q_0} = \exp\left(-\frac{i}{\hbar}y(q_0 + x/2)\right)\delta_{q_0+x}. \tag{26.6}$$

The $n = 2$ term therefore contributes nonzero masses at $q_0 \pm \ell\log 2$. No other term produces the same support point as the unique $n = 2$ contribution at either $q_0 - \ell\log 2$ or $q_0 + \ell\log 2$. Since $\mu(2) = -1$, the corresponding coefficient cannot cancel. Hence the support of the left side of (26.4) contains a point distinct from q_0, while the right side is supported only at q_0. □

Remark 26.3 (Relation to exact Heisenberg impossibility)*.* Theorem 3.9 is a universal consequence of the CCR. Proposition 26.2 is more specific: the Möbius Hamiltonian itself does not admit an exactly position-localized stationary mode. The reason is not a measurement uncertainty but the exact translation content of the arithmetic Weyl operators.

27. Mertens record laws and Heisenberg jets

This section develops a Mertens-side consequence of classical oscillation theory and then couples it exactly to an auxiliary Heisenberg lift. The analytic inputs are classical and are identified explicitly. The deductions from those inputs and the Weyl-area identities are proved below. No claim is made that the record law itself is a new theorem about the zero set.

27.1. **Historical maxima and the zero supremum.**

Definition 27.1 (Historical maximum and record times)**.** For $x \ge 1$, define

$$A^*(x) := \max_{1\le n\le x} |M(n)|, \qquad M(x) := \sum_{n\le x}\mu(n). \tag{27.1}$$

Let

$$\Theta := \sup\{\Re\rho : \zeta(\rho) = 0,\ 0 < \Re\rho < 1\}. \tag{27.2}$$

Since the oscillation theorem below implies that $A^*(x) \to \infty$, and since A^* is integer-valued and can increase by at most one at each integer because $|M(n) - M(n-1)| \le 1$, it cannot skip an integer height. Thus for each integer $k \ge 1$ we may define

$$r_k := \min\{n \ge 1 : A^*(n) = k\}. \tag{27.3}$$

Theorem 27.2 (Classical oscillation input)**.** *For every $\delta > 0$ and $\varepsilon > 0$, all sufficiently large Y have the following property: the interval $[Y^{1-\delta}, Y]$ contains a sign change of the Mertens function with combined oscillation of size $x^{\Theta-\varepsilon}$. In particular, there is a point $y \in [Y^{1-\delta}, Y]$ for which*

$$|M(y)| \gg_{\delta,\varepsilon} y^{\Theta-\varepsilon}. \tag{27.4}$$

Proof. This is the specialization $a = 0$ of Corollary 4 of Kaczorowski–Pintz [13], combined with the "combined oscillation" clause in their Theorem 1. Their interval is of the form $[Y^{1-\delta}, Y]$, and their singularity exponent for the Mertens Dirichlet integral is precisely the supremum Θ of the real parts of nontrivial zeros of ζ. The definition of combined oscillation gives values on the two sides of the sign change whose absolute values have the stated order; retaining either one gives (27.4). □

Lemma 27.3 (Standard upper bound at the zero supremum)**.** *For every $\eta > 0$,*

$$M(x) \ll_\eta x^{\Theta+\eta}, \qquad A^*(x) \ll_\eta x^{\Theta+\eta}. \tag{27.5}$$

Proof. The identity $\sum_{n\geq 1} \mu(n)n^{-s} = 1/\zeta(s)$ initially holds for $\Re s > 1$. The standard Littlewood–Perron argument continues the corresponding summatory estimate throughout every fixed half-plane $\Re s > \Theta + \eta$, because $1/\zeta(s)$ is holomorphic there and has the standard vertical growth available away from the zero set. This gives the first estimate in (27.5); see Titchmarsh [17]. Taking the maximum over $n \leq x$ gives the second estimate after enlarging the implied constant. □

Theorem 27.4 (Uniform growth exponent of the historical maximum)**.** *One has*

$$\boxed{A^*(x) = x^{\Theta+o(1)}.} \tag{27.6}$$

Equivalently,

$$\frac{\log A^*(x)}{\log x} \longrightarrow \Theta. \tag{27.7}$$

Proof. Fix $\eta > 0$. Choose $\delta, \varepsilon > 0$ so small that

$$(1-\delta)(\Theta-\varepsilon) > \Theta - \eta. \tag{27.8}$$

By Theorem 27.2, for every sufficiently large Y there is $y \in [Y^{1-\delta}, Y]$ satisfying $|M(y)| \gg y^{\Theta-\varepsilon}$. Hence

$$A^*(Y) \geq |M(y)| \gg Y^{(1-\delta)(\Theta-\varepsilon)} \gg Y^{\Theta-\eta}. \tag{27.9}$$

Lemma 27.3 gives

$$A^*(Y) \ll_\eta Y^{\Theta+\eta}. \tag{27.10}$$

Since $\eta > 0$ is arbitrary, (27.6) follows. □

27.2. **Uniform separation of logarithmic scales.**

Corollary 27.5 (Scale separation)**.** *For every fixed $0 < \alpha < 1$,*

$$\boxed{\frac{A^*(x^\alpha)}{A^*(x)} = x^{-(1-\alpha)\Theta+o(1)} \longrightarrow 0.} \tag{27.11}$$

Proof. Apply Theorem 27.4 at x and x^α:

$$A^*(x^\alpha) = x^{\alpha\Theta+o(1)}, \qquad A^*(x) = x^{\Theta+o(1)}. \tag{27.12}$$

Their quotient is the expression in (27.11). Since $\Theta \geq 1/2$ and $\alpha < 1$, its main exponent is strictly negative. □

Remark 27.6 (Relation to the earlier record criterion)*.* The sufficient condition isolated earlier in the paper was only the record-subsequence statement $A^*(r_k^\alpha) = o(A^*(r_k))$. Corollary 27.5 proves the stronger uniform statement for all $x \to \infty$. Thus no separate argument excluding long historical plateaus is needed.

27.3. **Logarithmic historical collapse.**

Definition 27.7 (Logarithmic history and energy)**.** For real $x \geq 1$, with $M(t) = M(\lfloor t \rfloor)$, put

$$I(x) := \int_1^x \frac{M(t)}{t}\, dt, \tag{27.13}$$

$$E(x) := \int_1^x \frac{M(t)^2}{t}\, dt. \tag{27.14}$$

Theorem 27.8 (Uniform logarithmic L^p-collapse)**.** *For every fixed $p > 0$,*

$$\boxed{\frac{1}{A^*(x)^p \log x} \int_1^x |M(t)|^p \frac{dt}{t} \longrightarrow 0.} \tag{27.15}$$

Consequently,

$$\boxed{I(x) = o(A^*(x) \log x).} \tag{27.16}$$

Proof. Fix $0 < \alpha < 1$. Split the integral at x^α. By the definition of A^*,

$$\begin{aligned} \int_1^x |M(t)|^p \frac{dt}{t} &\leq A^*(x^\alpha)^p \int_1^{x^\alpha} \frac{dt}{t} + A^*(x)^p \int_{x^\alpha}^x \frac{dt}{t} \\ &= \alpha A^*(x^\alpha)^p \log x + (1-\alpha) A^*(x)^p \log x. \end{aligned} \tag{27.17}$$

After division by $A^*(x)^p \log x$, Corollary 27.5 gives

$$\limsup_{x \to \infty} \frac{1}{A^*(x)^p \log x} \int_1^x |M(t)|^p \frac{dt}{t} \leq 1 - \alpha. \tag{27.18}$$

Letting $\alpha \uparrow 1$ proves (27.15). The case $p = 1$, together with $|I(x)| \leq \int_1^x |M(t)| dt/t$, gives (27.16). □

Corollary 27.9 (Record logarithmic law)**.** *At the record times r_k,*

$$\boxed{\frac{I(r_k)}{M(r_k) \log r_k} \longrightarrow 0.} \tag{27.19}$$

Moreover, for every $p > 0$,

$$\boxed{\frac{1}{\log r_k} \int_1^{r_k} \left| \frac{M(t)}{M(r_k)} \right|^p \frac{dt}{t} \longrightarrow 0.} \tag{27.20}$$

In particular,

$$\boxed{E(r_k) = o(k^2 \log r_k).} \tag{27.21}$$

Proof. By definition, $A^*(r_k) = |M(r_k)| = k$. Substitute $x = r_k$ in Theorem 27.8. Equation (27.19) follows from (27.16), and (27.21) is the case $p = 2$. □

Corollary 27.10 (Logarithmic rarity of values comparable with the record)**.** *For every fixed $c > 0$,*

$$\boxed{\frac{1}{\log r_k} \int_1^{r_k} \mathbf{1}_{\{|M(t)| \geq ck\}} \frac{dt}{t} \longrightarrow 0.} \tag{27.22}$$

Proof. For any fixed $p > 0$, Markov's inequality gives

$$\mathbf{1}_{\{|M(t)| \geq ck\}} \leq c^{-p} \left| \frac{M(t)}{k} \right|^p. \tag{27.23}$$

Integrate with respect to dt/t and apply (27.20). □

27.4. **The law of the record times.**

Theorem 27.11 (Record-time exponent)**.** *The first record times satisfy*

$$\boxed{r_k = k^{1/\Theta+o(1)}.} \tag{27.24}$$

Consequently, for every fixed $0 < \varepsilon < 1$*,*

$$\boxed{\frac{\log r_{\lceil \varepsilon k \rceil}}{\log r_k} \longrightarrow 1.} \tag{27.25}$$

Proof. At $x = r_k$, Theorem 27.4 gives

$$k = A^*(r_k) = r_k^{\Theta+o(1)}. \tag{27.26}$$

Taking logarithms yields $\log k = (\Theta + o(1)) \log r_k$, hence (27.24). Applying the same formula to $\lceil \varepsilon k \rceil$ gives

$$\frac{\log r_{\lceil \varepsilon k \rceil}}{\log r_k} = \frac{(1/\Theta + o(1)) \log(\lceil \varepsilon k \rceil)}{(1/\Theta + o(1)) \log k} \longrightarrow 1. \tag{27.27}$$

□

27.5. **An auxiliary Heisenberg lift of the Mertens path.** The record theorem above is a consequence of classical analytic number theory. We now connect it to the arithmetic cocycle by an exact auxiliary Weyl lift. This lift is not substituted for the original embedding $n \mapsto (\ell \log n, \kappa\Omega(n))$; rather, its first transverse jet recovers that original symplectic determinant.

Definition 27.12 (Transverse Mertens deformation)**.** For real t, define

$$a_t(n) := \mu(n) e^{t\Omega(n)}, \tag{27.28}$$

$$F_t(N) := \sum_{n \le N} a_t(n), \tag{27.29}$$

$$L(N) := \sum_{n \le N} \mu(n) \log n. \tag{27.30}$$

Let

$$v_n(t) := \big(a_t(n), \mu(n) \log n\big) \in \mathbb{R}^2 \tag{27.31}$$

and define the ordered area

$$\mathcal{A}_t(N) := \frac{1}{2} \sum_{m<n\le N} \det(v_m(t), v_n(t)). \tag{27.32}$$

Equivalently,

$$\mathcal{A}_t(N) = \frac{1}{2} \sum_{m<n\le N} \mu(m)\mu(n) \left(e^{t\Omega(m)} \log n - \log m \, e^{t\Omega(n)} \right). \tag{27.33}$$

Remark 27.13 (Weyl interpretation). For any fixed nonzero scale constant c, the ordered product of Weyl displacements with increments proportional to $v_n(t)$ has central phase $\exp(ic\mathcal{A}_t(N))$. Thus $\mathcal{A}_t$ is literally the central coordinate of the step-two Heisenberg lift of the auxiliary path.

Proposition 27.14 (Exact incremental equation)**.** *For every integer* $N \ge 2$*,*

$$\boxed{\Delta\mathcal{A}_t(N) = \frac{\mu(N)}{2} \left[F_t(N-1) \log N - e^{t\Omega(N)} L(N-1) \right],} \tag{27.34}$$

where $\Delta\mathcal{A}_t(N) = \mathcal{A}_t(N) - \mathcal{A}_t(N-1)$*.*

Proof. Only pairs whose second entry is N are added when the cutoff increases from $N-1$ to N. Hence

$$\begin{aligned} 2\Delta\mathcal{A}_t(N) &= \sum_{m<N} [a_t(m)\mu(N)\log N - \mu(m)\log m\, a_t(N)] \\ &= \mu(N)\log N\, F_t(N-1) - \mu(N)e^{t\Omega(N)}L(N-1), \end{aligned} \tag{27.35}$$

which is (27.34). □

Proposition 27.15 (Order-zero area–energy identity)**.** *For every integer $N \geq 1$,*

$$\boxed{2\mathcal{A}_0(N) = M(N)I(N) - E(N).} \tag{27.36}$$

Proof. At $t = 0$, Proposition 27.14 gives

$$\Delta\mathcal{A}_0(N) = \frac{\mu(N)}{2}\left[M(N-1)\log N - L(N-1)\right]. \tag{27.37}$$

Abel summation gives the exact identity

$$L(N-1) = M(N-1)\log(N-1) - I(N-1), \tag{27.38}$$

and the step-function convention in Definition 27.7 gives

$$I(N) - I(N-1) = M(N-1)\log\frac{N}{N-1}. \tag{27.39}$$

Therefore

$$\Delta\mathcal{A}_0(N) = \frac{1}{2}\mu(N)I(N). \tag{27.40}$$

Summing over N yields

$$2\mathcal{A}_0(N) = \sum_{n\leq N}\mu(n)I(n). \tag{27.41}$$

Discrete summation by parts now gives

$$\begin{aligned} \sum_{n\leq N}\mu(n)I(n) &= M(N)I(N) - \sum_{n<N} M(n)\big(I(n+1) - I(n)\big) \\ &= M(N)I(N) - \sum_{n<N} M(n)^2\log\frac{n+1}{n} \\ &= M(N)I(N) - E(N), \end{aligned} \tag{27.42}$$

which proves (27.36). □

Corollary 27.16 (Central area at record times)**.** *At the record times,*

$$\boxed{\mathcal{A}_0(r_k) = o(k^2\log r_k).} \tag{27.43}$$

Proof. By Corollary 27.9, $M(r_k)I(r_k) = o(k^2\log r_k)$ and $E(r_k) = o(k^2\log r_k)$. Apply Proposition 27.15. □

27.6. **The first transverse jet is the original arithmetic cocycle.**

Proposition 27.17 (Cocycle as first transverse jet)**.** *One has*

$$\boxed{\mathcal{A}_0'(N) = \frac{1}{2} \sum_{m<n\le N} \mu(m)\mu(n)\mathscr{A}(m,n),} \tag{27.44}$$

where

$$\mathscr{A}(m,n) = \Omega(m)\log n - \log m\,\Omega(n). \tag{27.45}$$

Moreover,

$$\boxed{\Delta\mathcal{A}_0'(N) = \frac{\mu(N)}{2}\left[M_\Omega(N-1)\log N - \Omega(N)L(N-1)\right],} \tag{27.46}$$

where

$$M_\Omega(x) := \sum_{n\le x} \mu(n)\Omega(n). \tag{27.47}$$

Proof. Differentiate the finite sum (27.33) at $t = 0$. This gives (27.44) immediately. Differentiating the finite increment identity (27.34) at $t = 0$ gives (27.46), because $F_0'(N-1) = M_\Omega(N-1)$. □

Lemma 27.18 (Selberg–Delange input for the weighted Mertens sum)**.** *As $x \to \infty$,*

$$\boxed{M_\Omega(x) \sim \frac{x}{(\log x)^2}.} \tag{27.48}$$

Proof. For complex z in a fixed neighborhood of 1, the Dirichlet series is

$$D(s,z) := \sum_{n\ge1} \frac{\mu(n)z^{\Omega(n)}}{n^s} = \prod_p (1 - zp^{-s}) = \zeta(s)^{-z}G(s,z), \tag{27.49}$$

where

$$G(s,z) := \prod_p (1 - zp^{-s})(1 - p^{-s})^{-z} \tag{27.50}$$

is holomorphic in the standard Selberg–Delange region near $(s,z) = (1,1)$, and $G(1,1) = 1$. The uniform Selberg–Delange theorem for compact z-sets gives the corresponding asymptotic family; see Tenenbaum [15]. Its leading coefficient is

$$\frac{G(1,z)}{\Gamma(-z)} \frac{x}{(\log x)^{z+1}}. \tag{27.51}$$

Since

$$\frac{1}{\Gamma(-z)} = (z-1) + O((z-1)^2) \qquad (z \to 1), \tag{27.52}$$

and the Selberg–Delange expansion is locally uniform and holomorphic in z, differentiating at $z = 1$ yields

$$\frac{\partial}{\partial z} \sum_{n\le x} \mu(n)z^{\Omega(n)}\Big|_{z=1} = \frac{x}{(\log x)^2}(1 + o(1)). \tag{27.53}$$

The derivative on the left is precisely $M_\Omega(x)$. □

Theorem 27.19 (Local first-jet reconstruction of Möbius)**.** *As $N \to \infty$,*

$$\boxed{\Delta\mathcal{A}_0'(N) = \frac{\mu(N)N}{2\log N} + o\Big(\frac{N}{\log N}\Big).} \tag{27.54}$$

If $\mu(N) \neq 0$, then equivalently

$$\Delta\mathcal{A}_0'(N) = \frac{\mu(N)}{2}\frac{N}{\log N}(1 + o(1)). \tag{27.55}$$

If $\mu(N) = 0$, the left side of (27.54) *is exactly zero.*

Proof. By Lemma 27.18,

$$M_\Omega(N-1)\log N = \frac{N}{\log N}(1 + o(1)). \tag{27.56}$$

The classical zero-free-region estimate for Mertens sums gives, with some $c > 0$,

$$M(x) \ll x \exp\Big\{-c(\log x)^{3/5}(\log\log x)^{-1/5}\Big\}. \tag{27.57}$$

By Abel summation,

$$L(x) = M(x)\log x - I(x) \ll x\log x \exp\Big\{-c'(\log x)^{3/5}(\log\log x)^{-1/5}\Big\} \tag{27.58}$$

for a possibly smaller $c' > 0$. Since $\Omega(N) \leq \log N/\log 2$,

$$\Omega(N)L(N-1) = o\Big(\frac{N}{\log N}\Big). \tag{27.59}$$

Insert (27.56) and (27.59) into (27.46). This proves (27.54). If $\mu(N) = 0$, the exact identity (27.46) shows that the increment vanishes exactly. □

Corollary 27.20 (First jet at record times)**.** *Let $\sigma_k := \operatorname{sgn} M(r_k)$. Then*

$$\mu(r_k) = \sigma_k \tag{27.60}$$

and

$$\boxed{\Delta\mathcal{A}_0'(r_k) \sim \frac{\sigma_k}{2}\frac{r_k}{\log r_k}.} \tag{27.61}$$

Consequently,

$$\boxed{\frac{\log|\Delta\mathcal{A}_0'(r_k)|}{\log k} \longrightarrow \frac{1}{\Theta}.} \tag{27.62}$$

Proof. By minimality of r_k, one has $|M(r_k - 1)| \leq k - 1$, whereas $|M(r_k)| = k$. Since $M(r_k) - M(r_k - 1) = \mu(r_k) \in \{-1, 0, 1\}$, the final increment must have the same sign as $M(r_k)$, proving (27.60). Apply Theorem 27.19 at $N = r_k$ to obtain (27.61). Finally use Theorem 27.11; the factors 2 and $\log r_k$ contribute only $o(\log k)$ to the logarithm. □

27.7. **The centered jet and the short-interval comparison.**

Definition 27.21 (Centered first-jet remainder)**.** Define

$$\mathcal{R}_{\mathrm{H}}(N) := \Delta\mathcal{A}_0'(N) - \frac{\mu(N)N}{2\log N}. \tag{27.63}$$

Corollary 27.22. *One has*

$$\boxed{\mathcal{R}_{\mathrm{H}}(N) = o(N/\log N).} \tag{27.64}$$

Proof. This is exactly Theorem 27.19 with its leading term subtracted. □

Remark 27.23 (The 0.55 short-interval theorem). Matomäki–Teräväinen prove the independent uniform estimate

$$\sum_{x<n\le x+H}\mu(n)=o(H)\qquad (H\ge x^{\theta},\ \theta>0.55), \tag{27.65}$$

with a quantitative logarithmic saving in their proof [16]. This theorem is not needed for Theorems 27.4–27.19. Its role here is diagnostic: it constrains the ordinary-time width over which a fixed fraction of a record height can be accumulated, but it does not replace the Kaczorowski–Pintz oscillation input or determine the centered remainder $\mathcal{R}_{\mathrm{H}}$.

28. Squarefree critical deformation, full jet control, and a global quadratic transform

This section imports only the useful arithmetic content of the earlier squarefree-polynomial experiment and removes its matrix terminology. The finite polynomial, its roots, and its logarithmic derivatives are retained; the previous numerical Taylor truncations are replaced by compact-uniform Selberg–Delange estimates and Rouché arguments. The section then returns to the Heisenberg deformation and derives a global Perron transform for the quadratic first jet.

28.1. **The squarefree polynomial and its signed section.**

Definition 28.1 (Squarefree generating polynomial)**.** For $x\ge 2$, set

$$P_x(z):=\sum_{n\le x}\mu(n)^2 z^{\Omega(n)}. \tag{28.1}$$

Because $\mu(n)^2$ restricts to squarefree integers, $\Omega(n)=\omega(n)$ on the support of the sum. Put

$$Q_{\mathrm{sf}}(x):=P_x(1)=\sum_{n\le x}\mu(n)^2. \tag{28.2}$$

Proposition 28.2 (Exact signed section and deformation)**.** *For every real or complex t,*

$$P_x(-1)=M(x), \tag{28.3}$$

$$P_x(-e^t)=F_t(x):=\sum_{n\le x}\mu(n)e^{t\Omega(n)}. \tag{28.4}$$

Moreover, for $\Re s>1$,

$$\sum_{n\ge 1}\frac{\mu(n)^2 z^{\Omega(n)}}{n^s}=\prod_p(1+zp^{-s}). \tag{28.5}$$

Proof. If $\mu(n)^2=1$, then n is squarefree and $\mu(n)=(-1)^{\Omega(n)}$. This gives (28.3) and (28.4) term by term. Absolute convergence for $\Re s>1$ gives the Euler product (28.5) because each prime may occur at most once. □

Proposition 28.3 (Exact factorial moments at the signed point)**.** *For every integer $j\ge 1$,*

$$P_x^{(j)}(-1)=(-1)^j\sum_{n\le x}\mu(n)(\Omega(n))_j, \tag{28.6}$$

where $(y)_j=y(y-1)\cdots(y-j+1)$.

Proof. Differentiate the finite polynomial (28.1). On squarefree integers, $(-1)^{\Omega(n)-j}=(-1)^j\mu(n)$, which gives (28.6). □

28.2. **Uniform Selberg–Delange and the unique critical root.** For z in a fixed compact neighborhood of -1, write

$$\prod_p (1+zp^{-s}) = \zeta(s)^z H(s,z), \qquad H(s,z) := \prod_p (1+zp^{-s})(1-p^{-s})^z. \tag{28.7}$$

The linear terms in p^{-s} cancel in $\log H$, so H is holomorphic near $(1,-1)$, with $H(1,-1)=1$. Define

$$A(z) := \frac{H(1,z)}{\Gamma(z)}. \tag{28.8}$$

Then

$$A(-1)=0, \qquad A'(-1)=-1. \tag{28.9}$$

The uniform Selberg–Delange theorem gives, on a fixed closed disk $D=\{|z+1|\le \delta_0\}$,

$$P_x(z) = x(\log x)^{z-1}(A(z)+E_x(z)), \qquad \sup_{z\in D}|E_x(z)| \ll \frac{1}{\log x}. \tag{28.10}$$

We use this standard input in the form proved for compact complex-parameter sets in Selberg–Delange theory [15, 14].

Lemma 28.4 (Cauchy control of the Selberg–Delange error)**.** *If $D' \Subset D$, then*

$$\sup_{z\in D'} |E_x'(z)| \ll_{D',D} \frac{1}{\log x}. \tag{28.11}$$

Proof. The error in (28.10) is holomorphic in z on the fixed parameter disk. Cauchy's integral formula on a circle contained in D and surrounding each point of D' gives (28.11) from the uniform bound in (28.10). □

Theorem 28.5 (Unique critical root near -1)**.** *There exists $\delta_1>0$ such that for all sufficiently large x, P_x has exactly one zero, counted with multiplicity, in $|z+1|<\delta_1$. Denote it by $\xi_*(x)$. Then*

$$\xi_*(x)\in\mathbb{R}, \qquad \xi_*(x)\to -1, \tag{28.12}$$

and the zero is simple.

Proof. Choose δ_1 so that A has no zero in $0<|z+1|\le\delta_1$. On $|z+1|=\delta\le\delta_1$, $|A(z)|\ge c_\delta>0$. For sufficiently large x, $|E_x(z)|<|A(z)|$ on that circle. After division by the nonvanishing factor $x(\log x)^{z-1}$, Rouché's theorem shows that $A+E_x$, and hence P_x, has exactly one zero in the disk. Taking arbitrarily small fixed δ yields convergence to -1. The coefficients of P_x are real, so uniqueness forces the zero to be real; counting multiplicity forces simplicity. □

Theorem 28.6 (Sharp critical-root localization without a Taylor-tail assumption)**.** *Assume $M(x)\neq 0$ and put*

$$h_x := 1+\xi_*(x). \tag{28.13}$$

Then

$$\boxed{h_x = \frac{M(x)(\log x)^2}{x}\,(1+o(1)).} \tag{28.14}$$

Equivalently,

$$\boxed{M(x) = \frac{x}{(\log x)^2} h_x\,(1+o(1)).} \tag{28.15}$$

Proof. Set

$$e_x := E_x(-1). \tag{28.16}$$

Evaluating (28.10) at $z = -1$ and using $A(-1) = 0$ gives the exact identity

$$e_x = \frac{M(x)(\log x)^2}{x}. \tag{28.17}$$

The classical zero-free-region estimate for M implies $e_x \to 0$. For w in a fixed small disk, (28.9) and Lemma 28.4 give

$$A(-1+w) + E_x(-1+w) = -w + e_x + R_x(w), \tag{28.18}$$

with

$$|R_x(w)| \ll |w|^2 + \frac{|w|}{\log x}. \tag{28.19}$$

Fix $0 < \varepsilon < 1$ and consider the circle $|w - e_x| = \varepsilon|e_x|$. On that circle, $|w| \le (1+\varepsilon)|e_x|$, so

$$\sup |R_x(w)| = o(|e_x|), \tag{28.20}$$

whereas $|-w + e_x| = \varepsilon|e_x|$. Rouché's theorem therefore gives exactly one zero inside the circle. By Theorem 28.5 it must be $h_x = 1 + \xi_*(x)$. Since ε is arbitrary, $h_x = e_x(1 + o(1))$. Use (28.17) to obtain (28.14); inversion gives (28.15). □

Corollary 28.7 (Record exponent in the critical root)**.** *At the first absolute record time r_k,*

$$\boxed{|1 + \xi_*(r_k)| = r_k^{\Theta - 1 + o(1)}.} \tag{28.21}$$

Hence

$$\boxed{\lim_{k\to\infty} \frac{\log|1 + \xi_*(r_k)|}{\log r_k} = \Theta - 1.} \tag{28.22}$$

In particular,

$$\mathrm{RH} \iff \lim_{k\to\infty} \frac{\log|1 + \xi_*(r_k)|}{\log r_k} = -\frac{1}{2}. \tag{28.23}$$

Proof. At r_k, $|M(r_k)| = k = r_k^{\Theta + o(1)}$ by Theorem 27.11. Insert this into (28.14); the logarithmic factor contributes only $o(\log r_k)$. The equivalence follows because RH is equivalent to $\Theta = 1/2$. □

28.3. **What the remaining roots contribute.** Write the roots of P_x, counted with multiplicity, as $\xi_{x,1}, \dots, \xi_{x,d_x}$, with $\xi_{x,1} = \xi_*(x)$. Since

$$Q_{\mathrm{sf}}(x) = \frac{6}{\pi^2}x + O(\sqrt{x}), \tag{28.24}$$

we can separate the Mertens-dependent critical root from the rest.

Proposition 28.8 (Exact cancellation product)**.** *If $M(x) \neq 0$, then*

$$\frac{|M(x)|}{Q_{\mathrm{sf}}(x)} = \prod_{j=1}^{d_x} \frac{|1 + \xi_{x,j}|}{|1 - \xi_{x,j}|}. \tag{28.25}$$

Proof. Factor $P_x(z) = c_x \prod_j (z - \xi_{x,j})$. Evaluate at $z = -1$ and $z = 1$, take absolute values, and divide. □

Theorem 28.9 (Deterministic residual product of the noncritical roots)**.** *As $x \to \infty$ through values with $M(x) \neq 0$,*

$$\boxed{\prod_{j=2}^{d_x} \frac{|1+\xi_{x,j}|}{|1-\xi_{x,j}|} \sim \frac{\pi^2}{3(\log x)^2}.} \tag{28.26}$$

Thus the dependence on the size of $M(x)$ is asymptotically concentrated in the single critical factor.

Proof. From Theorem 28.6,

$$\frac{|1+\xi_*(x)|}{|1-\xi_*(x)|} \sim \frac{|M(x)|(\log x)^2}{2x}, \tag{28.27}$$

because $\xi_*(x) \to -1$. Divide (28.25) by (28.27) and use (28.24). □

Corollary 28.10 (Additive cancellation coordinate of the noncritical roots)**.** *Define*

$$c_{x,j} := -\log \frac{|1+\xi_{x,j}|}{|1-\xi_{x,j}|}. \tag{28.28}$$

Then

$$\boxed{\sum_{j=2}^{d_x} c_{x,j} = 2\log\log x + \log\frac{3}{\pi^2} + o(1).} \tag{28.29}$$

Proof. Take minus logarithms in (28.26). □

28.4. **Logarithmic-derivative traces: the previous trace obstruction collapses.**

Definition 28.11 (Root traces)**.** For $M(x) \neq 0$ and integer $m \geq 1$, define

$$T_m(x) := \sum_{j=1}^{d_x} (1+\xi_{x,j})^{-m}. \tag{28.30}$$

Proposition 28.12 (Exact logarithmic-derivative identity)**.** *For every $m \geq 1$,*

$$\boxed{T_m(x) = -\frac{1}{(m-1)!}\,\frac{d^m}{dz^m}\log P_x(z)\Big|_{z=-1}.} \tag{28.31}$$

Proof. From the root factorization, $P_x'(z)/P_x(z) = \sum_j (z-\xi_{x,j})^{-1}$. Differentiate $m-1$ times and evaluate at $z=-1$. The two sign factors combine to give exactly (28.31). □

Theorem 28.13 (Uniform critical-root dominance of all traces)**.** *Let $h_x = 1+\xi_*(x)$. Then*

$$\boxed{\sup_{m\geq 1} |T_m(x)h_x^m - 1| \longrightarrow 0} \tag{28.32}$$

as $x \to \infty$ through $M(x) \neq 0$. Consequently, uniformly for all integers $m \geq 1$,

$$T_m(x) = \left(\frac{x}{M(x)(\log x)^2}\right)^m (1+o(1)). \tag{28.33}$$

Proof. By Theorem 28.5, choose a fixed $\delta > 0$ such that for all sufficiently large x, every noncritical root satisfies $|1+\xi_{x,j}| \geq \delta$. Hence

$$T_m(x) = h_x^{-m} + R_m(x), \qquad |R_m(x)| \leq d_x \delta^{-m}. \tag{28.34}$$

Therefore, once $|h_x| < \delta$,

$$|R_m(x)h_x^m| \leq d_x \left(\frac{|h_x|}{\delta}\right)^m \leq \frac{d_x|h_x|}{\delta} \qquad (m \geq 1). \tag{28.35}$$

The degree d_x is the largest number of distinct prime factors possible below x, so $d_x = O(\log x/\log\log x)$. The classical Vinogradov–Korobov-type bound for M, together with (28.14), gives $d_x|h_x| \to 0$. This proves (28.32). Equation (28.33) follows from (28.14). □

Corollary 28.14 (Consecutive traces reconstruct the critical displacement)**.** *Uniformly for integers $m \geq 1$,*

$$\frac{T_m(x)}{T_{m+1}(x)} = h_x(1+o(1)), \tag{28.36}$$

$$M(x) = \frac{x}{(\log x)^2}\frac{T_m(x)}{T_{m+1}(x)}(1+o(1)). \tag{28.37}$$

Proof. Apply Theorem 28.13 to m and $m+1$ and divide. Then use (28.15). □

Remark 28.15 (What is closed from the earlier trace programme). The exact traces do not suffer asymptotically from persistent cancellation among the noncritical roots: the unique root near -1 dominates every power simultaneously. This closes the earlier structural "trace obstruction" at the level of reconstructing M. It does not by itself improve any bound on M, because the dominant root displacement is already asymptotically equivalent to $M(x)(\log x)^2/x$.

28.5. **Critical-scale limit and uniform control of the full jet tower.** Put

$$L_x := \log\log x, \qquad \Phi_x(u) := \frac{(\log x)^2 L_x}{x} F_{u/L_x}(x). \tag{28.38}$$

Theorem 28.16 (Compact-uniform critical-scale limit)**.** *For every compact $K \subset \mathbb{C}$,*

$$\boxed{\Phi_x(u) = ue^{-u} + O_K\Big(\frac{1}{L_x} + \frac{L_x}{\log x}\Big)} \tag{28.39}$$

uniformly for $u \in K$. In particular,

$$\boxed{\Phi_x(u) \longrightarrow ue^{-u}} \tag{28.40}$$

uniformly on compact subsets of $\mathbb{C}$.

Proof. By (28.4), $F_{u/L_x}(x) = P_x(z_x(u))$ with

$$z_x(u) := -e^{u/L_x}. \tag{28.41}$$

Uniformly for u in a fixed compact set,

$$z_x(u) + 1 = -\frac{u}{L_x} + O_K(L_x^{-2}), \tag{28.42}$$

$$A(z_x(u)) = \frac{u}{L_x} + O_K(L_x^{-2}), \tag{28.43}$$

$$(\log x)^{z_x(u)-1} = \frac{e^{-u}}{(\log x)^2}\left(1 + O_K(L_x^{-1})\right). \tag{28.44}$$

Insert these formulas into (28.10). The main term gives ue^{-u} after multiplication by $(\log x)^2 L_x/x$, while the Selberg–Delange error contributes $O_K(L_x/\log x)$. This proves (28.39). □

Corollary 28.17 (Uniform Cauchy control of growing jets)**.** *Define*

$$M_j(x) := \sum_{n\leq x} \mu(n)\Omega(n)^j = \partial_t^j F_t(x)\Big|_{t=0}. \tag{28.45}$$

Fix $R > 0$. There is $C_R > 0$ such that for every $j \geq 1$,

$$\boxed{\left|\frac{(\log x)^2 M_j(x)}{x L_x^{j-1}} - j(-1)^{j-1}\right| \leq \frac{j!}{(2R)^j} C_R\left(\frac{1}{L_x} + \frac{L_x}{\log x}\right).} \tag{28.46}$$

In particular, for every fixed $\varepsilon > 0$, the asymptotic

$$M_j(x) = j(-1)^{j-1} \frac{xL_x^{j-1}}{(\log x)^2}(1 + o(1)) \tag{28.47}$$

holds uniformly throughout any range of $j = j(x)$ for which the right side of (28.46) is $o(j)$; for example the safe range

$$j \le (1 - \varepsilon)\frac{\log L_x}{\log\log L_x} \tag{28.48}$$

is admissible for all sufficiently large x.

Proof. The function $\Phi_x(u) - ue^{-u}$ is holomorphic. Apply Cauchy's derivative estimate on $|u| \le R$ using the compact-uniform error bound on $|u| \le 2R$. Since

$$\Phi_x^{(j)}(0) = \frac{(\log x)^2}{xL_x^{j-1}} M_j(x) \tag{28.49}$$

and $(ue^{-u})^{(j)}|_{u=0} = j(-1)^{j-1}$, we obtain (28.46). Stirling's formula gives the displayed safe range. □

Corollary 28.18 (Uniform control of the whole Taylor tail)**.** *For every $R > 0$ there is $C'_R > 0$ such that for $|u| \le R$,*

$$\boxed{\left| \sum_{j>J} \frac{M_j(x)}{j!} \left(\frac{u}{L_x}\right)^j \right| \le C'_R \frac{x}{(\log x)^2 L_x}\, 2^{-J}} \tag{28.50}$$

for all sufficiently large x, uniformly in $J \ge 0$.

Proof. By Theorem 28.16, Φ_x is uniformly bounded on $|u| \le 2R$. Cauchy's coefficient estimate therefore bounds the coefficient of u^j in Φ_x by $C_R(2R)^{-j}$. Summing the resulting geometric tail for $|u| \le R$ and undoing the normalization in (28.38) gives (28.50). □

Corollary 28.19 (Effective limiting differential equation)**.** *On every compact set $K \subset \mathbb{C}$,*

$$(\partial_u + 1)^2 \Phi_x(u) \longrightarrow 0 \tag{28.51}$$

uniformly on K.

Proof. Compact-uniform convergence of holomorphic functions implies compact-uniform convergence of all derivatives on smaller compact sets. Apply this to (28.40) and use $(\partial_u + 1)^2(ue^{-u}) = 0$. □

Remark 28.20 (Gamma-shape coincidence)*.* For positive real u, the function ue^{-u} is the density of a $\Gamma(2, 1)$ law. Here, however, u is a complex deformation parameter. No probability measure is inferred from this formal coincidence.

28.6. **Perturbative spectral blindness of every fixed transverse jet.** The preceding compact-uniform theorem yields the corresponding first-jet calculation. Passing to the second transverse derivative does not leave the Selberg–Delange regime; neither does any fixed derivative order. The statement below is deliberately about the *leading term*. It does not assert that lower-order remainders contain no information about zeta zeros.

Theorem 28.21 (Universal leading term for every fixed Heisenberg jet)**.** *For each fixed integer $j \ge 1$, define*

$$\mathcal{J}_j(N) := \partial_t^j \mathcal{A}_t(N)\Big|_{t=0}. \tag{28.52}$$

Then

$$\boxed{\Delta\mathcal{J}_j(N) = \frac{j(-1)^{j-1}}{2}\,\mu(N)\frac{N(\log\log N)^{j-1}}{\log N} + o_j\!\left(\frac{N(\log\log N)^{j-1}}{\log N}\right).} \tag{28.53}$$

If $\mu(N) = 0$, the left side vanishes exactly.

Proof. Differentiate the finite exact identity (27.34) exactly j times at $t = 0$. Since $\partial_t^j F_t(N-1)|_{t=0} = M_j(N-1)$, one obtains

$$\Delta\mathcal{J}_j(N) = \frac{\mu(N)}{2}\Big[M_j(N-1)\log N - \Omega(N)^j L(N-1)\Big]. \tag{28.54}$$

For fixed j, Corollary 28.17 gives

$$M_j(x) = j(-1)^{j-1}\frac{x(\log\log x)^{j-1}}{(\log x)^2}(1 + o_j(1)). \tag{28.55}$$

The Vinogradov–Korobov-type estimate already used for $L(x)$ implies, for every fixed $B > 0$,

$$L(x) = O_B\!\left(\frac{x}{(\log x)^B}\right). \tag{28.56}$$

Since $\Omega(N) \le \log N/\log 2$, choose $B > j + 2$. Then

$$\Omega(N)^j L(N-1) = o_j\!\left(\frac{N(\log\log N)^{j-1}}{\log N}\right). \tag{28.57}$$

Insert (28.55) and (28.57) into (28.54). The exact factor $\mu(N)$ also proves the final assertion. □

Corollary 28.22 (The second jet is still Selberg–Delange)**.** *One has*

$$\boxed{\Delta\mathcal{A}_0''(N) = -\mu(N)\frac{N\log\log N}{\log N} + o\!\left(\frac{N\log\log N}{\log N}\right).} \tag{28.58}$$

Thus the second transverse jet is not the first place where a new exponent or a new zero-sensitive leading term appears.

Proof. Take $j = 2$ in Theorem 28.21. □

Corollary 28.23 (Common record exponent for the entire fixed tower)**.** *At the first absolute record times r_k, for every fixed $j \ge 1$,*

$$\boxed{\frac{\log|\Delta\mathcal{J}_j(r_k)|}{\log k} \longrightarrow \frac{1}{\Theta}.} \tag{28.59}$$

Proof. At a first absolute record, $\mu(r_k) = \operatorname{sgn} M(r_k) \neq 0$. Theorem 28.21 gives

$$|\Delta\mathcal{J}_j(r_k)| = r_k^{1+o(1)}, \tag{28.60}$$

because fixed powers of $\log r_k$ and $\log\log r_k$ contribute $r_k^{o(1)}$. The record law $r_k = k^{1/\Theta+o(1)}$ completes the proof. □

Remark 28.24 (What "spectrally blind" means here). The leading asymptotics in (28.53), including the entire compact critical-scale profile ue^{-u}, are obtained from the local Selberg–Delange structure at $s = 1$. Their coefficients do not involve Θ or individual nontrivial zeros. Consequently, evaluating those universal asymptotics at $N = r_k$ can reflect Θ only through the already zero-sensitive location of the record time. This is the precise, limited sense in which the fixed perturbative tower is called spectrally blind. No assertion is made about beyond-all-orders remainders or nonperturbative values of t.

Warning 28.25 (Closed route: increasing the fixed jet order). Moving from $\mathcal{A}_0'$ to $\mathcal{A}_0''$, or to any other fixed $\partial_t^j \mathcal{A}_t|_{t=0}$, does not by itself create a new route toward Θ: every such leading term is already forced by Theorem 28.21. The remaining candidates are genuinely nonlocal or nonperturbative: the accumulated quantities $\mathcal{J}_j(N)$, the complete $\mathcal{A}_t(N)$ outside the $t = O(1/\log\log N)$ window, the remainder after subtraction of the complete local Selberg–Delange expansion, and the global Perron–Mellin transform of Section 28.10.

28.7. **The inner Mertens scale and a second complete tail estimate.** Define, whenever $M(x) \neq 0$,

$$a_x := \frac{M(x)}{M_\Omega(x)}. \tag{28.61}$$

Since $M_\Omega(x) \sim x/(\log x)^2$ and the classical zero-free-region bound gives $M(x) = o_A(x/(\log x)^A)$ for every fixed $A > 0$, one has $a_x \to 0$.

Theorem 28.26 (Inner-scale linear limit). *For every compact $K \subset \mathbb{C}$,*

$$\boxed{\frac{F_{va_x}(x)}{M(x)} = 1 + v + o_K(1)} \tag{28.62}$$

uniformly for $v \in K$.

Proof. The exact exponential remainder gives

$$F_t(x) = M(x) + tM_\Omega(x) + R_x(t), \tag{28.63}$$

where

$$|R_x(t)| \le \frac{|t|^2}{2} \sum_{n \le x} \mu(n)^2 \Omega(n)^2 e^{|t|\Omega(n)} \tag{28.64}$$

$$\ll |t|^2 x (\log x)^2 \exp\left(\frac{|t|\log x}{\log 2}\right). \tag{28.65}$$

For $t = va_x$, uniformly on compact v-sets,

$$|t|\log x \ll_K \frac{|M(x)|(\log x)^3}{x} \to 0. \tag{28.66}$$

Hence

$$\frac{|R_x(va_x)|}{|M(x)|} \ll_K \frac{|M(x)|(\log x)^6}{x} \to 0. \tag{28.67}$$

Finally $a_x M_\Omega(x) = M(x)$, giving (28.62). □

Corollary 28.27 (Logarithmic coordinate of the critical root). *Let*

$$t_c(x) := \log(-\xi_*(x)), \tag{28.68}$$

which is real for all sufficiently large x. Then

$$\boxed{t_c(x) = -a_x(1 + o(1)) = -\frac{M(x)(\log x)^2}{x}(1 + o(1)).} \tag{28.69}$$

Proof. By definition $F_{t_c(x)}(x) = P_x(\xi_*(x)) = 0$. Theorem 28.26 has its unique limiting zero at $v = -1$; Rouché on a fixed small circle around -1 gives $t_c/a_x \to -1$. The second formula uses $M_\Omega(x) \sim x/(\log x)^2$. □

28.8. **The complete critical-scale law for the Heisenberg increment.**

Theorem 28.28 (Heisenberg critical-scale profile)**.** *Let K be a compact subset of $\{u \in \mathbb{C} : \Re u \le 0\}$. Then uniformly for $u \in K$,*

$$\boxed{\Delta\mathcal{A}_{u/L_N}(N) = \frac{\mu(N)N}{2\log N\, L_N} u e^{-u} + o_K\!\left(\frac{N}{\log N\, L_N}\right),} \tag{28.70}$$

where $L_N = \log\log N$.

Proof. Use the exact increment identity (27.34). By Theorem 28.16, replacing N by $N-1$ changes the renormalized first term by $o_K(1)$, and

$$F_{u/L_N}(N-1)\log N = \frac{N}{\log N\, L_N} u e^{-u} + o_K\!\left(\frac{N}{\log N\, L_N}\right). \tag{28.71}$$

For $\Re u \le 0$, $|e^{(u/L_N)\Omega(N)}| \le 1$. The Vinogradov–Korobov-type estimate used in the proof of Theorem 27.19 gives

$$L(N-1) = o\!\left(\frac{N}{\log N\, L_N}\right). \tag{28.72}$$

Insert these estimates into (27.34). □

Corollary 28.29 (Critical-scale profile at record times)**.** *If $\sigma_k = \operatorname{sgn} M(r_k)$, then for compact $K \subset \{\Re u \le 0\}$,*

$$\boxed{\frac{2\log r_k\, \log\log r_k}{\sigma_k r_k}\Delta\mathcal{A}_{u/\log\log r_k}(r_k) \longrightarrow u e^{-u}} \tag{28.73}$$

uniformly for $u \in K$.

Proof. Use $\mu(r_k) = \sigma_k$ from Corollary 27.20 in Theorem 28.28. □

Theorem 28.30 (Heisenberg increment at the critical root deformation)**.** *At $t = t_c(N)$, one has the exact identity*

$$\boxed{\Delta\mathcal{A}_{t_c(N)}(N) = -\frac{e^{t_c(N)\Omega(N)}}{2}\Big(\mu(N)^2\log N + \mu(N)L(N-1)\Big).} \tag{28.74}$$

At the first absolute record times,

$$\boxed{\Delta\mathcal{A}_{t_c(r_k)}(r_k) \sim -\frac{1}{2}k\log r_k.} \tag{28.75}$$

Proof. The equation $F_{t_c(N)}(N) = 0$ gives

$$F_{t_c(N)}(N-1) = -\mu(N)e^{t_c(N)\Omega(N)}. \tag{28.76}$$

Insert this into (27.34) to obtain (28.74). At $N = r_k$, $\mu(r_k) = \sigma_k$ and $M(r_k - 1) = \sigma_k(k-1)$. The global historical-average estimate (27.16) gives

$$L(r_k - 1) = \sigma_k(k-1)\log r_k + o(k\log r_k). \tag{28.77}$$

Also (28.69) and the classical zero-free-region bound imply $t_c(r_k)\Omega(r_k) \to 0$. Therefore the exponential factor tends to one and the bracket in (28.74) is $k\log r_k(1 + o(1))$. □

28.9. **The record-jump inversion law.**

Corollary 28.31 (A pure-arithmetic RH-equivalent record-jump exponent)**.** *The following are equivalent:*

$$\text{RH}, \tag{28.78}$$

and

$$\boxed{|\Delta\mathcal{A}_0'(r_k)| = k^{2+o(1)}.} \tag{28.79}$$

Proof. Corollary 27.20 gives

$$\frac{\log|\Delta\mathcal{A}_0'(r_k)|}{\log k} \to \frac{1}{\Theta}. \tag{28.80}$$

Thus (28.79) is equivalent to $1/\Theta = 2$, hence to $\Theta = 1/2$, which is RH. □

28.10. **A global Perron–Mellin transform for the quadratic first jet.** Define

$$a(n) := \mu(n)\Omega(n), \qquad b(n) := \mu(n)\log n. \tag{28.81}$$

Then the first-jet increment can be written

$$2\Delta\mathcal{A}_0'(n) = \sum_{m<n} \big(a(m)b(n) - b(m)a(n)\big). \tag{28.82}$$

For $\Re s > 1$, define

$$U(s) := \sum_{n\geq 1} \frac{a(n)}{n^s} = -\frac{1}{\zeta(s)} \sum_p \frac{1}{p^s - 1}, \tag{28.83}$$

$$V(s) := \sum_{n\geq 1} \frac{b(n)}{n^s} = \frac{\zeta'(s)}{\zeta(s)^2}. \tag{28.84}$$

Lemma 28.32 (Local audit at $s = 1$)**.** *As $s \to 1$ inside a fixed sector avoiding the negative real direction from 1,*

$$\sum_p \frac{1}{p^s - 1} = \log\frac{1}{s-1} + O(1), \tag{28.85}$$

$$U(s) = -(s-1)\log\frac{1}{s-1} + O(s-1), \tag{28.86}$$

$$V(s) = -1 + O(s-1). \tag{28.87}$$

Proof. The difference

$$\sum_p \left(\frac{1}{p^s - 1} - p^{-s}\right) \tag{28.88}$$

is normally convergent in a neighborhood of $s = 1$, because each summand is $O(p^{-2\Re s})$. The standard prime-zeta expansion gives $\sum_p p^{-s} = \log(1/(s-1)) + O(1)$ in the stated sector, proving (28.85). The Laurent expansion $\zeta(s) = (s-1)^{-1} + O(1)$ gives $1/\zeta(s) = (s-1) + O((s-1)^2)$, hence (28.86). Finally $V(s) = -(1/\zeta(s))' = -1 + O(s-1)$, which proves (28.87). □

Remark 28.33 (Why the local jet and the global transform are different)*.* Lemma 28.32 makes the dichotomy explicit. The local weighted sum M_Ω and every fixed transverse jet inherit their leading asymptotics from the logarithmic singular structure at $s = 1$. The Perron integrand below, however, contains both w and $s - w$ and can in principle develop singularities when nontrivial singularities of the two factors interact. Determining whether those candidate spectral terms survive the antisymmetrization is therefore a genuinely different problem from computing another fixed jet.

Proposition 28.34 (Perron representation of the global first-jet increments)**.** *Let $s \in \mathbb{C}$ with $\sigma := \Re s > 2$, and choose c with $1 < c < \sigma - 1$. Then the absolutely convergent Dirichlet series*

$$\mathscr{D}_{\mathcal{A}}(s) := \sum_{n\geq 1} \frac{2\Delta\mathcal{A}_0'(n)}{n^s} \tag{28.89}$$

admits the Perron-limit representation

$$\boxed{\mathscr{D}_{\mathcal{A}}(s) = \lim_{T\to\infty} \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{U(w)V(s-w) - V(w)U(s-w)}{w}\, dw.} \tag{28.90}$$

The limit is understood in the usual Perron sense; no contour displacement is asserted in this proposition.

Proof. For $m \neq n$ and $c > 0$, Perron's kernel gives

$$\mathbf{1}_{m<n} = \lim_{T\to\infty} \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \left(\frac{n}{m}\right)^w \frac{dw}{w}. \tag{28.91}$$

For fixed T, the double sums obtained after inserting (28.91) converge absolutely because $c > 1$ and $\sigma - c > 1$. Their products are exactly $U(w)V(s-w)$ and $V(w)U(s-w)$. The coefficient series in (28.89) is absolutely convergent for $\sigma > 2$, since $|2\Delta\mathcal{A}_0'(n)| \ll n(\log n)^2$. Passing through finite partial sums and then taking the standard Perron limit gives (28.90). □

Remark 28.35 (Explicit zeta content of the integrand)*.* If

$$\mathcal{P}_1(s) := \sum_p \frac{1}{p^s - 1}, \tag{28.92}$$

then

$$U(w)V(s-w) - V(w)U(s-w) = \frac{\mathcal{P}_1(s-w)\zeta'(w)\zeta(s-w) - \mathcal{P}_1(w)\zeta'(s-w)\zeta(w)}{\zeta(w)^2\zeta(s-w)^2}. \tag{28.93}$$

Thus the accumulated quadratic geometry is connected directly to zeta and prime-zeta singularities before any record subsequence is selected. This is qualitatively different from the local record-jump identity.

28.11. **Exact prime-power centering and channel decomposition.** For $\Re s > 1$, let

$$\mathcal{P}_1(s) = \sum_p \frac{1}{p^s - 1}.$$

Using the prime-zeta identity and grouping by the product of the two indices,

$$\boxed{\mathcal{P}_1(s) = \sum_{m\geq 1} \frac{\varphi(m)}{m} \log\zeta(ms) = \log\zeta(s) + H(s),} \tag{28.94}$$

where

$$\boxed{H(s) = \sum_p \sum_{r\geq 2} \left(1 - \frac{1}{r}\right) p^{-rs}.} \tag{28.95}$$

The second series is normally convergent on every closed half-plane $\Re s \geq \frac{1}{2} + \varepsilon$, so H is holomorphic in $\Re s > 1/2$.

Accordingly,

$$U(s) = -\frac{\log\zeta(s)}{\zeta(s)} - \frac{H(s)}{\zeta(s)}, \qquad \boxed{U_c(s) = -\frac{H(s)}{\zeta(s)}.} \tag{28.96}$$

Define h by

$$h(p^r) = 1 - \frac{1}{r} \quad (r \geq 2), \qquad h(n) = 0 \quad \text{otherwise.}$$

Then the Dirichlet coefficients of U_c satisfy

$$\boxed{u_c = -h * \mu,} \tag{28.97}$$

that is,

$$\boxed{u_c(n) = -\sum_{\substack{p^r \mid n \\ r \geq 2}} \left(1 - \frac{1}{r}\right) \mu\left(\frac{n}{p^r}\right).} \tag{28.98}$$

Hence

$$\boxed{A_c(x) = \sum_{n \leq x} u_c(n) = -\sum_{\substack{p^r \leq x \\ r \geq 2}} \left(1 - \frac{1}{r}\right) M\left(\left\lfloor \frac{x}{p^r} \right\rfloor\right).} \tag{28.99}$$

Every Mertens argument on the right is contracted by a prime power at least 4.

For each fixed integer $r \geq 2$, define

$$\boxed{U_r(s) = -\frac{P(rs)}{\zeta(s)}.} \tag{28.100}$$

Its coefficients and summatory function are

$$u_r(n) = -\sum_{p^r \mid n} \mu(n/p^r), \qquad A_r(x) = -\sum_{p^r \leq x} M\left(\left\lfloor \frac{x}{p^r} \right\rfloor\right). \tag{28.101}$$

Theorem 28.36 (Prime-power channels separate points)**.** *Let s_1, s_2 satisfy $\Re s_j > 1/2$. If*

$$P(rs_1) = P(rs_2) \qquad \textit{for every integer } r \geq 2,$$

then $s_1 = s_2$.

Proof. As $r \to \infty$,

$$P(rs) = 2^{-rs}\left(1 + O((2/3)^{r\Re s})\right).$$

Equality for all sufficiently large integers r implies $e^{-(s_1 - s_2)\log 2} = 1$. Subtracting the leading prime contribution and using

$$P(rs) - 2^{-rs} = 3^{-rs}\left(1 + O((3/5)^{r\Re s})\right)$$

gives $e^{-(s_1 - s_2)\log 3} = 1$. Since $\log 2/\log 3 \notin \mathbb{Q}$, both relations force $s_1 = s_2$. □

At a simple zero ρ of ζ,

$$U_c(\rho + z) = -\frac{H(\rho)}{\zeta'(\rho)}\frac{1}{z} + O(1), \qquad V(\rho + z) = \frac{1}{\zeta'(\rho)}\frac{1}{z^2} + O(1).$$

Thus for two simple zeros ρ_1, ρ_2 the leading antisymmetric local determinant is

$$\boxed{\frac{H(\rho_2) - H(\rho_1)}{\zeta'(\rho_1)\zeta'(\rho_2)}.} \tag{28.102}$$

The diagonal $\rho_1 = \rho_2$ therefore cancels identically. Channel by channel the corresponding determinant is proportional to $P(r\rho_2) - P(r\rho_1)$, and Theorem 28.36 shows that the full family of prime-power channels distinguishes every pair of distinct points in $\Re s > 1/2$.

29. Summary of principal results

(1) The arithmetic Weyl embedding produces the cocycle

$$\chi_\vartheta(m,n)=\exp\Big[\frac{i\vartheta}{2}\big(\Omega(m)\log n-\log m\,\Omega(n)\big)\Big]$$

and an associative twisted Dirichlet algebra.

(2) The twisted Möbius inverse has nonnegative second deformation curvature

$$-z_0''(n)=\frac{1}{2}\sum_{i<j}T(a_i)T(a_j)\log^2(p_j/p_i),$$

with a prime-log variance representation and multiplicative superadditivity.

(3) The curvature Dirichlet series is a prime-log covariance determinant, and grouped prime-zeta continuation gives exact replica filtering and local monodromy formulas.

(4) The Mertens historical maximum satisfies

$$A^*(x)=x^{\Theta+o(1)},\qquad r_k=k^{1/\Theta+o(1)},$$

and the logarithmic past collapses in every fixed L^p scale at record times.

(5) The auxiliary Heisenberg area satisfies

$$2\mathcal{A}_0(N)=M(N)I(N)-\int_1^N\frac{M(t)^2}{t}\,dt,$$

and its first transverse derivative is the arithmetic cocycle.

(6) Uniform Selberg–Delange theory gives

$$1+\xi_*(x)=M(x)\frac{(\log x)^2}{x}(1+o(1))$$

and

$$\frac{(\log x)^2\log\log x}{x}F_{u/\log\log x}(x)\longrightarrow ue^{-u}$$

uniformly on compact subsets of the complex u-plane. The fixed-jet tower and its Taylor tails follow by Cauchy estimates.

(7) The accumulated first jet admits the Perron–Mellin representation

$$\mathscr{D}_{\mathcal{A}}(s)=\frac{1}{2\pi i}\int\frac{U(w)V(s-w)-V(w)U(s-w)}{w}\,dw$$

in its initial domain.

(8) The decomposition $\mathcal{P}_1=\log\zeta+H$ produces the exact centered factor $U_c=-H/\zeta$, the coefficients (28.98), the contracted Mertens representation (28.99), and the separating family $U_r=-P(rs)/\zeta$.

(9) In the centered local kernel the diagonal spectral contribution cancels exactly; the remaining pair interaction is governed by $H(\rho_2)-H(\rho_1)$, or channelwise by $P(r\rho_2)-P(r\rho_1)$.

References

[1] H. P. Robertson, *The uncertainty principle*, Physical Review **34** (1929), 163–164. doi:10.1103/PhysRev.34.163.

[2] A. Kleppner, *Multipliers on Abelian groups*, Mathematische Annalen **158** (1965), 11–34.

[3] L. Baggett and A. Kleppner, *Multiplier representations of Abelian groups*, Journal of Functional Analysis **14** (1973), 299–324. doi:10.1016/0022-1236(73)90075-X.

[4] L. Rigal and P. Zadunaisky, *Twisted semigroup algebras*, Algebras and Representation Theory **18** (2015), no. 5, 1155–1186. doi:10.1007/s10468-015-9525-z; arXiv:1406.2985.

[5] W. Heisenberg, *Über den anschaulichen Inhalt der quantentheoretischen Kinematik und Mechanik*, Zeitschrift für Physik **43** (1927), 172–198. doi:10.1007/BF01397280.

[6] G. B. Folland, *Harmonic Analysis in Phase Space*, Annals of Mathematics Studies 122, Princeton University Press, Princeton, 1989.

[7] M. Reed and B. Simon, *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, revised and enlarged ed., Academic Press, 1980.
[8] C.-E. Fröberg, *On the prime zeta function*, BIT Numerical Mathematics **8** (1968), 187–202. doi:10.1007/BF01933420.
[9] J. A. Packer and I. Raeburn, *On the structure of twisted group C^*-algebras*, Transactions of the American Mathematical Society **334** (1992), 685–718. doi:10.2307/2154478.
[10] T. M. Apostol, *Introduction to Analytic Number Theory*, Springer, New York, 1976. doi:10.1007/978-1-4757-5579-4.
[11] G. J. Székely and M. L. Rizzo, *Energy statistics: A class of statistics based on distances*, Journal of Statistical Planning and Inference **143** (2013), no. 8, 1249–1272. doi:10.1016/j.jspi.2013.03.018.
[12] H. Herichi and M. L. Lapidus, *Truncated infinitesimal shifts, spectral operators and quantized universality of the Riemann zeta function*, Annales de la Faculté des Sciences de Toulouse, Série 6, **23** (2014), no. 3, 621–664. doi:10.5802/afst.1419.
[13] J. Kaczorowski and J. Pintz, *Oscillatory properties of arithmetical functions. I*, Acta Mathematica Hungarica **48** (1986), no. 1–2, 173–185. doi:10.1007/BF01949062.
[14] R. de la Bretèche and G. Tenenbaum, *Remarks on the Selberg–Delange method*, Acta Arithmetica **200** (2021), 349–369. doi:10.4064/aa201024-26-5.
[15] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, 3rd ed., Graduate Studies in Mathematics, vol. 163, American Mathematical Society, Providence, RI, 2015. doi:10.1090/gsm/163.
[16] K. Matomäki and J. Teräväinen, *On the Möbius function in all short intervals*, Journal of the European Mathematical Society **25** (2023), no. 4, 1207–1225. doi:10.4171/JEMS/1205; arXiv:1911.09076.
[17] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford University Press, Oxford, 1986.
[18] A. Weil, *Sur les "formules explicites" de la théorie des nombres premiers*, Comm. Sém. Math. Univ. Lund [Medd. Lunds Univ. Mat. Sem.], Tome Supplémentaire (1952), 252–265.
[19] E. Bombieri, *Remarks on Weil's quadratic functional in the theory of prime numbers, I*, Atti Accad. Naz. Lincei Cl. Sci. Fis. Mat. Natur. Rend. Lincei Mat. Appl. (9) **11** (2000), no. 3, 183–233. MR1841692; Zbl 1008.11034.